Energy System Models

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Contents

1 Introduction

This note deals with arguably *the* main tool that researchers, governments, AI agents etc. reach for when asked to quantitatively assess practically any aspect of the green transition: Energy system models. We use it to construct detailed pathways to net zero emissions, to analyse the effects of carbon pricing, evaluate the value of offshore wind, whether there is a case to invest in more interconnectors, how much a climate target costs and so on.

While detailed energy system models are inherently deeply dependent on technologies and insights from engineering, it can be fascinating and indeed incredibly useful to look at it through the lens of economics. Each model is a central planner's problem: minimise the cost of serving energy demand, subject to physics (e.g. energy balances, capacity limits, weather patterns) and policy (e.g. carbon budgets, land constraints). For economists, the level of technical detail may at times feel discomforting, but by relying on the good old fundamental welfare theorems, the planner's solution can be interpreted as a competitive market equilibrium.[1]

The note presumes that the reader is somewhat familiar with various optimisation techniques and types of problems including Lagrange, Karush-Kuhn-Tucker, and linear programming. Appendix A provides a refresher on the basics. As the note focuses on the models and applications, it also presumes that the reader has access to some supplementary material e.g. textbooks like Cretì and Fontini (2019) or Mulder (2023). All data, figures, tables, and models are collected in a repository here. The repository is equipped with instructions for how to use the models to replicate tables and figures from the note and how to explore further. It is also equipped with a set of instructions that allow you to engage with the repository using your favourite AI assistant. However, **nothing in this note requires you to run anything.**

1.1 Where these models sit

Before we start building, it is worth knowing what kind of animal we are dealing with, because you have met its rivals in other courses.

Energy economics is served by two modelling traditions. A **top-down** model writes aggregate energy use as a behavioural relationship, $E_t = F(y_t; \gamma_t)$, where y_t collects prices and income and the parameters γ_t are *estimated from observed behaviour* (the CGE and econometric models you have seen elsewhere). A **bottom-up** model, which is what this note is about, writes the same quantity as a sum over individual pieces of equipment, $E_t = \sum_i E_{i,t}(x_t; \theta_{i,t})$, where the parameters $\theta_{i,t}$ (efficiencies, capacities, costs) come from *engineering data* rather than from regressions, and the decision rule $E_{i,t}(\cdot)$ comes from a central optimisation rather than from estimated behaviour.

Each buys something and pays for it. Bottom-up models carry enormous technological detail: they know that a combined-cycle plant is 58% efficient, that the wind blew hard on 14 January, and that a cable to Norway has 1.7 GW of capacity. That detail is exactly what you need to

[1]For students with an engineering background, the note may feel rudimentary in terms of technical detail, but hopefully provide plenty of new insights from the economics interpretation.

answer a question about *technologies*, and it is forward-looking in a way estimated models cannot be: A technology that has never been deployed has no history to estimate from. What they pay is scope and discipline. There is no individual optimisation in a planner's problem, so there is no misaligned incentive, no strategic behaviour, no market power and no policy failure; and because the parameters are not identified from observed behaviour, validating the model is genuinely hard. Throughout the note, we will attempt to validate things empirically holding model output against market data or refer to academic literature that does.

1.2 How the note is built

The note is organised as one model growing up. Section 2 is the seed: a single hour, ten technologies, and the entire dual vocabulary: merit order, price, rents, the value of lost load, tax versus cap. Section 3 adds time and weather, and with them the economics of intermittency: prices become distributions, and a technology's value comes to depend on *when* it produces. Section 4 couples the hours through a battery and through a ramping constraint; Section 5 couples electricity to heat through a heat pump; Section 6 couples places through interconnectors. Every coupling is one new component and one new dual, never a new theory. Section 7 then makes capacity itself a choice: investment, screening curves, and the long-run zero-profit condition that ties the rents of earlier sections to cost recovery. Section 8 runs the same machinery over Denmark's whole neighbourhood: one carbon budget across twelve zones, interconnectors that can be built, flexible vehicle charging and a hydrogen sector, and asks which of the Danish lessons survive the change of scope. Section 9 closes by turning the tools on the model itself: how much of a capacity expansion answer is an artefact of the weather year it saw, and what the linear planner structurally cannot say.

Five appendices support the text. Appendix A is a refresher on the Lagrangian-KKT-duality toolkit the main sections use freely; Appendix B covers annuities, levelised costs and screening curves; Appendix C records where every number comes from; Appendix D is an inventory of what ingredients the models cover and what they do not; and Appendix E places the note's results against the recent literature on the European system.

Finally, some comments on the structure:

i. Most sections open with a complete optimisation problem: objective, constraints, and every multiplier named. Because each model is the previous one plus a little, whatever is new is printed in blue.

ii. Each model section closes with the same small table: constraint, dual, what it measures, and its market counterpart. By Section 9 the tables stack into one mental map from optimisation to market. Learn to fill this table in for any model you meet and you have learned what these models are for.

iii. A handful of important results are set off and numbered (the merit order, tax-cap equivalence, the arbitrage band, the ramping wedge, long-run zero profit).

iv. Every model simplifies, and it is important to be able to know which direction it pushes the model. Most sections set up boxes that take one assumption at a time and reason out the sign of its error, usually without solving anything.

v. Wherever the model has a real-world counterpart we put the two side by side: model prices against Nord Pool prices, model congestion rents against the bottleneck income the TSOs actually collected. The disagreements teach as much as the fits, and each one is traceable to a named modelling assumption.

2 The Economic Dispatch

The first model is incredibly simple: Given a fixed set of power plants and a known level of demand for electricity, what is the cost-effective way to serve this demand. This is the *economic dispatch* problem. We will use this to very briefly talk about the marginal costs of operating different power plants, then practice setting up the linear program, and interpreting the planner solution in terms of more familiar terms like equilibrium prices, operating profits, and the cost of carbon constraints.

> **Reading alongside this section.** This section assumes only that you can set up a Lagrangian and read KKT conditions; Appendix A is a refresher if one is needed. For institutional background on how a wholesale electricity market is actually organised, who bids what, when, and into which auction, see e.g. Mulder (2023, ch. 1–3) or Cretì and Fontini (2019, ch. 3–5). The lecture note "Generation Technologies" (Berg 2026b) provides an introduction to different power plants and their characteristics. For the interested reader, Dechezleprêtre et al. (2023) holds some interesting evidence on carbon-pricing in practice (focus on EU ETS).

2.1 Technologies and marginal cost

In this simple linear model, we characterize power plants – here indexed g – using only a handful of fixed parameters: First, they have a fixed size, represented by their *generating capacity* $\bar{q}_g$ expressed in MW, which states what the total amount of power (MWh) the plant can deliver in a single hour. Second, we assume that they have a constant *conversion efficiency* η_g, meaning that for each MWh of primary energy inputs, it produces η_g units of MWh electricity. Third, they have constant other non-energy *variable operating costs* o_g per MWh of power output (measured in €/MWh). Fourth, if the power plant relies on fuel injection as the primary energy input and φ_g measures the CO_2 content in tonnes/MWh, the plant emits $e_g = \varphi_g/\eta_g$ tonnes CO_2 per MWh of electricity it produces; This is the plant's *emission intensity.*

Formally, this means that the *marginal costs* c_g and the *emission intensity* e_g are given by:

$$c_g = \frac{p_g^F}{\eta_g} + o_g. \tag{1}$$

Table 1: The small technology table. Marginal costs and emission intensities are derived, not assumed; fuel prices are stylised 2025 forward-market levels.

Technology	Fuel	p_g^F €/MWh$_{th}$	η_g	o_g €/MWh	φ_g t/MWh$_{th}$	c_g €/MWh	e_g t/MWh
Solar PV[a]	–	–	–	0.0	–	0.0	0.00
Onshore wind[b]	–	–	–	1.5	–	1.5	0.00
Offshore wind[c]	–	–	–	0.0	–	0.0	0.00
Waste CHP[d]	waste	0	0.21	28.5	0.15	28.5	0.73
Coal[e]	coal	12	0.36	3.3	0.34	37.0	0.94
Wood-chip CHP[f]	solid biomass	25	0.27	4.9	0.00	97.7	0.00
Wood-pellet CHP[g]	wood pellets	38	0.27	4.9	0.00	145.9	0.00
Gas CCGT[h]	gas	35	0.57	4.6	0.20	66.0	0.35
Gas OCGT[i]	gas	35	0.41	4.8	0.20	91.2	0.49
Oil peaker[j]	oil	55	0.35	6.3	0.26	163.5	0.73

[a] Utility-scale photovoltaics. No fuel, and no variable operating cost in the source. Technology-data key `solar-utility`..
[b] Onshore wind turbines. Technology-data key `onwind`..
[c] Offshore wind turbines. The variable operating cost is nominal. Technology-data key `offwind`..
[d] Waste-to-energy plant. The fuel is free at the gate (gate-fee revenue ignored) but the plant carries a large variable operating cost; only the fossil share of municipal waste is counted as emitting. Technology-data key `waste CHP`..
[e] Hard-coal plant. Technology-data key `coal`..
[f] District-heating biomass plant, priced on wood chips; biogenic CO_2 counted as zero. Technology-data key `central solid biomass CHP`..
[g] The same plant burning pellets: identical machine, more expensive fuel. Technology-data key `central solid biomass CHP`..
[h] Combined-cycle gas turbine. Technology-data key `CCGT`..
[i] Open-cycle gas turbine – the same fuel, a cheaper and less efficient machine. Technology-data key `OCGT`..
[j] Oil-fired peaking plant. Technology-data key `oil`..

$$e_g = \frac{\varphi_g}{\eta_g}. \tag{2}$$

Constant marginal costs up to a hard capacity limit is the modelling decision that makes everything in this note linear, and it is a good first approximation for some technologies.[2]

To get a sense of how different technologies compare, Table 1 presents main technologies from the Danish Energy Agency (DEA).[3] The table is deliberately small enough to redo every calculation in this section by hand, and it is generated from the same file the model reads, so the numbers here are the numbers behind every figure.

Wind and solar, naturally, have no fuel inputs and generally near-zero marginal costs. In our table, waste incineration plants also feature very low marginal costs, because they are essentially paid to handle the waste. Beyond these generator types, marginal costs ranks then (from low-to-high) coal, then gas, biomass, and finally oil peaker; unfortunately, this is the same ranking in terms of emission intensity (except for the oil peaker). That tension – the cheap technologies are either capacity-limited or dirty – is what makes the dispatch problem economically interesting, even in this highly stylized framework.

[2] It is worst for plants with minimum load levels, start-up costs and ramping limits — the *unit-commitment* features that make the true problem non-convex. We return to what this omission does to prices in Section 9.5.

[3] Parameters are drawn from PyPSA, Danish Energy Agency, et al. (2025), which draws chiefly on the Danish Energy Agency's catalogue (Danish Energy Agency and Energinet 2024), at a pinned version; the fuel prices and carbon contents are the note's own assumptions, and are used unchanged throughout the note.

2.2 The planner's problem

We now consider a static model of the electricity wholesale market, with a fixed set of generators $g \in \mathcal{G}$ with characteristics as outlined above, and a fixed demand or *load* D to serve. The planner's optimisation problem is to choose how much each generator produces $q_g \geq 0$ to minimise costs:

$$\min_{q} \quad \sum_{g \in \mathcal{G}} c_g \, q_g \tag{3a}$$

$$\text{s.t.} \quad \sum_{g \in \mathcal{G}} q_g = D \qquad\qquad (\lambda) \tag{3b}$$

$$0 \leq q_g \leq \bar{q}_g \quad \forall g \qquad\qquad (\underline{\mu}_g, \overline{\mu}_g), \tag{3c}$$

Here (3a) is the total system costs, which at this stage only includes the sum of marginal costs of generation. (3b) is the market clearing condition. (3c) is the capacity and non-negativity constraints for each g. Variables in parenthesis (λ, $\overline{\mu}_g$, $\underline{\mu}_g$) are dual variables/shadow variables/Lagrange multipliers on the relevant constraint; they are the subject of Section 2.3.

We note that because we take demand D as given (i.e. demand is perfectly inelastic), the objective of minimising the system costs follows trivially. However, once we start dealing with where this demand comes from, in particular if there are different types of consumers, it is no longer trivial to identify one "correct" objective function. Below, we discuss the simplest adjustment with a consumer that has a *value of lost load*; later we also return to this issue again to discuss winners and losers of various policies.

Before we turn to the formal part, it can be instructive to start with a visualisation of this model in Figure 2.1, as it becomes clearly evident what the optimal solution is. The demand side of this market is straightforward: There is simply a fixed load of $D = 5$ GW represented by the vertical dashed line. On the supply side, order the generators by marginal cost, $c_{(1)} \leq c_{(2)} \leq \dots$ and fill up demand from the bottom: run the cheapest plant at capacity, then the next cheapest, until load is met. This generation schedule is called *merit order dispatching.* In our case, solar, wind and waste run first, then coal, and the CCGT fleet tops up: the merit order dispatch is exactly the supply curve of a competitive market, and the intersection with the vertical demand curve at the marginal generator's cost is the market equilibrium. Let us now look at this formally.

2.3 Duality: the price and the rents

The multipliers named in parentheses in (3) are where the economics live, and extracting them is a routine application of the Kuhn–Tucker technique. Following the convention of Appendix A — the Lagrangian is always set up for a *maximisation* — we restate the problem as maximising *minus* the system costs and define the Lagrangian

$$\mathcal{L}^{\text{disp}} = -\sum_{g} c_g q_g + \lambda\Big(\sum_{g} q_g - D\Big) + \sum_{g} \overline{\mu}_g \, (\bar{q}_g - q_g) + \sum_{g} \underline{\mu}_g q_g, \tag{4}$$

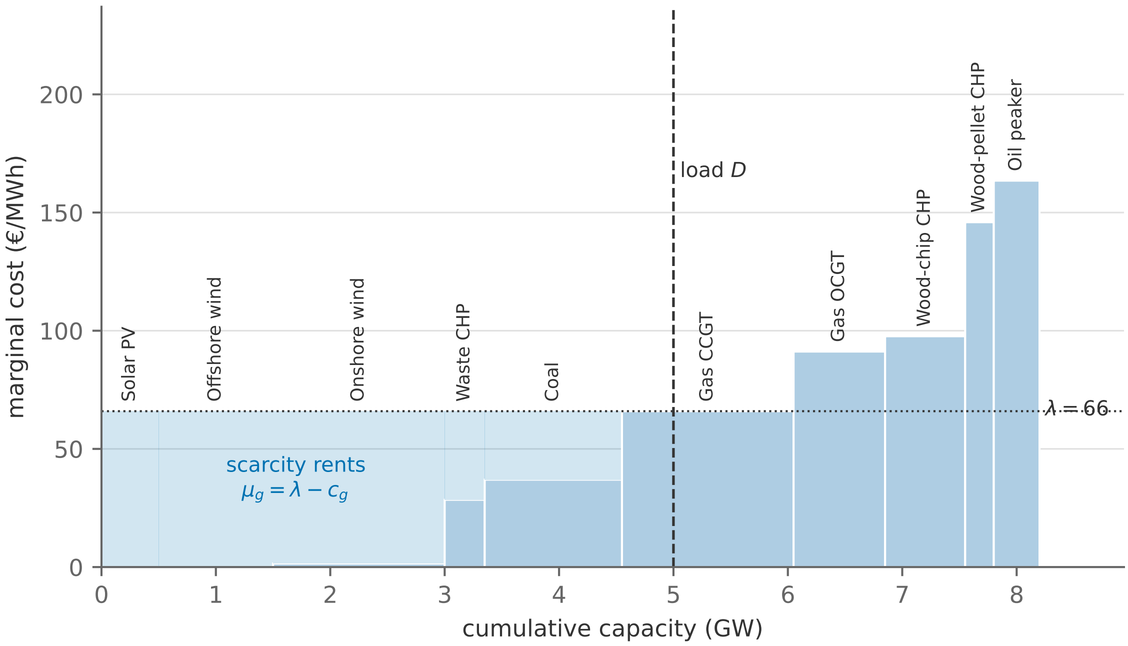

Figure 2.1: The merit order for the instance of Table 1, with load $D = 5$ GW. Each block is one technology, ordered by marginal cost; block width is capacity. The marginal generator is the CCGT, and the price is its marginal cost, $\lambda = 66.0$ €/MWh. The shaded area between the price and the supply curve is the sum of scarcity rents $\sum_g \bar{\mu}_g \bar{q}_g$ — the producer surplus.

where $\lambda, \bar{\mu}_g \geq 0, \underline{\mu}_g \geq 0$ are the shadow variables associated with the market clearing condition, the capacity constraint, and the non-negativity constraint, respectively.

A word on the superscript, which is a convention we keep for the rest of the note. Each section adds one piece of machinery to the model, and each therefore has its own Lagrangian; the superscript names which one. Because the Lagrangian is *linear* in the constraints, a section's Lagrangian is always the previous one plus the terms its new machinery contributes — never a fresh derivation. $\mathcal{L}^{\text{disp}}$ here is the base case; Section 3 repeats it over the hours of a year as $\mathcal{L}^{\text{vre}}$, and Sections 4 to 8 each add to that. When we say "the generators' conditions are unchanged", this additive structure is why.

The conditions for optimality are then

$$\lambda - c_g + \underline{\mu}_g - \bar{\mu}_g = 0 \qquad \forall g, \tag{5a}$$

as well as the complementary slackness conditions

$$\bar{\mu}_g(\bar{q}_g - q_g) = 0, \qquad \bar{q}_g - q_g \geq 0, \qquad \bar{\mu}_g \geq 0, \qquad \forall g, \tag{5b}$$

$$\underline{\mu}_g q_g = 0, \qquad q_g \geq 0, \qquad \underline{\mu}_g \geq 0, \qquad \forall g, \tag{5c}$$

One important and general insight when interpreting KKT conditions is related to the pairs of shadow variables of upper ($\bar{\mu}_g$) and lower ($\underline{\mu}_g$) bounds: Only one of the pairs of shadow variables can be active (> 0) at any point. To see this, consider our small example here: If $\bar{\mu}_g > 0$ the complementary slackness condition (5b) states that $q_g = \bar{q}_g > 0$. This in turn means that $\underline{\mu}_g = 0$ for the second complementary slackness condition (5c) to hold.

These conditions are worth dwelling on a bit more, at least as good practice, as they are the formalisation of the merit order dispatch solution.

Result 1 (Merit order and the price)**.** *At the cost-minimising dispatch, generators run in order of marginal cost: every generator with $c_g < \lambda$ runs at capacity, every generator with $c_g > \lambda$ is idle, and the multiplier on market clearing equals the marginal cost of the marginal generator, $\lambda = c_m$. Moreover, this dispatch and this λ are a competitive equilibrium: if every generator takes λ as a price and maximises profit, markets clear.*

To see it, run through the three positions a generator can occupy:

- $0 < q_g < \bar{q}_g$: both slackness conditions force $\overline{\mu}_g = \underline{\mu}_g = 0$, so eq. (5a) reads $c_g = \lambda$. A partially dispatched generator must be *marginal*, and it pins down λ.
- $q_g = \bar{q}_g$: now $\underline{\mu}_g = 0$ and $\overline{\mu}_g = \lambda - c_g \geq 0$. A generator runs at capacity exactly when it is cheaper than the marginal one.
- $q_g = 0$: now $\overline{\mu}_g = 0$ and $\underline{\mu}_g = c_g - \lambda \geq 0$. A generator sits idle exactly when it is dearer.

Collecting the multipliers also delivers the LP dual of (3),[4]

$$\min_{\lambda,\, \overline{\mu} \geq 0} \quad \sum_{g \in \mathcal{G}} \overline{\mu}_g \bar{q}_g - \lambda D \tag{6a}$$

$$\text{s.t.} \quad \lambda - \overline{\mu}_g \leq c_g \quad \forall g, \tag{6b}$$

whose optimum equals the primal's maximised objective — minus total system costs (strong duality; Appendix A). The dual objective will earn its keep below: its two terms are producer surplus and consumer expenditure. The economics enter through the interpretation of the multipliers.

λ is the price. Formally, λ is the shadow price of eq. (3b): the increase in minimised system cost from one additional MWh of load. By the argument above it equals the marginal generator's cost, $\lambda = c_m$. But it is also the market-clearing price of a competitive market. Suppose each generator takes a price λ as given and chooses output to maximise profit, $\max_{0 \leq q_g \leq \bar{q}_g} (\lambda - c_g)\, q_g$. Its supply correspondence is: nothing if $c_g > \lambda$, everything if $c_g < \lambda$, anything if $c_g = \lambda$. At $\lambda = c_m$ these choices can be made to sum exactly to D — the market clears, and the resulting dispatch is the planner's. This is the first welfare theorem in its smallest habitat, and it is the licence for the phrase used throughout this note: *the model's dual variable is the market's price.* A single LP therefore answers both a planning question (what should run) and a market question (what will the price be).

[4] Note that because lower bound on production is zero, $\underline{q}_g = 0$, the full dual problem actually includes the sum $\sum_g \underline{\mu}_g \underline{q}_g$, which by construction is always zero and can thus be dropped here.

$\overline{\mu}_g$ is a scarcity rent. The multiplier on eq. (3c) is the value of one more MW of capacity of type g: relaxing the constraint saves $\lambda - c_g$ by displacing the marginal generator. For every inframarginal plant, $\overline{\mu}_g = \lambda - c_g$ is exactly its operating profit per MWh — the margin it earns selling at λ what it produces at c_g. In our base run the onshore wind fleet earns $\mu = 64.4$ €/MWh and coal earns $\mu = 29.0$ €/MWh, while the marginal CCGT earns nothing. Summing over plants, producer surplus is $\sum_g \overline{\mu}_g \bar{q}_g$ — the shaded area in Figure 2.1, and precisely the first term of the dual objective (6a).

Two remarks on the rents. First, they are *rents* in the classical sense: returns to a scarce factor (capacity), not profits from any productive cleverness, and taxing them away would not change today's dispatch. But today's rents are tomorrow's investment signal — in Section 7 the condition that a technology's rents, summed over the year, exactly cover its annualised capital cost is the long-run equilibrium condition, an idea going back to Boiteux (1960). A plant that never earns a rent is a plant nobody would build. Second, the marginal plant earning zero operating profit in the single period is not a pathology of the model; it is the knife-edge that makes the level of the price determinate.

2.4 The value of lost load

There is an assumption buried in eq. (3b) that we should drag into the open: Writing market clearing as an *equality* says that the load will be served — full stop, whatever it costs. If total capacity were below D, the problem would simply have no feasible solution. In effect we have told the planner that electricity is infinitely valuable.

Real systems do not work that way. When supply cannot meet demand, some demand is cut: interruptible industrial contracts are called, then rolling blackouts. So let us put that possibility into the model and see what it does. Suppose consumers are willing to pay v euros per MWh to avoid having their load cut — the *value of lost load* (VoLL). Let $d \in [0, D]$ be the load actually served, so that $D - d$ is the load shed. The planner now maximises welfare rather than minimising cost:

$$\max_{q,\, d} \quad v\, d - \sum_g c_g\, q_g \tag{7a}$$

$$\text{s.t.} \quad \sum_g q_g = d \qquad (\lambda) \tag{7b}$$

$$0 \le q_g \le \bar{q}_g \quad \forall g \qquad (\underline{\mu}_g, \overline{\mu}_g)$$

$$0 \le d \le D \qquad (\underline{\theta},\ \overline{\theta}). \tag{7c}$$

Everything in blue is new; everything else is equations (3) unchanged. The generators' first-order condition (5a) is untouched. The new one is for d:

$$v - \lambda + \underline{\theta} - \overline{\theta} = 0, \tag{8}$$

with the usual complementary slackness on the two bounds. Read it exactly as we read the generators' condition — by asking, one at a time, where d can sit:

- Upper bound: no load shedding. There is enough cheap capacity to serve everything. Then $d = D$, $\underline{\theta} = 0$, and $\overline{\theta} = v - \lambda > 0$: consumers value the served load above what it cost to serve. The price is set by the marginal *generator*, exactly as before, and everything in Section 2.3 goes through word for word.
- Interior: partial load shedding. Some load is cut but not all. Then neither bound binds, $\underline{\theta} = \overline{\theta} = 0$, and eq. (8) gives $\boldsymbol{\lambda = v}$. The price is no longer any generator's marginal cost — it is the consumers' willingness to pay. This is **scarcity pricing**.
- Lower bound: everything is shed. Only if consumers value electricity below even the cheapest generator's cost, $v < \min_g c_g$, which is not a case anyone worries about in practice.

Two important consequences are worth noting here:

VoLL is the price cap, and the scarcity price. The price can never exceed v: if it tried, shedding load would be cheaper than serving it. And in any hour where capacity runs short, the price is *exactly* v, regardless of what any generator costs. This means that the demand curve of Figure 2.1 shifts from vertical to horizontal at v. Empirical estimates of VoLL for European systems run to roughly 10,000–25,000 €/MWh (Mulder 2023, Box 7.9) — two orders of magnitude above the 66.0 €/MWh of our base case. It is important to note, however, that this measures the costs that consumers are willing to pay to avoid having unexpected black-outs. The more general idea of having "flexible demand", i.e. where consumers react to real time prices throughout the year, is a completely different story that we will touch upon later in the course.

This is how capacity gets paid. Look again at the scarcity rent $\overline{\mu}_g = \lambda - c_g$. A peaking plant with high marginal cost earns nothing in a normal hour, because it is not running. It earns $v - c_g$ in a scarcity hour — an enormous margin, but collected rarely. That is not a market failure; it is the mechanism by which an energy-only market finances capacity that exists to be available rather than to generate.

> **Which way does it bias the answer?** *Assuming the load is always served ($v = \infty$).*
>
> The rest of Section 2 and all of Section 3 keep market clearing as an equality — the case $v \to \infty$. Which way does that bend the results? It removes the top of the price distribution: the model's highest possible price becomes the dearest plant's marginal cost instead of VoLL. So a fixed-demand model **understates** average prices, **understates** the revenue of peaking capacity, and **understates** the profitability of storage and demand flexibility, all of which earn disproportionately in the extreme hours. It does not bias the dispatch at all, as long as capacity is sufficient — which is why we can safely postpone it. The moment capacity becomes a *choice* (Section 7), the omission stops being harmless: a model that cannot price scarcity cannot explain why anyone builds a peaker. Watch this bias reappear, measured, in Section 3.6, where the real DK1 market spikes to many times any

plant's marginal cost and the model's prices cannot follow.

2.5 A carbon tax

Suppose the planner (or a government legislating on top of the market) prices carbon at τ euros per tonne. Each generator's effective marginal cost becomes

$$c_g(\tau) \;=\; \frac{p_g^F}{\eta_g} + o_g + \tau\, e_g, \tag{9}$$

and the dispatch problem is otherwise unchanged: the tax works entirely by re-sorting the merit order. Technologies are re-ranked by $c_g(\tau)$, and because the tax loads on e_g, the re-ranking is systematic: dirty-and-cheap technologies slide up the order, clean-and-dear ones slide down relative to them.

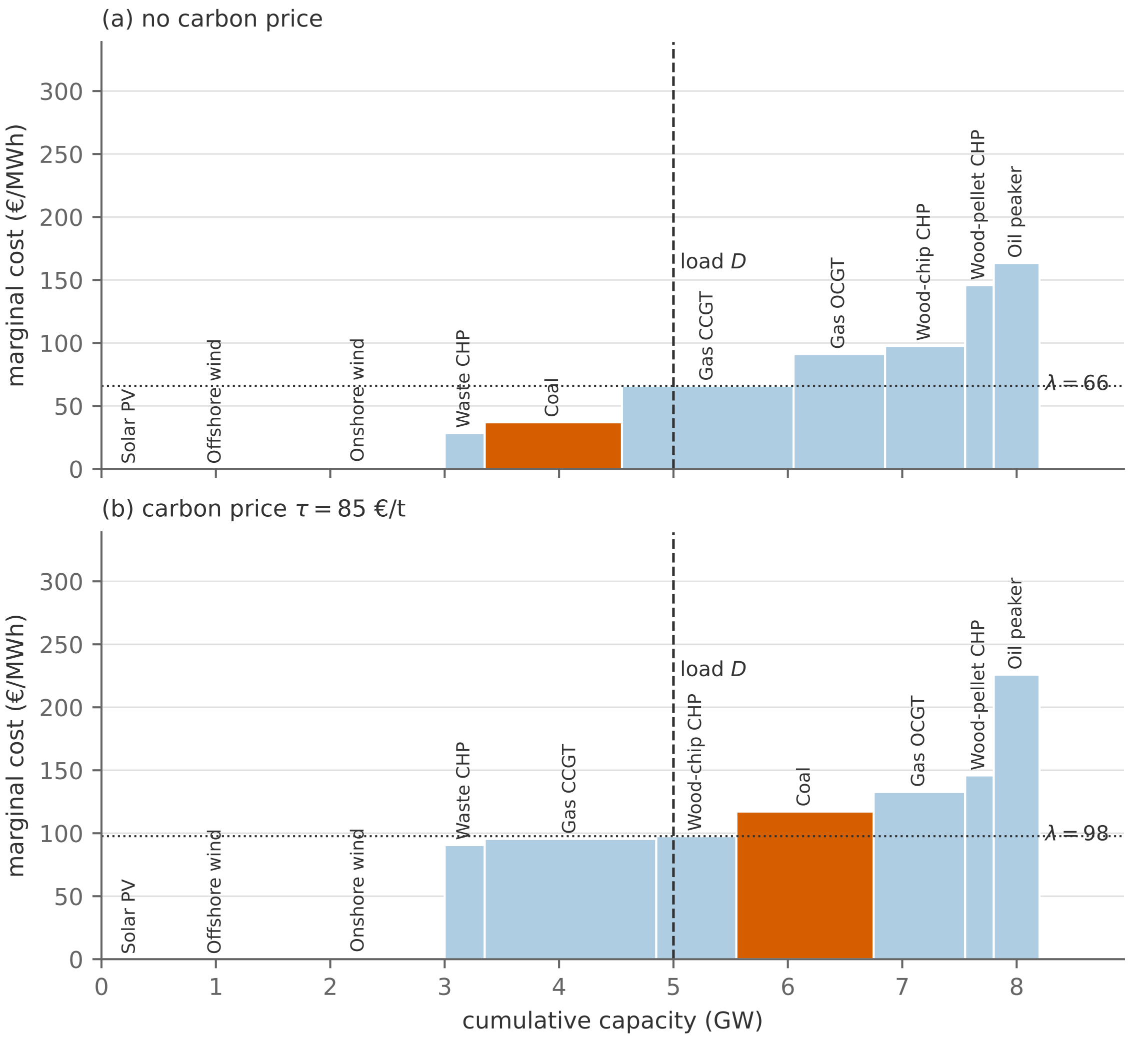

Figure 2.2: The merit order without (a) and with (b) a carbon price of $\tau = 85$ €/t

Figure 2.2 shows the experiment at $\tau = 85$ €/t, a level in the range of recent EU ETS prices: coal's marginal cost rises from 37.0 to 117.2 €/MWh, the CCGT's only from 66.0 to 95.5 €/MWh.

System emissions fall from 1545 t to 777 t, by 50%, without any change in demand or in the capital stock.

The price effects repay close reading:

i. *Pass-through.* The electricity price rises by 29.3 €/MWh, which is exactly $\tau \cdot e_m$ for the marginal technology (the CCGT, 85×0.34). Consumers pay the carbon cost *of the marginal unit*, not the average carbon content of what they consume (with a clean marginal technology the pass-through would be zero even if inframarginal generation were emission intensive).

ii. *Windfalls.* Every clean inframarginal generator pockets the price increase: onshore wind's rent rises from 64.4 to 96.2 €/MWh without wind doing anything differently. Carbon pricing is, among other things, a transfer to clean capacity, which is precisely the investment signal at work, and a recurring political-economy complaint about the instrument.

iii. *The tax falls on rents too.* Coal, which previously earned 29.0 €/MWh, now earns nothing. The burden of the tax is shared between consumers (higher λ) and dirty producers (lost rents), in proportions the merit order determines.

2.6 An emissions cap

The other canonical instrument fixes the quantity instead of the price: total emissions may not exceed a cap E. Append to the planner's problem (3) the constraint

$$\sum_{g\in\mathcal{G}} e_g\, q_g \;\leq\; E \qquad (\sigma), \tag{10}$$

with multiplier $\sigma \geq 0$. This is the additive structure at work for the first time: The new constraint adds the term $\sigma\left(E - \sum_g e_g q_g\right)$ to $\mathcal{L}^{\text{disp}}$ in (4), contributing $-\sigma e_g$ to its derivative with respect to q_g, so for any active generator, the optimality condition now reads

$$\lambda \;=\; c_g + \sigma e_g + \overline{\mu}_g, \tag{11}$$

which is eq. (9) with σ in the role of τ: the shadow price of the cap is an *endogenous carbon price.* Equally useful is the reading of σ as *marginal abatement cost*: it measures what the last tonne of abatement cost the system, i.e. the cost premium of the cleaner generation that replaced dirtier generation at the margin.

An old friend, arriving from a new direction. It is worth pausing on that last sentence, because you have met σ before. In the lecture note "A Simple Model of Abatement Costs" (Berg 2026a), we started from an aggregate relationship $F(E)$ with emissions $M = \varphi E$, and derived the marginal abatement cost curve

$$MAC \;\equiv\; -\frac{\partial C}{\partial A} \;=\; \frac{F'(E) - p^E}{\varphi}, \tag{12}$$

where $A = M_0 - M$ is abatement and C is net output. The resulting MAC curve is upward sloping, and the optimal policy ensures that this equals the marginal damage, $MAC = D'(M)$.

What eq. (11) says is that a bottom-up model *derives* the same object: We can get the marginal abatement cost curve by resolving the system model for a range of different caps E and recording $\sigma(E)$. Each point on the curve is a named pair of technologies swapping places in the merit order. Figure 2.3 presents the two side by side (so you don't have to fetch the other lecture).

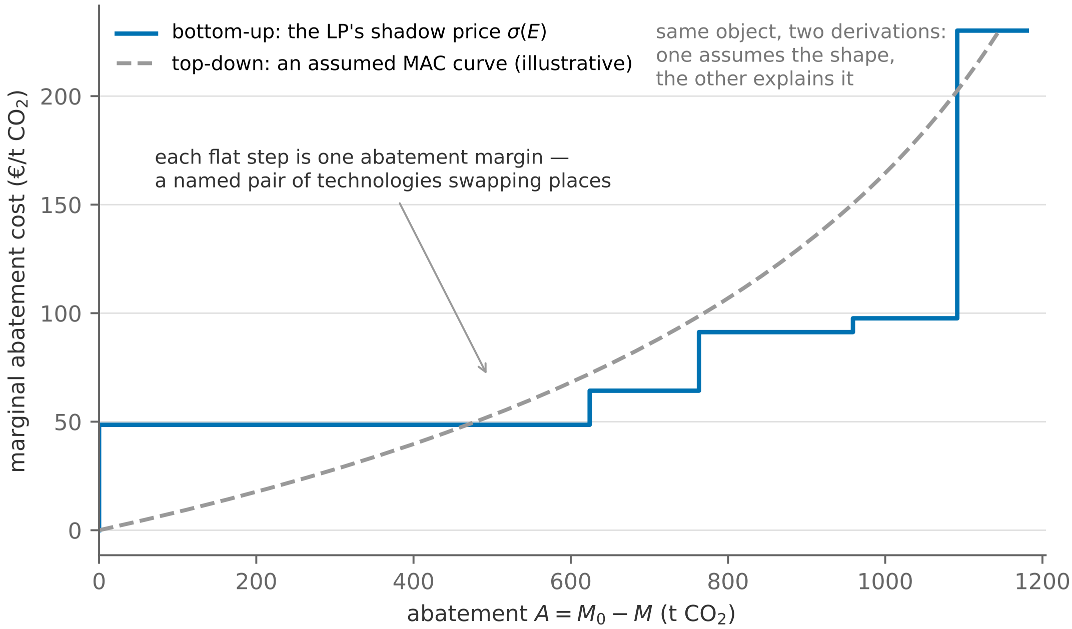

Figure 2.3: The marginal abatement cost curve, derived and assumed. The staircase is this section's LP: $\sigma(E)$ from the cap sweep, plotted against abatement $A = M_0 - M$. The dashed curve is the smooth, convex MAC of the one-good top-down model (12), calibrated only so that the two span the same range.

We note that in this illustrative example next to a very stylized bottom-up model, the two shapes seem very different. In reality, however, the more realism we add to the bottom-up curve, the MAC curve often converge to a more smooth function. Thus, the *shape of the two* are not necessarily the big result here, but rather where they come from and how they may be interpreted.

Result 2 (Tax–cap equivalence, and its limits)**.** *A cap at level E with shadow price $\sigma(E)$ induces the same dispatch, electricity price and emissions as a carbon tax $\tau = \sigma(E)$; and a cap set at the realised emissions of any tax replicates that tax's outcome. However, it is important to note up front that the equivalence breaks in many different settings, for instance by introducing uncertainty in the model.*

The first point about equivalence is straightforward, while the second point on uncertainty deserves a short note: If demand or fuel prices are uncertain, a tax fixes the marginal cost of abatement and lets emissions fluctuate, while a cap fixes emissions and lets the carbon price fluctuate; which is better depends on the relative slopes of marginal abatement cost and marginal damage, the classic argument of Weitzman (1974).

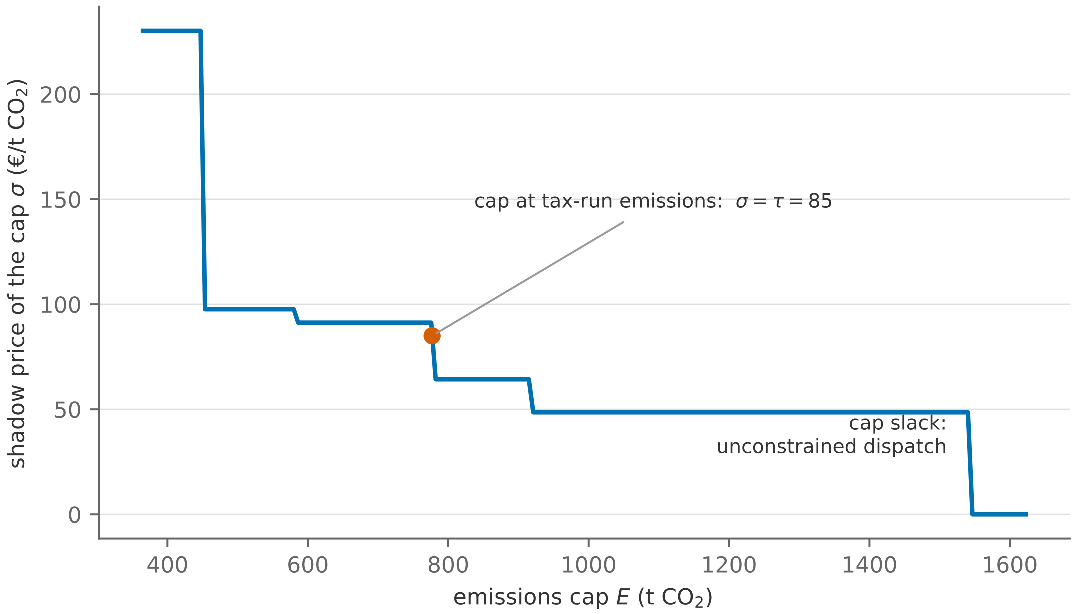

Figure 2.4: The shadow price of the emissions cap as the cap is tightened from unconstrained emissions (1545 t) towards the feasibility floor (365 t, the cleanest way to serve load at all). Each flat segment is one margin of abatement — one pair of technologies trading places in the dispatch; each vertical segment is a cap level at which that margin is exhausted.

From duals to markets. Every model section of this note closes with the same small table: each constraint, its dual, what the dual measures, and the real-world object that measures the same thing:

Constraint	Dual	What it measures	Market counterpart
market clearing (3b)	λ	marginal cost of serving one more MWh	the day-ahead spot price
capacity limit (3c)	$\overline{\mu}_g$	value of one more MW of g; scarcity rent	an inframarginal plant's operating margin
served load $d \leq D$ (7c)	$\overline{\theta}$	surplus on load that is served; $\lambda = v$ when it is not	scarcity pricing; the market's price cap
emissions cap (10)	σ	marginal abatement cost	the EU ETS allowance price

The current section shows very clearly that given existing capacities, nothing beats running renewables at full capacity; solar and wind has by far the lowest marginal costs of operation. The setup is very instructive in terms of understanding the basics and being able to read the economics from the duals of the planner solution, but very clearly requires a lot more details before we inevitable still have to conclude that solar and onshore wind are really cost-effective technologies (but we'll learn a lot from the journey there).

3 Variable generation and demand

Section 2 presented an incredibly stylized energy system, and so will this section. However, we will introduce an essential feature of the energy system, in particular for a course that deals with the green transition: Hourly variability in generation capacities and demand. This section introduces a time dimension h and adds an *availability profile* to each technology — the fraction of its capacity the weather offers in each hour. Because nothing yet links the hours together, the model simply remains one merit order per hour. However, even though the formal setup is virtually identical, this small change produces plenty of insights from model results and empirical evidence: the price is no longer a number but a distribution, and the value of a technology stops being a property of the technology alone and becomes a property of correlations between availability of generators and demand.

> **Reading alongside this section.** This section requires no supplementary reading to understand the concepts presented throughout. Instead, we point to relevant literature that deals with the same topics that we touch upon here. Hirth (2013) defines the value factor we derive below and estimates it across European systems. Joskow (2011) is three pages on why comparing technologies by levelised cost is a mistake. Borenstein (2012) is a gentle full treatment of the economics of renewable generation. López Prol et al. (2020) redo Hirth's exercise and draw the relative-versus-absolute distinction we take up in Section 3.4; Heptonstall and Gross (2021) is a systematic review that puts a number on total integration costs; and Reichenberg et al. (2023) recasts cannibalisation as *revenue risk*, which is where Section 9 picks it up.

3.1 The case of Denmark

Before we turn to the model and empirical evidence, let us briefly take a look at the relevant data and use Denmark as an example.

Figure 3.1 collects hourly data on demand, onshore wind, offshore wind, and solar generation.[5] Panels (a) and (b) shows how availability of particularly solar changes between a week in January and July, and panel (c) summarizes the three technologies' yearly availability with *duration curves*. Duration curves shows for what share of hours is availability above x for the three (we'll return to this below). Importantly, panels (a) and (b) also add the shape of hourly demand in the same hours so we can inspect how demand and variable generation correlates.

The panels show that (i) offshore and onshore wind are strongly correlated, (ii) solar is strongly cyclical both daily and seasonally, (iii) wind is less cyclical (spread out), but comes in multi-day weather system, and (iv) the capacity factors (average yearly availability) are far below one in particular for solar. These patterns are crucial, in particular when we start looking into deeply

[5] All data are from the Danish TSO Energinet's portal Energi Data Service Openly licensed (CC-BY 4.0). Appendix C records the dataset. The note's profiles are the production of each fleet divided by its maximum over the year. That normalisation is a classroom proxy for the true availability per unit of capacity (it ignores capacity additions within the year and assumes the fleet's best hour reaches nameplate), but it preserves the shape.

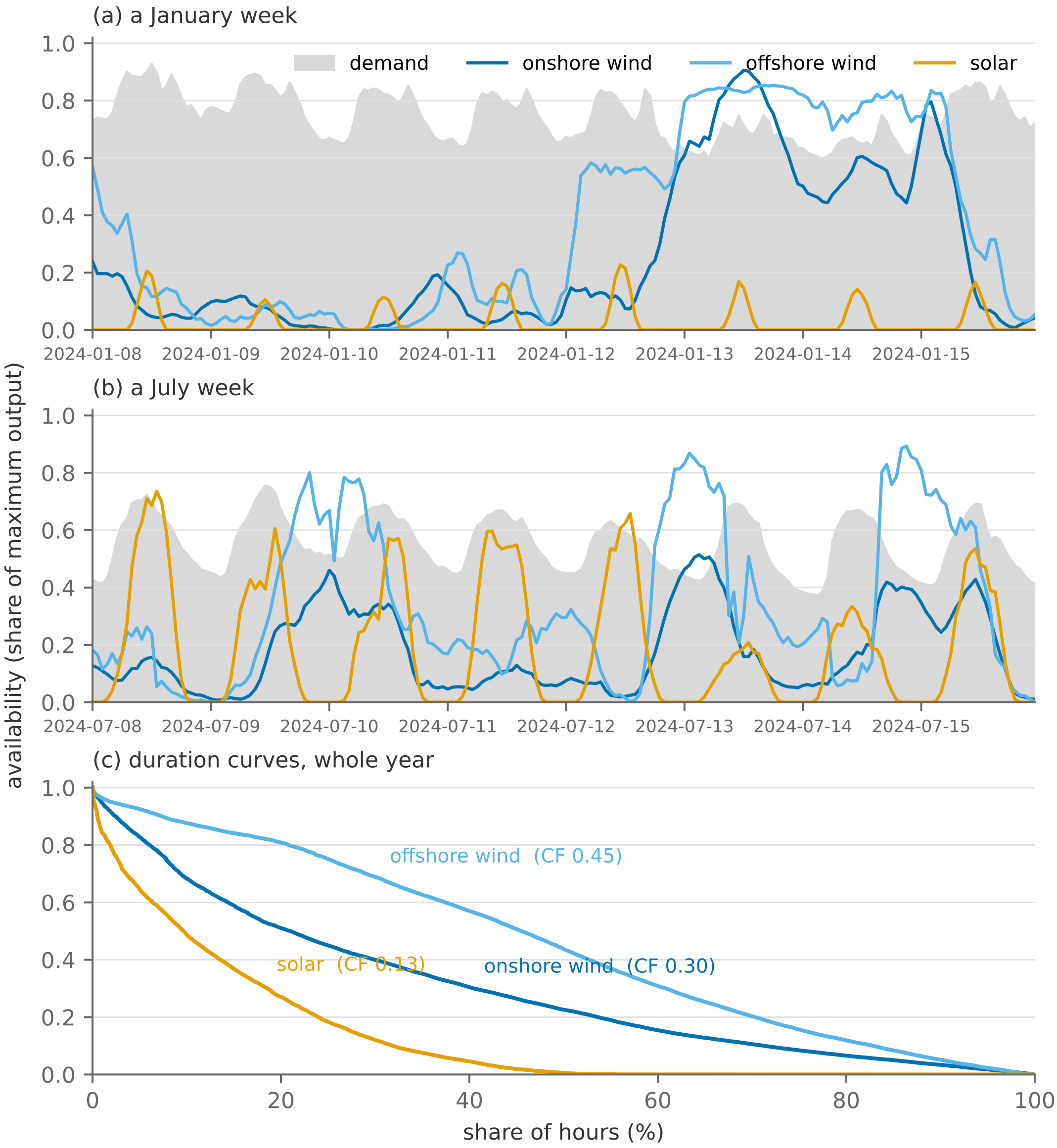

Figure 3.1: DK1 availability profiles and demand, 2024. (a): a winter week — wind-dominated, storms and lulls lasting days, solar nearly absent. (b): a summer week — a strong daily solar rhythm, weaker wind. Demand is the grey band, scaled by its own annual maximum so that it shares the axis. (c): the year as duration curves; the area under each curve is the capacity factor. Source: Energi Data Service, CC-BY 4.0.

decarbonised systems, because they govern what kind of complementary technologies have to be deployed for each of them (section 7 returns to this).

On the demand side, DK1 features clear daily and seasonal patterns, with higher levels of demand during winter and the middle of the day. It is worth noting that these swings are strong and varies by a factor of nearly three, from 1.5 GW to 4.2 GW in the the our base year 2024.

Table 2 presents the evidence in a different way by computing correlations between availability profiles and hourly demand. The positive correlation of the first column tells a positive story about wind and solar resources being available when demand is high. The other two columns nuance this a bit: Danish wind and demand are both generally higher during winter, reflected by a strong seasonal correlation (+0.89 onshore, +0.92 offshore). Solar is the opposite at that scale (-0.93),

Correlation of availability with demand	hour by hour	across months	across hours of the day
Onshore wind	+0.44	+0.89	+0.58
Offshore wind	+0.29	+0.92	-0.80
Solar	+0.25	-0.93	+0.94

Table 2: Correlation between availability profiles and hourly demand. Source: Energi Data Service, CC-BY 4.0.

and nearly perfect at the daily one (+0.94): it peaks at midday, and so, at 11:00, does demand. Solar's mild hourly correlation of +0.25 is not mild alignment. It is one strong alignment and one strong misalignment, cancelling.

Now one additional factor complicates the picture here: We are using *observed* consumption as a proxy for the pure demand variation. However, it is worth noting here that to the extend that demand *can* respond to price signals, the correlations above (in particular the "hour by hour" column) overestimates the true positive correlation: When wind and solar generation is large, the equilibrium price drops (all things equal), which drives up demand.

Finally, before we proceed with the model and simulations comparing it to empirical evidence, it is worth dwelling shortly on how the Danish merit order curve looks like as this will help us understand the results later on. Figure 3.2 collects the fleet of 2024: panel (a) is what stands installed in DK1, panel (b) what each plant costs to run — which is to say, the order they are called in.[6] Denmark has no oil or open-cycle peakers to speak of, and none appear here.

That fleet of power plants has a feature worth stopping at: its firm capacity — everything that is not weather-driven — comes to 2.6 GW, against a peak demand of 4.2 GW. This means that without either imports or significant variable generation, DK1 cannot serve its own peak demand. As a relatively small open economy in a coupled European market, imports and exports are essential for the Danish system: in 2024 DK1 imported 3.9 TWh and exported 3.8 TWh across its six interconnectors. Note what those two numbers do *not* say: gross trade of 32% of consumption nets out to almost nothing (0.1 TWh, or 0.2% of consumption). The wires are busy in both directions — DK1 was a net importer in 53% of hours and a net exporter in the rest — and it is that hour-by-hour swinging, not the annual balance, that the rest of this section is about.

The implication is that, as an artefact of us trying to keep the model simple and ignore transmission lines and exports/imports, we have essentially created an infeasible problem in roughly 450 hours of the year where demand cannot be met with domestic generation. Thus, for the model to run and give *somewhat* reasonable results, we fix this by adding one synthetic "import" plant on top of the true Danish fleet of power plants. In our results later on, this "import" plant will be presented as a part of the Danish merit order curve and be important for the price formation.[7]

[6] Wind and solar capacities are the maxima the profiles were normalised by, so profile times capacity reproduces DK1's own hourly production. The thermal capacities are the PyPSA-Eur snapshot of the same zone that Section 6 uses, so the two sections describe the same country; Appendix C documents both.

[7] Specifically, we set $\bar{q} = 5.5$ GW — the sum of DK1's interconnectors to Norway, Sweden, Germany and DK2 — available in every hour at a marginal cost equal to the capacity-weighted average of those zones' day-ahead prices, which averaged 63.7 €/MWh over the year. Four of the six wires, then: the cables to the Netherlands and to Great Britain are left out because this section prices imports off its neighbours' published day-ahead prices, and the twelve-zone network of Section 6 does not extend that far west. It is a stand-in, and a deliberately crude one:

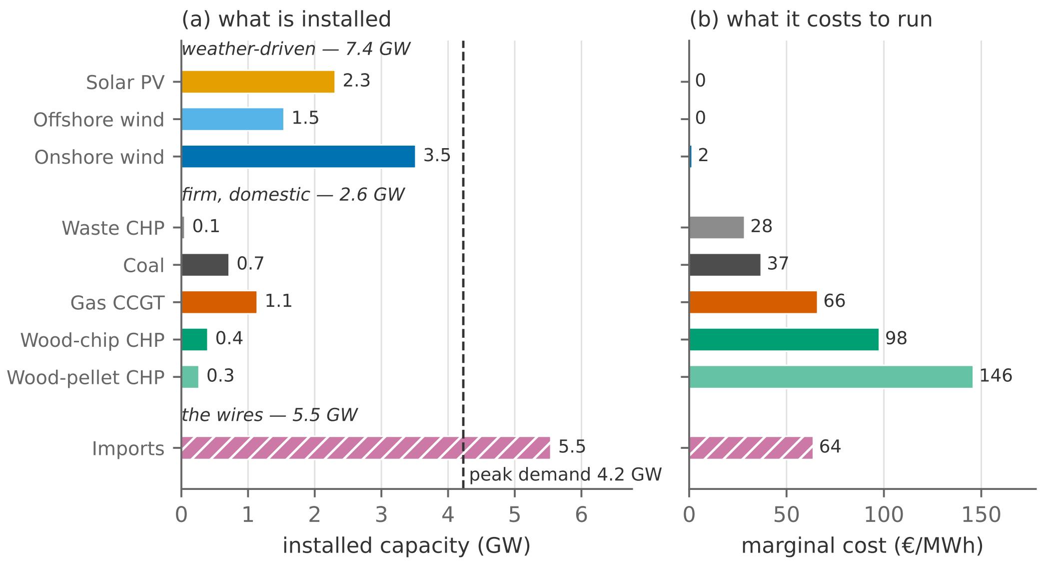

Figure 3.2: The DK1 fleet of 2024 that this section dispatches, grouped as the text reasons about it and ordered within each group by marginal cost. (a): installed capacity, against the year's peak demand. (b): marginal cost. The hatched row is not a Danish power station but the synthetic import plant introduced at the end of this subsection; its cost is the hourly price of the neighbouring zones, and panel (b) shows the annual mean of that price. Sources: Energi Data Service and PyPSA-Eur; Appendix C documents both.

3.2 The model, repeated over the year

Index hours by $h = 1, \ldots, T$ and let $\gamma_{g,h} \in [0,1]$ be the availability of technology g in hour h. With fixed load D_h, the planner now solves

$$\min_{q} \quad \sum_{h} \sum_{g \in \mathcal{G}} c_{g,h}\, q_{g,h} \tag{13a}$$

$$\text{s.t.} \quad \sum_{g \in \mathcal{G}} q_{g,h} = D_h \qquad (\lambda_h) \qquad \forall h \tag{13b}$$

$$0 \le q_{g,h} \le \gamma_{g,h}\, \bar{q}_g \qquad (\underline{\mu}_{g,h}, \overline{\mu}_{g,h}) \qquad \forall g, h \tag{13c}$$

This is identical to the model in Section 2, except for most symbols now carrying an hourly index h as well.[8]

Because neither the objective nor any constraint links two different hours, the problem *separates*: it is T copies of Section 2's LP, and every result from there applies hour by hour. The Lagrangian says the same thing: It is Section 2's, one copy per hour,

$$\mathcal{L}^{\text{vre}} = \sum_h \mathcal{L}_h^{\text{disp}} = \sum_h \Big[-\sum_g c_{g,h}\, q_{g,h} + \lambda_h \Big(\sum_g q_{g,h} - D_h \Big) + \sum_g \overline{\mu}_{g,h} \big(\gamma_{g,h} \bar{q}_g - q_{g,h} \big) + \sum_g \underline{\mu}_{g,h}\, q_{g,h} \Big], \tag{14}$$

Denmark may *buy* at what its neighbours charge but may not *sell* to them, and their prices are read from the market rather than computed. Section 6 replaces it with a network of twelve zones that computes those prices, and the difference between the two is worth keeping in mind for the rest of this section.

[8] The marginal cost $c_{g,h}$ only varies because of our synthetic "import" plant, where this cost reflects the actual import price for DK1 in a given hour in 2024.

and differentiating it hour by hour returns eq. (5) with a h subscript throughout; λ_h is the price in hour h, equal to the marginal cost of that hour's marginal technology; $\overline{\mu}_{g,h}$ is generator g's scarcity rent in hour h. This is the object the rest of the note builds on: Every section from here adds terms to $\mathcal{L}^{\text{vre}}$ rather than starting again. The supply and demand curves now move with weather and habits: Wind entering at near-zero marginal cost shifts the whole curve rightward in windy hours and vanishes in calm ones, so the same load can meet a cheap curve or an expensive one depending on the hour.

3.3 Residual load

Since variable renewable energy (VRE, wind and solar) bids in at near zero marginal cost, it is dispatched whenever available, and the conventional fleet faces what is left: the *residual load*

$$R_h \;=\; D_h - \sum_{g\in\text{VRE}} \gamma_{g,h}\, \bar{q}_g. \tag{15}$$

This curve represents what other plans and imports need to cover in different hours across the year. Figure 3.3 sorts R_h into a duration curve, for the actual fleet and for counterfactual fleets with wind capacity doubled and quadrupled.

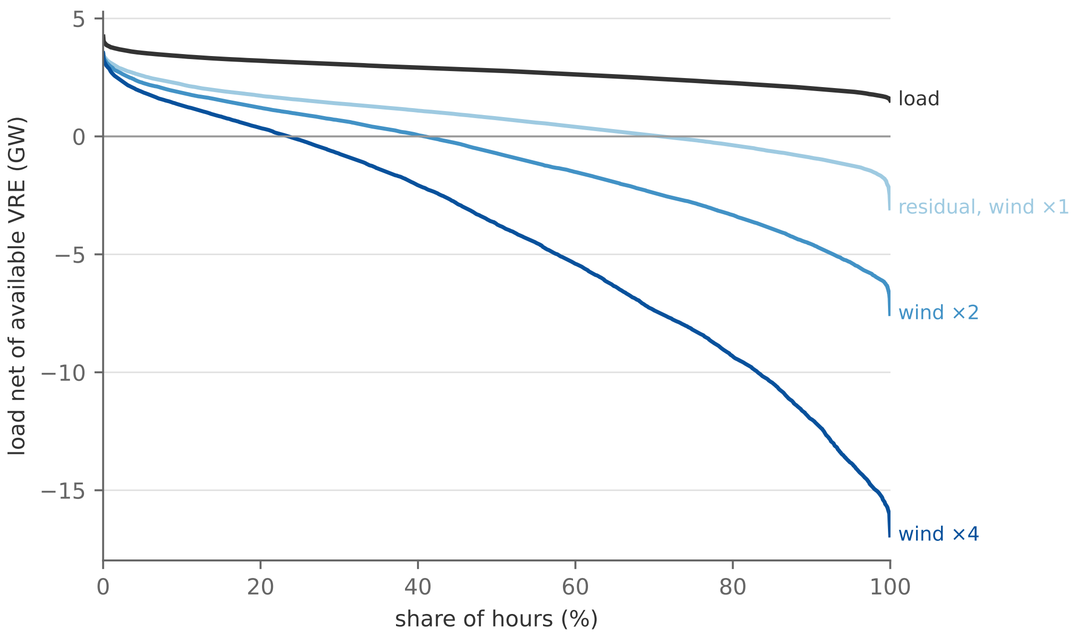

Figure 3.3: Load duration curve and residual load duration curves for DK1 2024, at actual wind capacity and with wind scaled ×2 and ×4 (solar at actual capacity throughout).

As more VRE is added, the overall levels of residual demand naturally drop, but the curve also steepens: For the hours where residual load is largest, new wind resources barely has any impact (left tail), whereas the hours that already has plenty of wind generation becomes oversaturated (right tail). The curve shows that at current DK1 capacity levels, a significant number of hours feature negative residual demand, i.e. surplus production of VRE; at ×4 the surplus exceeds total load for most of the year and to a severe degree. These hours of structural surplus require some mechanism to soak up the excess energy: exports, storage, flexible demand, or nothing at all. When the available renewable energy produces more than the system can absorb on the demand side, is has to be *curtailed* for the system to balance. It is not a malfunction: it is the optimal response to energy that is worth less than nothing to the system in that hour, and a well-designed renewable system may curtail a great deal. Curtailment lowers a technology's realised capacity factor below what the weather would allow, which raises its levelized cost, and in real markets it is the mechanism behind negative prices and the reason power purchase agreements are written the way they are.

3.4 Price distributions and value definitions

The price distribution. Figure 3.4 sorts the equilibrium prices. Absent, the synthetic "import" plant that has time-varying marginal costs to reflect the state of neighbouring markets, this curve would simply be a staircase with each step reflecting the different plant's marginal costs. The smooth parts represent hours where imports are price setting.

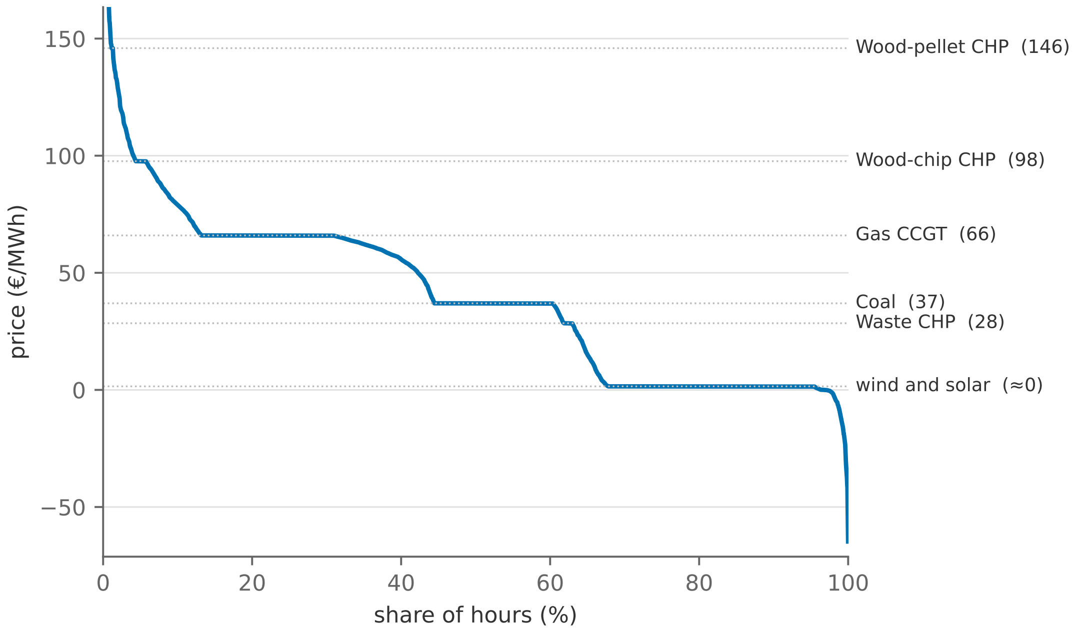

Figure 3.4: The price duration curve: the year's λ_h, sorted. The price only ever takes values in the fleet's menu of marginal costs (right margin), because in every hour some technology is marginal and eq. (5a) sets λ_h equal to its cost. The axis is cut just above the most expensive domestic plant so that the staircase stays legible: the tail is not shown, and it runs to 894 €/MWh in the year's scarcest hour, where the price has left the domestic menu behind and the wires are marginal.

Which technology depends on the hour's residual load: in the surplus hours at the right the marginal unit is *wind itself*, sending the price to its near-zero marginal cost (onshore wind alone sets it in 28% of hours); coal sets it in the mid-residual hours (16%), gas in the tighter ones (18%), and wood in the tightest few.

Value definitions. Three statistics summarise how a technology meets the price distribution.

The *capacity factor* is average output per unit of capacity,

$$CF_g = \frac{1}{T\,\bar{q}_g} \sum_h q_{g,h}, \tag{16}$$

a purely physical number. The *capture price* is the average price the technology actually sells at — the generation-weighted mean,

$$\bar{p}_g = \frac{\sum_h \lambda_h \, q_{g,h}}{\sum_h q_{g,h}}, \tag{17}$$

and the *value factor* normalises it by the *base price* $\bar{p}$,

$$VF_g = \frac{\bar{p}_g}{\bar{p}}, \qquad \bar{p} \equiv \frac{\sum_h \lambda_h D_h}{\sum_h D_h}, \tag{18}$$

so that $VF_g > 1$ means the technology produces disproportionately in expensive hours and $VF_g < 1$ the opposite (Hirth 2013). The value factor therefore captures a revenue channel that tells us something important about the technology beyond its cost structure: It measures the relative value of the MWh that a technology produces relative to the average technology and thus prices the intertemporal scarcity of the system (see also e.g. Joskow 2011).

Absolute and relative cannibalisation in Danish data. Figure 3.5 simulates our small toy model with the Danish weather, demand patterns, fleet, and import prices. Panel (a) shows the capture prices across different energy shares for wind and solar and panel (b) shows the corresponding value factors. It is worth stressing that the shapes of the two curves naturally depends on the system configuration, but they replicate the empirical evidence we have so far (see e.g. Hirth 2013; López Prol et al. 2020): Solar's value factor is above one for very low market shares and drops sharply, and wind's starts out around roughly 1, but drops more slowly across the market share.

Figure 3.6 shows the interaction between the wind and value and the solar market share. Panel (a) shows wind's value factors as a function of solar's energy share and panel (b) the same thing only in levels. Here, it shows that the value factor of wind clearly increases in the solar market share, while the level of the wind capture price clearly decreases.

The figures present (at least) three important conclusions worth carrying with us: First, for a wind developer: the revenue per MWh is not the market price but the capture price, and the capture price is a *decreasing function of everyone else's wind.* Second, for a policy maker: the first units of a VRE technology are the most valuable ones, so support schemes calibrated to today's prices systematically overpay tomorrow's entrants unless they are indexed to something like the capture price. Both readings, however, take the fleet as fixed. Nothing in this section can absorb surplus energy or shift it into the tight hours — the value collapse is partly a statement about missing *flexibility*, not only about wind. The following sections discuss this further.

The third and final conclusion we want to stress here is the warning in López Prol et al. (2020) about using the value factor alone to measure the profitability and complementarity of technologies:

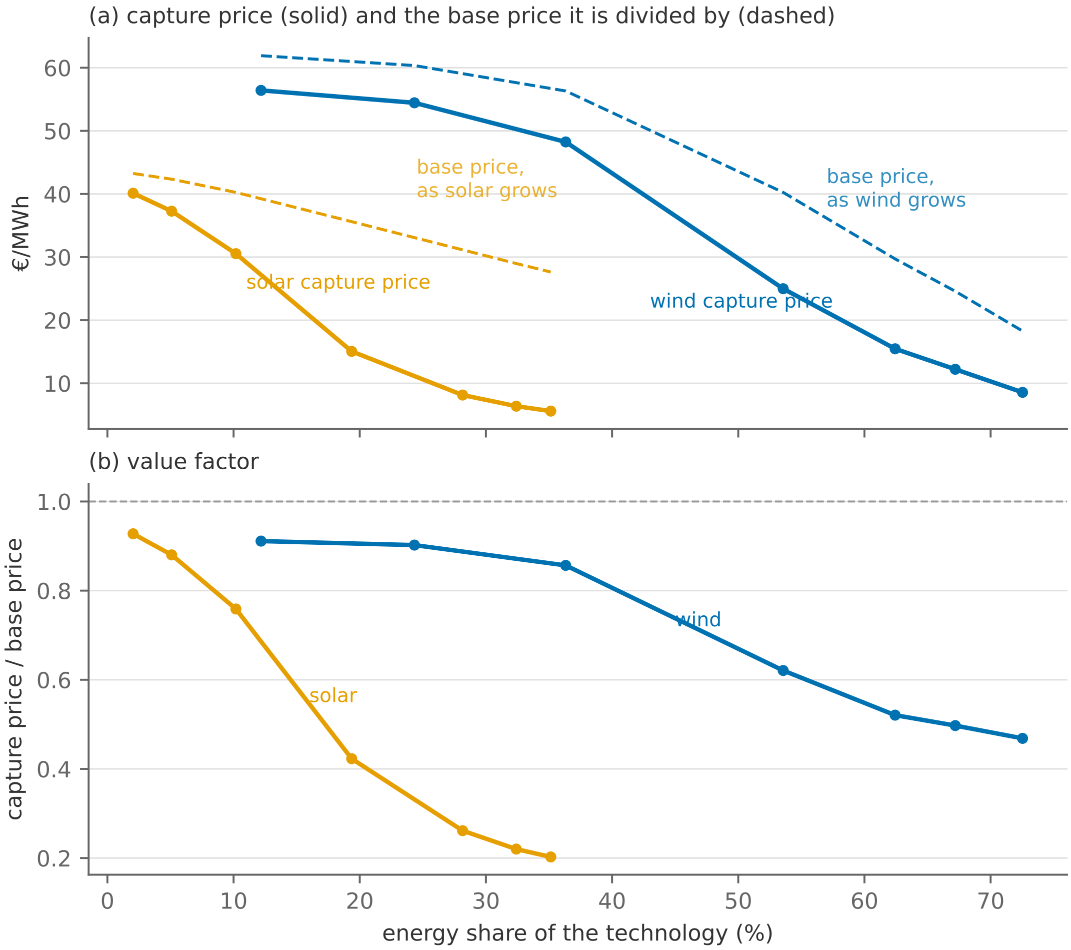

Figure 3.5: Cannibalisation as a curve: capture price (a) and value factor (b) against the technology's own energy share, scaling wind and solar capacity separately. Each point is a full re-solve of the year. The dashed line in (a) is the base price $\bar{p}$ *along that same sweep* — the denominator panel (b) divides by. It is drawn because it does not stand still: as zero-cost capacity enters, the whole price level falls with it (the merit-order effect), so the value factor in (b) is a ratio of two quantities that are both moving. Panel (b) is therefore what is left of the decline once the fall in the general price level is divided out: the part specific to the technology's own timing.

On the surface, the value factor in Figure 3.6 in panel (a) indicates that wind and solar may be complementary technologies in the sense that using more solar increases the value of wind (at least the value *factor*). However, panel (b) clearly rejects this story, showing that the capture price of wind still decreases with more solar in the system, only slower than the *average* price. This is precisely the distinction López Prol et al. (2020) draw between *relative* and *absolute* cannibalisation on Californian data, and it is why they insist on reporting both.

3.5 Is cannibalisation a market failure?

We have just derived a result that sounds alarming: build more wind and wind becomes less valuable. It is tempting to jump from there to "therefore renewables need support to overcome cannibalisation". While there are real arguments for support schemes for renewables, this one does not hold up to scrutiny.

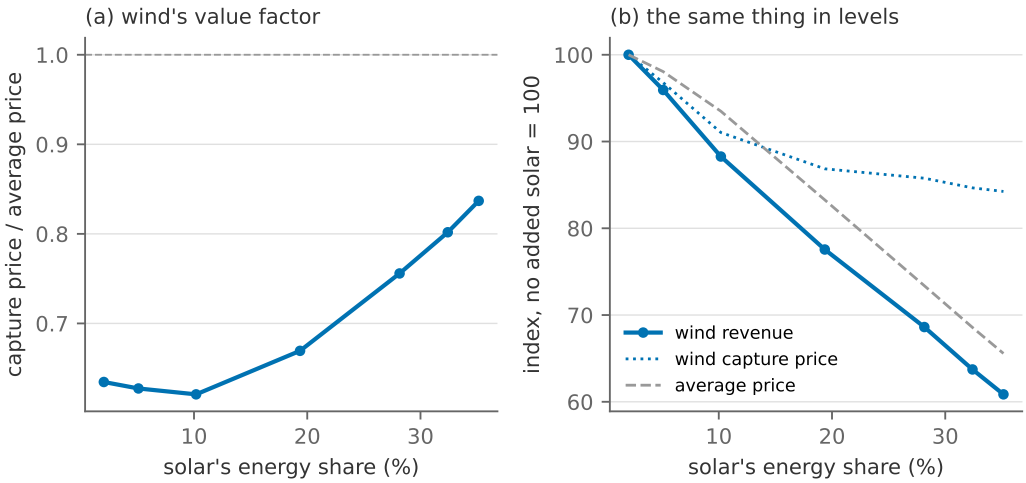

Figure 3.6: The same six runs, read two ways. Solar capacity is scaled up; wind is untouched. (a): wind's value factor *rises*, from 0.63 to 0.84. (b): wind's capture price and its total revenue both *fall* — revenue by 39%. Nothing good has happened to wind. The value factor improved only because its denominator, the base price, fell faster than its numerator.

Recall the standard rule for when regulation is warranted on *efficiency* grounds: intervene when private actions impose costs on others *that do not pass through prices.* Economists call a cost onto others that travels *through* the price system a **pecuniary externality**, and a cost that travels outside it a **technological** (or real) externality. Only the second is a market failure. When a new bakery opens and drives down the price of bread, the incumbent baker is harmed, but nothing inefficient has happened — the price fell because bread became genuinely more abundant, and the price change is the mechanism by which that information reaches everybody. Compensating the incumbent would make things worse, not better.

Before we end with a real confrontation with data, let us dwell on this rather important topic for just a moment longer and reflect on how the variable nature of certain renewables may be a good reason for interventions:

Pecuniary — no case for intervention.

- *Low prices in windy hours.* Electricity genuinely *is* less scarce when the wind blows. The low price is correct information, not a distortion, and it is what tells a heat pump, a battery or a smelter to consume then.
- *Cannibalisation itself.* A wind farm that depresses the price other wind farms receive is doing exactly what the new bakery does. The efficient amount of wind is precisely the amount at which the marginal farm's capture price covers its cost — which is Result 6, and which the market reaches on its own.
- *Higher price volatility.* Unpleasant for an investor, but risk that can in principle be traded: fixed-price contracts, PPAs and hedges exist. However, if insurance markets are missing or incomplete, this can be a real argument for second-best type of regulation; the market failure

is in a separate market, but cannot be addressed, which warrants regulation of volatility in other markets.

Potentially technological — a case worth examining.

- *Network congestion.* The grid is a common-pool resource. If a generator connects where it aggravates congestion and does not pay for the congestion it causes — because tariffs are flat, or because zonal pricing hides the constraint (Section 6.1) — that cost genuinely falls on others outside the price it faces.
- *Reliability and reserve requirements.* More forecast error means the system operator must hold more reserve. If reserve is procured centrally and its cost socialised into a flat tariff rather than charged to whoever created the imbalance, the externality is real. Note that this is an argument about *uncertainty*, not variability — a distinction Section 9 takes seriously.
- *Local amenity effects.* Noise, landscape, wildlife. These are genuine external costs — but they attach to *siting*, and have nothing to do with variability.

Two conclusions follow: First, most of what is loosely called “the integration cost of renewables” is not a market failure: Hirth (2013) decomposes the gap between a technology’s levelised cost and its market value into *profile* costs (our value factor, in €/MWh), *balancing* costs (the cost of forecast error) and *grid* costs, and is explicit that these represent the genuinely lower value of electricity delivered at the wrong time, in the wrong place, or with uncertainty attached — not a distortion to be corrected. Heptonstall and Gross (2021) put the total at roughly 20–35 €/MWh at high penetration, rising with the renewable share and dominated by the profile component — which is to say, dominated by exactly the mechanism this section derived.

Second, when regulation *is* warranted, the targeting principle applies: one instrument per externality, applied as close to the source as possible. If congestion is the problem, price congestion — that is what nodal pricing does. If reserve is the problem, charge imbalance to those who cause it. If carbon is the problem, price carbon, as Section 2.5 did.

While cannibalisation can be a poor justification for renewable support schemes, other arguments do survive contact with the targeting principle. Here, we briefly touch upon three different arguments.

The first is *innovation*: Deploying a technology teaches the industry how to build it more cheaply, and the firm doing the deploying captures only part of that learning: the rest spills over to its competitors and its successors, so private deployment falls short of what is collectively worthwhile. Gillingham and Stock (2018) summarise it as follows: the dynamic, innovation-driven case for support is real and hard to quantify, and it is a different case from the static one this section deals with.[9]

[9] For other relevant literature on this see e.g. Fischer and Newell (2008), Acemoglu et al. (2012), Popp (2019), and Nemet (2019).

The second is *second best*. Everything above assumed carbon is priced at its damage; where it is not, a subsidy to clean generation stands in for the missing carbon price — a poor substitute, but not nothing (Kalkuhl et al. 2013). Abrell et al. (2019) solve for the optimal support in exactly that setting and find that how the subsidy is *financed* matters far more for welfare than how finely it is differentiated across wind and solar. Helm and Mier (2021) add this section's own mechanism: because the value of clean generation falls as the fossil share falls, the optimal subsidy is a declining schedule rather than a rate — support calibrated to today's system overpays tomorrow's.

The third is *risk*. The volatility bullet above assumed hedges can be bought; where they cannot, an investor facing uninsurable revenue risk demands a higher return and the same capacity costs more to build. Newbery (2023) argues that a long-term contract that removes that risk lowers the cost of capital by enough to pay for itself — provided it is written so that it does not also switch off the price signal the generator dispatches against.

3.6 Comparing model outcomes and market data

Finally, let us compare how our stylized model fare when compared to market data. Recalling that the shadow value λ_h is the market equilibrium price, we put the simulated λ_h beside the published day-ahead prices for DK1 in Figure 3.7.

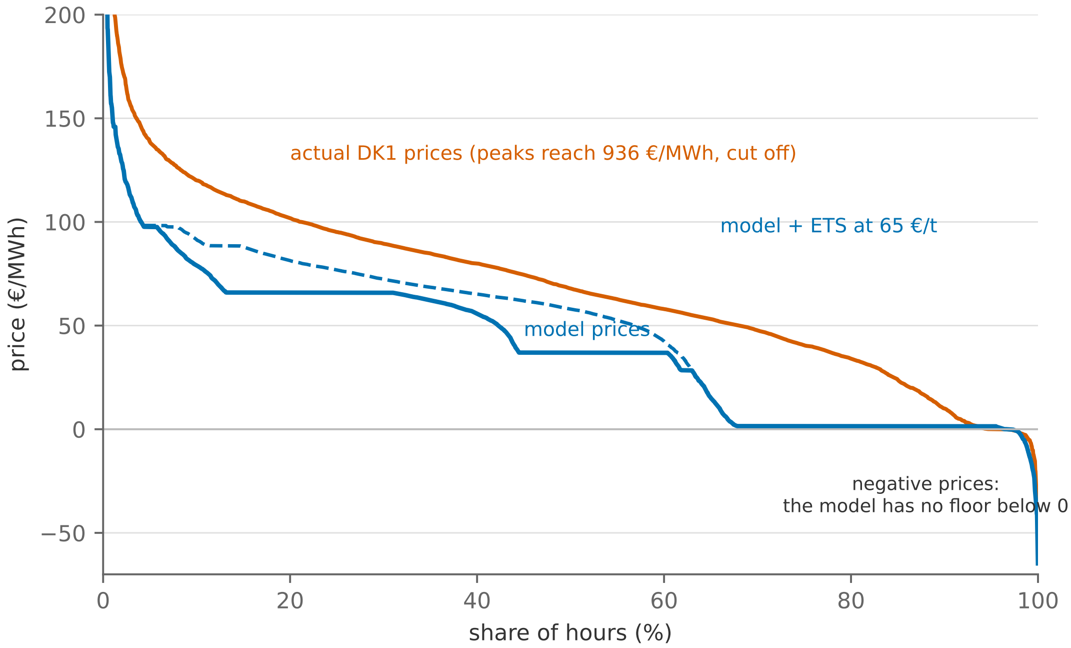

Figure 3.7: The model's price duration curve against the actual DK1 day-ahead prices of 2024. The dashed curve is the same model with the year's average ETS price (65 €/t) charged on every emitting technology. The shape agrees; the level does not, and the gap has one cause.

The shape is right. The model's price is at or below 37 €/MWh in half of all hours and reaches 98 €/MWh in the top twentieth, against 68 and 139 €/MWh in the market. It is worth noting that the tail with high prices comes directly from our way of accounting for expensive imports.

The level is not. The model's base price is 40.2 €/MWh against the market's 70.6, and adding the year's ETS price to every emitting plant lifts it only to 48.5. The model is roughly a third too cheap, and the reason is the boundary rather than the fleet: **Denmark may buy but not sell**. In the real 2024 DK1 exported 3.8 TWh, and every one of those MWh was sold into a market that wanted it at a price. Here they have nowhere to go, so they are curtailed and the price falls to wind's marginal cost instead of settling at what a neighbour would have paid. Cheap hours are therefore too cheap and too frequent, which drags the whole distribution down.

The market values, and where they miss. Ratios cancel much of that level gap, so the market-value statistics are the sharper test. In the actual 2024 market the Danish wind fleet earned a capture price of 55.4 €/MWh, a value factor of 0.80; the model produces 0.62 at the same capacity. It misses in the direction the missing exports predict: without an outlet, wind cannibalises itself harder in the model than it does in reality, and the difference between 0.62 and 0.80 is a measurement of what DK1's interconnectors are worth to a Danish wind operator.

Solar is the mirror image, and the more interesting one. The market gave Danish solar a value factor of 0.68; the model gives 0.76, at a domestic solar share of 10.2%. With the neighbours' hourly prices in the merit order, cheap German midday power now undercuts Danish solar in exactly the hours Danish solar produces. The cannibalisation of Danish solar is substantially *foreign* cannibalisation, and it is now inside the model rather than in a paragraph explaining a discrepancy.

A week, side by side. Duration curves compare distributions after reordering hours neatly. Before we proceed to more complex models, let us briefly look at the model versus data throughout a chronological week. Figure 3.8 compares the model dispatch one January week with market data, both as a share of that hour's load, so that the comparison is about *composition* rather than levels. The panels are collected into comparable blocks (variable renewables, dispatchables, and net trade) to make categories across Energinet's settlement data and model technologies compatible.

First, because the modell cannot export excess supply, the model stack closes at 100%. The real system exports instead, most visibly during the storm from 15 January, when DK1 generated 160% of its own load from wind and solar alone and sent the difference abroad. The magnitude of this channel is significant: net trade swings from +50% of load to −96% within the single week. Second, dispatchable plants never fully switches off in the market data. In panel (b), it never falls below 24% of load (757 MW), even in hours with huge variable renewable generation and prices at or below zero, where the model has already shut its thermal fleet down entirely. If we dive into the underlying data, this comes from local CHP plants. In the Danish context, this is likely due to the fact that the plants are committed to produce district heating in a cold January month in Denmark.[10] What this link to district heating means for the model more broadly is something we will discuss further in section Section 5.

[10] There are other potential reasons for this type of behavior, including generation subsidies or technical constraints such as ramping or unit commitment. We revisit the latter in the next section.

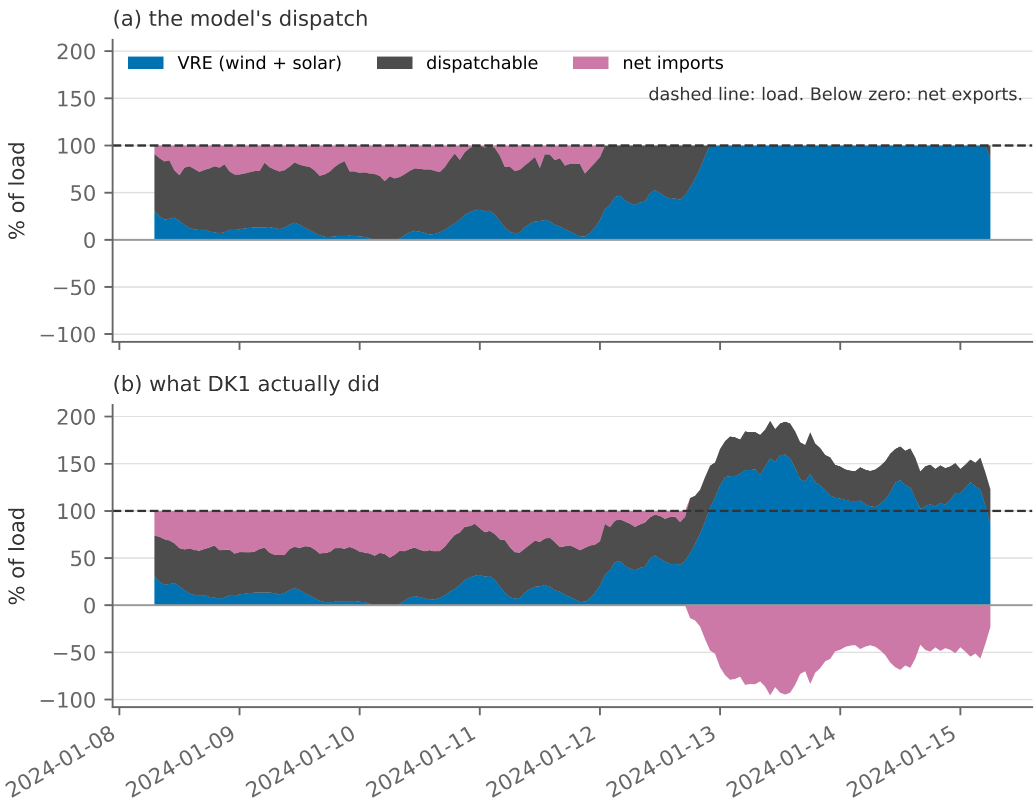

Figure 3.8: One week (8–15 January 2024), modelled and realised, each hour as a percentage of that hour's load, in three blocks: variable renewables (wind and solar), dispatchable plant, and net imports — which stack downwards in the hours the zone is a net exporter. The week is chosen automatically as the year's most variable in residual load.

What is still missing. Three things, and each returns later:

i. *No plant here will pay to produce.* The market goes negative in 4% of hours and the model in 3%, but the model's negative hours are borrowed rather than earned: they are hours when the neighbours' price is negative and the import plant carries it in. No Danish generator in this fleet ever bids below zero, because none of the reasons to do so is in the model — subsidised generation, must-run CHP, inflexible plants avoiding a shutdown cost. Section 4.3 puts the first of them in.

ii. *Trade runs one way.* Section 6 makes the neighbours endogenous and lets power flow both directions; comparing that section's prices with this will shed light on what the exporting channel is worth.

iii. *Missing flexibility.* Demand is exogenous in every hour, and there are no intertemporal flexibility that allows supply or demand to move across hours from a surplus hour to a tight one.

Which way does it bias the answer? *No negative bids, one direction of trade, no flexibility.*

The three omissions above do not all push the answer the same way.

- *No negative bids.* A fleet that shuts down the moment the price falls below its marginal cost is more flexible than the real one: **understates** curtailment, **overstates** what wind and solar capture in the surplus hours.
- *Trade one way.* Surplus is curtailed rather than exported, which **understates** the value of Danish wind — the gap between 0.62 and 0.80 above is a measurement of it. But the neighbours' prices are data, not outcomes, so the model cannot let Denmark move them: that **overstates** what the wires are worth.
- *No flexibility.* Nothing carries energy between hours, so a surplus can only be thrown away and a tight hour met by the most expensive plant available: **overstates** the spread of prices and, with it, the cannibalisation derived above.

The second is the largest, which is why this model's wind is worth less than the market's rather than more. Sections 4 and 6 take on all three.

From duals to markets. As usual, let us summarise the bridges we have used to cross from model outcomes to market data.

Constraint	Dual	What it measures	Market counterpart
market clearing per hour (13b)	λ_h	that hour's system marginal cost	the hourly day-ahead price (Elspot DK1)
availability-scaled capacity (13c)	$\overline{\mu}_{g,h}$	the hour's scarcity rent on g	a plant's hourly gross margin
derived: capture price (17)		generation-weighted price earned	a wind farm's realised revenue per MWh
derived: value factor (18)		market value relative to what the average MWh costs	the "profile discount" in PPA pricing

Where this goes next. From Section 4 onwards the models grow: storage, a second energy carrier, a network of twelve zones. They stop being small enough to write out constraint by constraint, and are assembled instead from standard components in an established modelling framework. That buys realism — the simulations below are closer to the system they describe than anything in this section — at the price of a little distance between the blocks of algebra we write down and the machinery that solves them. Nothing about the economics changes: every result that follows is still read off the same duals, the same λ_h and the same $\overline{\mu}_{g,h}$, and Appendix D.1 translates each component back into the constraints it stands for.

4 Storage and flexibility

Previous models treat hours independently: the LP of Section 3 separates into one merit order per hour, and no decision in one hour constrains any other. Storage ends that. A battery that charges at noon and discharges at eight in the evening ties the two hours into a single decision problem, and the economics acquire a genuinely intertemporal object: the value of energy *held*, not consumed.

Section 3 ended with the observation that the collapse in the value of wind was partly a statement about a system with no flexibility at all: nothing could absorb a surplus hour or move energy into a tight one. The next few sections add the standard responses to intermittency one at a time:

i. *Storage*, and the flexibility of the thermal fleet itself (this section);

ii. *Sector coupling and flexible demand*: a consumer that can respond to the price (Section 5);

iii. *Trade*: moving energy in space rather than time (Section 6);

iv. *Backup capacity*, which is an investment question and waits for Section 7.

For each of the first three we will ask the same three questions at the end of the section: what does the new technology do to the two tails of the price distribution, what does it do to the value of wind and solar, and what does it do to system cost. Comparing the answers across the three sections gives a compact summary of the economics of flexibility.

> **Reading alongside this section.** Zerrahn et al. (2018) treat the economics of storage in renewable systems, in particular the question of *how much* storage a decarbonising system needs, which is where the result in Section 4.4 comes from. Butters et al. (2025) study battery investment in market equilibrium, which connects this section forward to Section 7. For the intertemporal optimisation technique itself, Appendix A covers everything used here. The lecture note "Generation Technologies" (Berg 2026b) describes the storage technologies themselves.

4.1 The model

Before writing down the model, let us briefly state what characterises a battery; "Generation Technologies" has the longer description. A battery is not a generator: it buys electricity in one hour and sells it back in another, and some of the energy is lost on the way. Four numbers describe it. First, a *power rating* $\bar{s}$ in MW, the most it can charge or discharge in an hour. Second, a *duration* $\bar{\delta}$ in hours; the combination $\bar{\delta}\bar{s}$ measures the energy a battery can hold in MWh. Third, one-way efficiencies η_c for charging and η_d for discharging, whose product is referenced *round-trip efficiency*. Fourth, a small operating cost c^s per MWh charged or discharged, which is added to capture small wear on battery cells from usage. The stored energy at the end of hour h is the *state of charge*, soc_h.

The model is then Section 3's problem with three additions, all of them in blue: a charge variable, a discharge variable, and the bookkeeping that connects them across hours. The planner chooses generation $q_{g,h}$, charging g^c_h, discharging g^d_h and the stored energy soc_h to solve

$$\begin{aligned}
\min \quad & \sum_h \sum_{g\in\mathcal{G}} c_g\, q_{g,h} \;+\; \sum_h c^s\left(g^c_h + g^d_h\right) && && \text{(19a)}\\
\text{s.t.} \quad & \sum_g q_{g,h} + g^d_h - g^c_h \;=\; D_h && (\lambda_h) \quad \forall h && \text{(19b)}\\
& 0 \le q_{g,h} \;\le\; \gamma_{g,h}\,\bar{q}_g && (\underline{\mu}_{g,h}, \overline{\mu}_{g,h}) \quad \forall g,h && \text{(19c)}\\
& soc_h \;=\; soc_{h-1} + \eta_c\, g^c_h - g^d_h/\eta_d && (\theta_h) \quad \forall h && \text{(19d)}\\
& 0 \le soc_h \;\le\; \bar{\delta}\,\bar{s} && (\underline{\psi}_h, \overline{\psi}_h) \quad \forall h && \text{(19e)}\\
& 0 \le g^c_h \le \bar{s}, \quad 0 \le g^d_h \le \bar{s} && (\underline{\phi}^c_h, \overline{\phi}^c_h;\ \underline{\phi}^d_h, \overline{\phi}^d_h) \quad \forall h && \text{(19f)}\\
& soc_T \;=\; soc_0 && (\kappa) && \text{(19g)}
\end{aligned}$$

Equation (19a) now has marginal costs from the usual generators $g \in \mathcal{G}$ and the battery, where marginal costs of charge and discharge are assumed the same c^s. Equation (19b) now has the battery on the supply side when it discharges and the demand side when it charges. The battery is not a generator, it is both a producer and a consumer. Equation (19d) is the "law of motion" for stored energy: charging one MWh from the grid puts η_c MWh in the tank, and delivering one MWh to the grid draws $1/\eta_d$ MWh out of it, so a round trip loses $1 - \eta_c\eta_d$. Equations (19e) and (19f) are two *separate* capacity limits: storage technology is characterised by a power rating $\bar{s}$ (how fast) and an energy capacity, written as a duration $\bar{\delta}$ (how long). Finally eq. (19g) requires the tank to end where it started, without which the optimiser would cheerfully empty it on the last day and book the proceeds as free energy.

Duality: the value of stored energy. Let us run through the technique of Section 2.3 once more. Following the convention of Appendix A we maximise minus the system costs. Every term of Section 3's Lagrangian (14) survives unchanged — the battery's contribution $+g^d_h - g^c_h$ to market clearing splits off as a term of its own, because the Lagrangian is linear in the constraint — so we may write the new one as the old one plus what the battery adds,

$$\begin{aligned}
\mathcal{L}^{\text{sto}} = \mathcal{L}^{\text{vre}} &+ \sum_h \lambda_h \left(g^d_h - g^c_h\right) - \sum_h c^s\left(g^c_h + g^d_h\right)\\
&+ \sum_h \theta_h \Big(soc_{h-1} + \eta_c g^c_h - g^d_h/\eta_d - soc_h\Big) + \sum_h \overline{\psi}_h \left(\bar{\delta}\bar{s} - soc_h\right) + \sum_h \underline{\psi}_h\, soc_h\\
&+ \sum_h \overline{\phi}^c_h \left(\bar{s} - g^c_h\right) + \sum_h \underline{\phi}^c_h\, g^c_h + \sum_h \overline{\phi}^d_h \left(\bar{s} - g^d_h\right) + \sum_h \underline{\phi}^d_h\, g^d_h.
\end{aligned} \tag{20}$$

Because $\mathcal{L}^{\text{vre}}$ enters whole, the generators' conditions are unchanged from Section 3 — differentiating with respect to $q_{g,h}$ touches nothing in blue. The three new variables give three new optimality conditions,

$$
\begin{aligned}
g_h^c: &\quad -c^s - \lambda_h + \eta_c\,\theta_h + \underline{\phi}_h^c - \overline{\phi}_h^c = 0, && \text{(21a)}\\
g_h^d: &\quad -c^s + \lambda_h - \theta_h/\eta_d + \underline{\phi}_h^d - \overline{\phi}_h^d = 0, && \text{(21b)}\\
soc_h: &\quad -\theta_h + \theta_{h+1} + \underline{\psi}_h - \overline{\psi}_h = 0, && \text{(21c)}
\end{aligned}
$$

together with complementary slackness on every bound, exactly as in (5b)–(5c).[11]

The new dual θ_h on eq. (19d) is the *shadow price of stored energy*: what one more MWh in the tank at the end of hour h is worth to the system. Every dual so far has priced a *flow*; this one prices a *stock*. Read eq. (21a) by asking, as in Section 2, where g_h^c can sit:

- Interior: charging, below the power rating. Both bound multipliers are zero, so $\lambda_h = \eta_c\,\theta_h - c^s$: the price paid for a MWh, plus the operating charge, equals what the η_c MWh that reach the tank are worth.
- Upper bound: charging flat out. Now $\overline{\phi}_h^c = \eta_c\theta_h - c^s - \lambda_h > 0$: the price is below the threshold, and the inverter's rating is what stops the battery from buying more. $\overline{\phi}_h^c$ is the scarcity rent on the power rating.
- Lower bound: not charging. Now $\underline{\phi}_h^c = \lambda_h + c^s - \eta_c\theta_h > 0$: the price is above the threshold.

Equation (21b) reads the same way with the thresholds mirrored: discharging pays when $\lambda_h > \theta_h/\eta_d + c^s$. Finally, eq. (21c) says how θ_h moves through the year: when neither the empty-tank nor the full-tank bound binds, $\theta_{h+1} = \theta_h$. Moving a marginal MWh of stored energy from one hour to the next is then feasible in both directions, so its value cannot differ between them. The value steps only in hours where a storage constraint is active. Collecting the three conditions gives the following optimality rule:

Result 3 (The arbitrage band)**.** *Optimal storage operation is a threshold rule around the stored-energy value: charge when the market price is below $\eta_c\,\theta_h - c^s$, discharge when it is above $\theta_h/\eta_d + c^s$, and hold in between. The width of the band is the round-trip loss plus twice the operating charge c^s, and θ_h itself moves only in hours where a storage constraint binds: a full or empty tank, or the power rating.*

Storage is thus an arbitrageur with a cost wedge, and θ_h is its private valuation of inventory. For a hydro reservoir this valuation has a name in the industry, the *water value*, and Section 6 will meet it as an equilibrium object.

4.2 What arbitrage does to prices

To investigate the effect of adding batteries to the model, the following section reuses the model from Section 3 and adds a single battery with a power rating of 1000 MW and a duration of 4 hours, so that it can charge or discharge at full power for four hours before it is full or empty. Its

[11] In the last hour the cyclic condition (19g) takes the place of θ_{T+1}; its multiplier κ plays that role.

one-way efficiencies give a round-trip efficiency of 90%, a standard value for lithium-ion batteries (see e.g. the short note on technologies, Berg 2026b), and its operating cost c^s is set so small that it only stops the battery from cycling between hours of equal price. Because the state of charge links every hour to the next, the model is solved over all hours of the year in sequence rather than hour by hour.

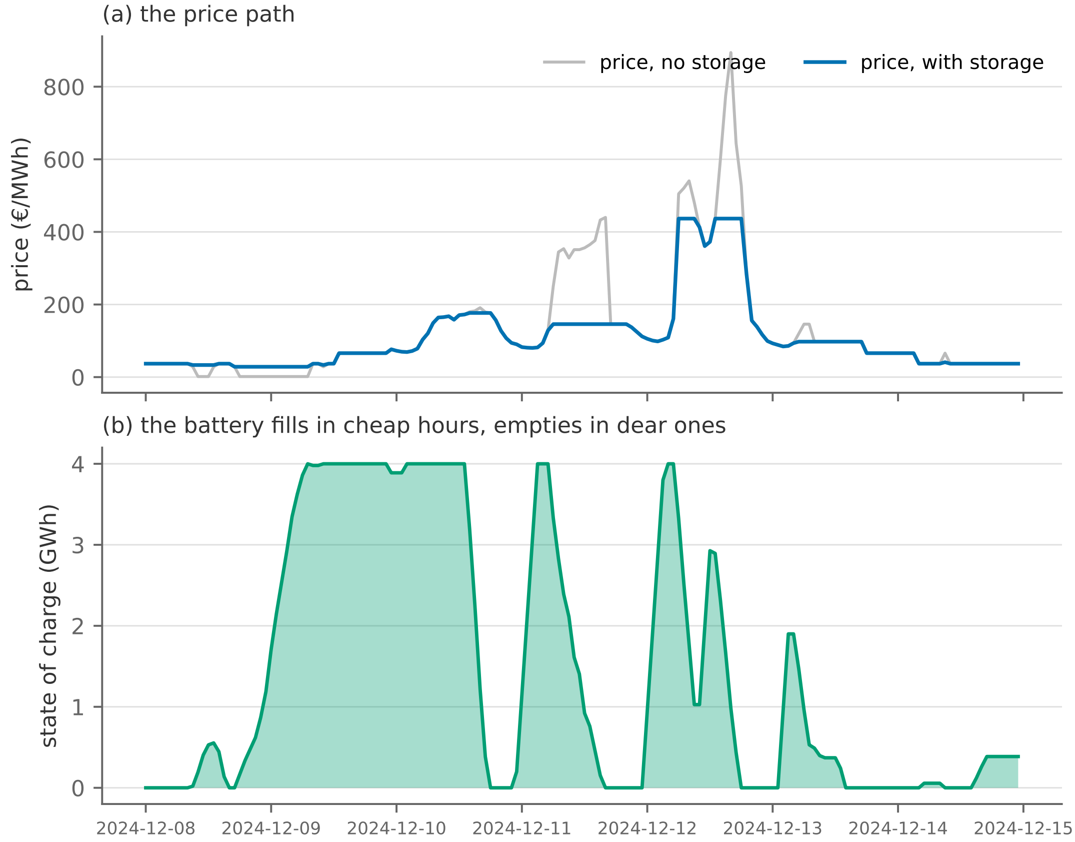

Figure 4.1: The week of the year with the most volatile prices, without and with the battery. The state of charge (b) is the mirror image of the price path (a): full just before the expensive hours, empty just after the cheap ones.

Figure 4.1 shows the battery at work in the most volatile week of the year. The pattern is exactly the threshold rule above: the state of charge rises in the cheap hours and is spent into the expensive ones. And because charging adds demand in cheap hours and discharging adds supply in dear ones, the price path itself is compressed — arbitrage moves the prices it trades on. Figure 4.2 shows the same effect for the whole year at once, zoomed in on the two tails of the price duration curve. Two things stand out. First, in the expensive tail the curve with storage sits below the no-storage curve, and the difference runs through the 10–15% most expensive hours of the year, not just the few scarcity peaks. Second, in the cheap tail the two curves coincide. In these hours the price is set either by curtailed wind, at its variable cost of about zero, or by imports at the neighbours' price, which is data in this model; the battery adds 1000 MW of demand in such an hour, but it exhausts neither the surplus nor the import capacity, so the price does not move. Over the year, the average price falls a little (41.0 to 38.6 €/MWh) while its standard deviation falls from 43.1 to 35.5 €/MWh. Storage does not necessarily make electricity cheaper on average, but it makes the price *distribution* narrower, and in this system it does so from the top.

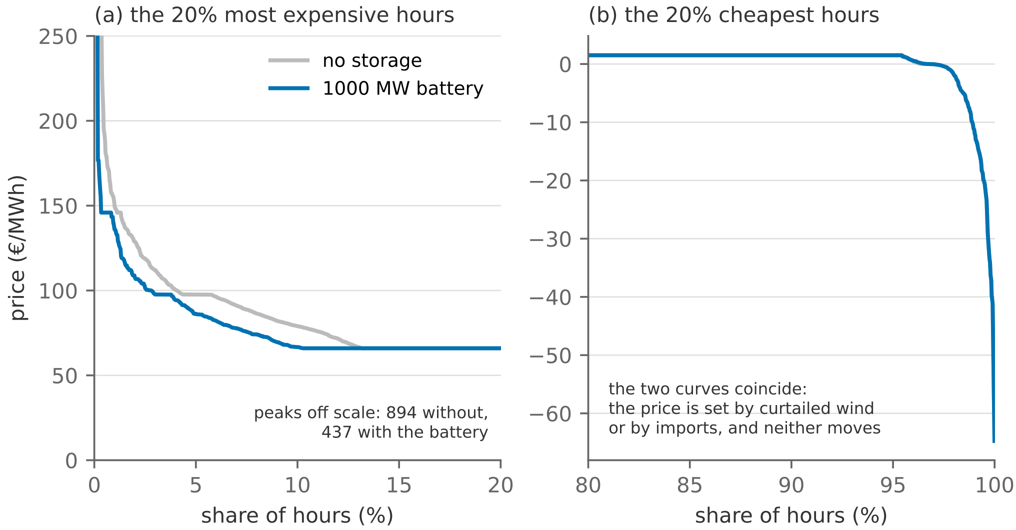

Figure 4.2: The two tails of the price duration curve for the year, with and without the battery: (a) the 20% most expensive hours, with the scarcity peaks off scale, and (b) the 20% cheapest hours, where the two curves coincide.

Storage's own profit is the flip side of that compression. The battery's arbitrage revenue over the year is 37.8 M€, while total system cost falls by 51.2 M€. However, this measure of system costs does not cover the investments required to stand up the battery. Priced at the investment costs of Section 7, with the inverter and the cells each annuitised over their own lifetime (Appendix C documents the sources), the battery costs 65.3 M€ a year to own, so this year's spreads alone would not pay for it.

Two things qualify that conclusion. First, batteries are the technology whose cost has fallen fastest: the price of lithium-ion cells has dropped by more than 90% since their commercial introduction in the early 1990s, at a learning rate of around 20% for every doubling of cumulative production (Ziegler and Trancik 2021; Way et al. 2022), and the investment cost above is a 2030 vintage that may well prove conservative. Second, the spread the battery earns on depends on the system it sits in. We will revisit the profitability of batteries in Section 7 in a 2050 system with far more solar and wind than DK1 has today, and there the optimal plan builds 2.7 GW of batteries at the 8-hour duration; the European system of Section 8 builds far more still.

In the next few subsections, we will explore further how the overall system configuration affects the profitability of batteries and what this tells us about complementarities between different technologies.

4.3 Adding more rigidities to the system – ramping constraints

We note that beyond the technical specifications of the battery, including the investment costs, what matters for the profitability of the battery is how flexible the system already is as it determines the price spread that it can take advantage of. So, our first extension here is to add some realistic rigidities to the system, to see if this changes our conclusion for the batteries – ramping constraints:

A large thermal plant cannot jump from zero to full output between two hours: boilers have thermal inertia, turbines have mechanical limits, and pushing them hard costs money and lifetime. The simplest way to model this is to include simple limits on how much a generator can adjust its output from hour to hour.

Let generator g face a *ramp rate* $\rho_g \in (0, 1]$: the share of its capacity by which it can increase output from one hour to the next. Add to the model

$$q_{g,h} - q_{g,h-1} \;\leq\; \rho_g\, \bar{q}_g, \qquad (\xi_{g,h}), \tag{22}$$

with $\xi_{g,h} \geq 0$ its multiplier. Now the generation $q_{g,h}$ appears in two ramping constraints, once as the new level in hour h, and once as the old level in hour $h+1$; the optimality condition now reads:

$$-c_g + \lambda_h + \underline{\mu}_{g,h} - \overline{\mu}_{g,h} - \xi_{g,h} + \xi_{g,h+1} \;=\; 0. \tag{23}$$

Rearranging gives the section's second result.

Result 4 (The ramping wedge)**.** *With ramping constraints, generator g enters the merit order in hour h not at its marginal cost c_g but at an* effective *marginal cost*

$$c_{g,h} \;=\; c_g \;+\; \underbrace{\xi_{g,h} - \xi_{g,h+1}}_{\textit{flexibility cost}}. \tag{24}$$

The merit order therefore becomes time-dependent*: the same two plants can swap places from one hour to the next without any cost or price changing. A plant may run when $c_g > \lambda_h$ (accepting a loss now to be positioned for an expensive hour later), and may withhold when $c_g < \lambda_h$ (because ramping up now would strand it in a cheap hour).*

Here, $\xi_{g,h}$ is what the plant paid in hour $h-1$ in terms of output it had to produce, at a loss, purely to be high enough to reach its hour-h level. $\xi_{g,h+1}$ is what it is willing to pay *now* in order to reach a desired level in hour $h+1$. A peaker anticipating an expensive evening starts climbing in the afternoon and takes a loss doing it. Let us dig a bit deeper into what this tells us:

i. *"Flexibility" is now a measurable cost:* The quantity $\xi_{g,h} - \xi_{g,h+1}$ is in €/MWh and can be summed over the year. This is the missing third column in any comparison of generating technologies: investment cost, marginal cost, *and* flexibility cost.

ii. *It reorders which technologies are attractive.* A coal or nuclear plant with a low ρ is penalised precisely in the volatile hours that a renewable-heavy system produces in abundance, while a gas peaker with $\rho \approx 1$ is not. This is a large part of the reason Section 7's optimal system builds OCGT rather than anything cheaper per MWh, and a large part of the reason nuclear and variable renewables tend to be *substitutes* rather than complements: an inflexible baseload plant and a fluctuating one compete for the same residual hours.

iii. *The price no longer equals anybody's marginal cost.* In an hour where a ramping constraint binds, $\lambda_h = c_{g,h} \neq c_g$ for the marginal plant. Section 2's crisp identity survives — the price is still the marginal cost of serving one more MWh — but the marginal cost is now a shadow-price-adjusted object.

Figure 4.3 puts numbers on the wedge for DK1's own fleet. One ramp rate is imposed on the slow thermal plant – the coal, wood and waste CHP boilers, 1.4 GW in all – and tightened from none to 0.1 per hour, while the gas turbine and the 5.5 GW of interconnectors stay free. At the catalogue rate of 0.9 per hour, which is what the network sections of Sections 6 to 8 impose, the summed flexibility cost is 0.9 M€ a year and the battery's value does not move; at 0.1 the rigidity costs the system 15.4 M€ and the battery's value rises by 15%. The rigidity that matters in DK1 is not in its boilers: flexibility arrives through the wires, and it is the wires the battery competes with.

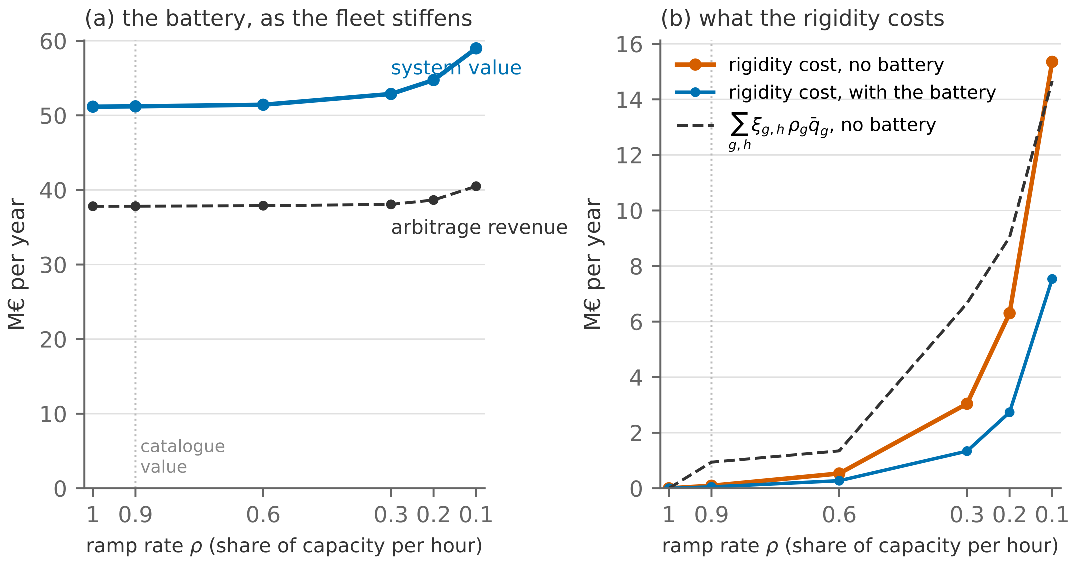

Figure 4.3: The ramping wedge in DK1. A ramp rate ρ on the slow thermal plant, tightened from none ($\rho = 1$) to 0.1 per hour. (a) The 1000 MW battery's arbitrage revenue and its system value. (b) What the rigidity costs the system relative to no limit, with and without the battery, and the summed shadow value of the ramp constraints $\sum_{g,h} \xi_{g,h}\, \rho_g\, \bar{q}_g$. The dotted line marks the catalogue rate.

Which way does it bias the answer? *Frictionless ramping.*

- *No ramp limits.* A fleet that can follow any residual load shape **overstates** the flexibility of the conventional plants, and therefore **understates** the value of everything that competes to provide flexibility: storage, interconnection, demand response, and gas peakers relative to baseload. It also **understates** price volatility, because a fleet that can follow any residual load shape produces fewer extreme hours than a real one.
- *Perfect foresight.* The planner never ramps in the wrong direction, which a real operator sometimes does, so this too **overstates** how easily the system can be run.

4.4 Storage and the value of wind

Now we turn to exploring the complementarity of wind and storage. Figure 4.4 re-runs the deep-cannibalisation world — wind at 1.5 times DK1's own capacity — with the battery swept from zero to 8 GW.

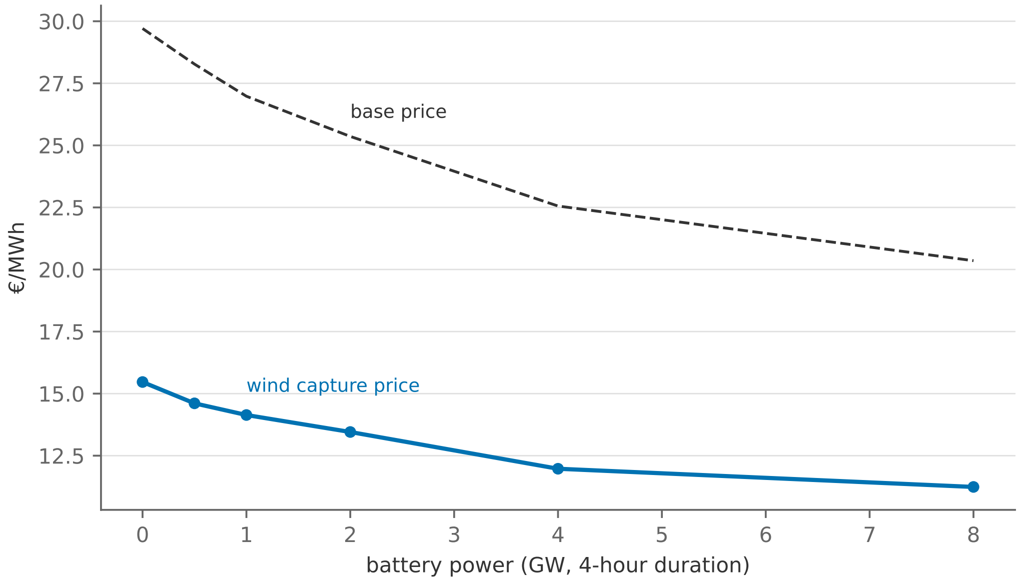

Figure 4.4: Wind's capture price and the base price $\bar{p}$ (eq. (18)) as battery capacity grows, with wind at 1.5 times DK1's actual capacity. Even 8 GW of 4-hour batteries barely moves wind's relative value: the surpluses it would need to absorb last for days, not hours.

The figure shows that the effect of the battery, even at the huge scale of 8 GW, is moderate: While prices generally drop, the value factor for wind only lifts from 0.52 to 0.55, with the capture price actually dropping. The main reason for this is a mismatch of time scales between the 4-hour battery and the wind with availability profiles with peaks and calms lasting potentially for days: Against a week of surplus the battery fills in its first four hours and then stands full and useless; against a week of calm it is empty by breakfast. (Solar is the opposite case: its cycle is exactly diurnal, so the same battery is a near-perfect complement to solar — which is why the two arrive as a pair in Section 7's optimal mixes.) The general lesson: "storage" is not one technology but a family indexed by *duration*, and each VRE technology needs the flexibility that matches its own spectrum of fluctuations. What can absorb a Danish wind week is long-duration storage (hydrogen, reservoirs) — or the trade of Section 6, which is storage in space rather than time, with Norwegian reservoirs playing the tank.

One more lesson hides in the exercise, to be collected in Section 9: everything above assumes the battery knows the week's prices in advance. The LP's operator has perfect foresight, and part of the arbitrage value computed here is the value of that clairvoyance; Section 9.4 revisits this.

From duals to markets.

Constraint	Dual	What it measures	Market counterpart
storage balance (19d)	θ_h	value of a stored MWh at hour h	a hydro reservoir's *water value*; a battery trader's inventory valuation
energy capacity $soc_h \leq \bar{\delta}\bar{s}$	$\overline{\psi}_h$	value of a larger tank	the price premium of storage *duration*
power rating $g^c_h \leq \bar{s}$, $g^d_h \leq \bar{s}$	$\overline{\phi}^c_h, \overline{\phi}^d_h$	value of faster charge/discharge	an inverter's capacity value
ramp limit (22)	$\xi_{g,h}$	the flexibility cost of moving output	the loss a plant accepts to be positioned for a peak

The three questions. As promised at the start of the section, the same three questions will be answered at the end of Sections 5 and 6.

Adding storage to the system...	
...does what to the two tails of the price distribution?	Compresses the expensive tail; the cheap tail does not move, because its price is set by curtailed wind or by imports at the neighbours' price. The standard deviation falls from 43.1 to 35.5 €/MWh while the mean barely moves.
...does what to the value of wind?	Very little at 4-hour duration (0.52 → 0.55 even at 8 GW): the timescales do not match. It would do a great deal for solar.
...does what to system cost?	Lowers it, by 51.2 M€ a year at the base battery — and with strongly decreasing returns, since each MW narrows the spread the next would earn on. This *does not*, however, include investment costs.

5 Electrification and sector coupling

The green transition is not only a story about electricity. To a large extent it is a story about everything else becoming electricity: heating, transport, and large parts of industry. Heat is the natural first case for this note. Denmark's district heating systems are large, centrally dispatched, and increasingly served by heat pumps drawing power from the grid. This section adds a heat sector to the power model and asks what electrification does to load, prices and emissions.

> **Reading alongside this section.** This section requires no supplementary reading. Ruhnau et al. (2020) study heat pumps, wind and the interaction between them in a European system model, and Section 5.3 ends by taking their central objection seriously. The heat pump is also this note's first price-responsive consumer, which makes it the natural place to meet the demand-response literature: Ambec and Crampes (2021) on how real-time pricing can balance intermittent supply in theory, and Fabra et al. (2021) on how much consumers actually respond in practice. Andersen and Dietrich

(2026) is the Danish counterpart to the latter, on household smart meter data, and Section 5.2 takes its central finding — that flexibility follows the appliance — as the qualification to everything this section makes the heat pump do. Joskow and Wolfram (2012) is a short overview of dynamic pricing if the topic is new to you.

5.1 A heat market and a heat pump

The model adds a second market, for heat, with a district heating load D_h^H and two technologies to serve it. The first is a gas boiler, which produces heat at a constant marginal cost $c^B = p^{gas}/\eta^B + o^B$ per MWh of heat. It is the marginal heat technology throughout, so whenever it runs it sets the heat price: merit order logic transplanted to a new carrier. The second is the heat pump, and this is where the coupling lives. It is a conversion from the electricity market to the heat market, whose efficiency is the hour's **coefficient of performance** (COP),

$$q_h^H \;=\; COP_h \cdot q_h^E, \qquad COP_h = 6.81 - 0.121\,\Delta T_h + 0.00063\,\Delta T_h^2, \tag{25}$$

where q_h^E is the electricity the pump draws, q_h^H the heat it delivers, and ΔT_h the temperature lift from the outdoor air to the 50 °C supply of the district heating network. The curve is the empirical air-source relation of Staffell et al. (2012). For a reader meeting the COP for the first time: A COP of 3 means that one MWh of electricity delivers three MWh of heat. No thermodynamic laws are harmed, because a heat pump does not *create* heat, it *moves* it from outdoors to indoors, and moving energy is cheaper than making it. The catch is that the cost of moving heat rises with the temperature lift, which is exactly what eq. (25) encodes.

Figure 5.1 shows the outdoor temperature over the year and the COP it implies.

The figure shows that (i) the COP averages a comfortable 3.0 over the year, a shade below the 3.2–3.6 the Danish Energy Agency's catalogue assumes for large district-heating heat pumps (Danish Energy Agency and Energinet 2024), which exploit lower supply temperatures and better heat sources than our deliberately simple setup; (ii) it ranges from 1.8 on the coldest hours to 4.3 in summer; and (iii) it falls to its lowest values exactly when it is coldest, which is precisely when heat demand peaks. This adverse correlation is the signature fact of heating electrification: The technology is least efficient exactly when it is most needed. Electrified heat therefore adds a load that is not just large but *peaked and weather-correlated*, arriving on top of a power system whose tight hours are the cold, dark, calm ones (Section 3).

The instance is the Section 3 power system plus a stylised 20 TWh/yr heat load whose shape follows Danish degree-hours (Aarhus temperature, 2024) with a flat hot-water base; the base case gives the heat pump 800 MW of electric capacity.[12] Written out in full, with the heat side in blue, the planner chooses generation $q_{g,h}$, boiler heat q_h^B and the heat pump's electricity intake q_h^E to

[12] The heat side is deliberately coarse: no combined heat and power plants, no thermal storage in the network, one representative temperature. Each would add realism, Danish CHP in particular, without changing the mechanism this section teaches; Section 5.4 discusses the first of them.

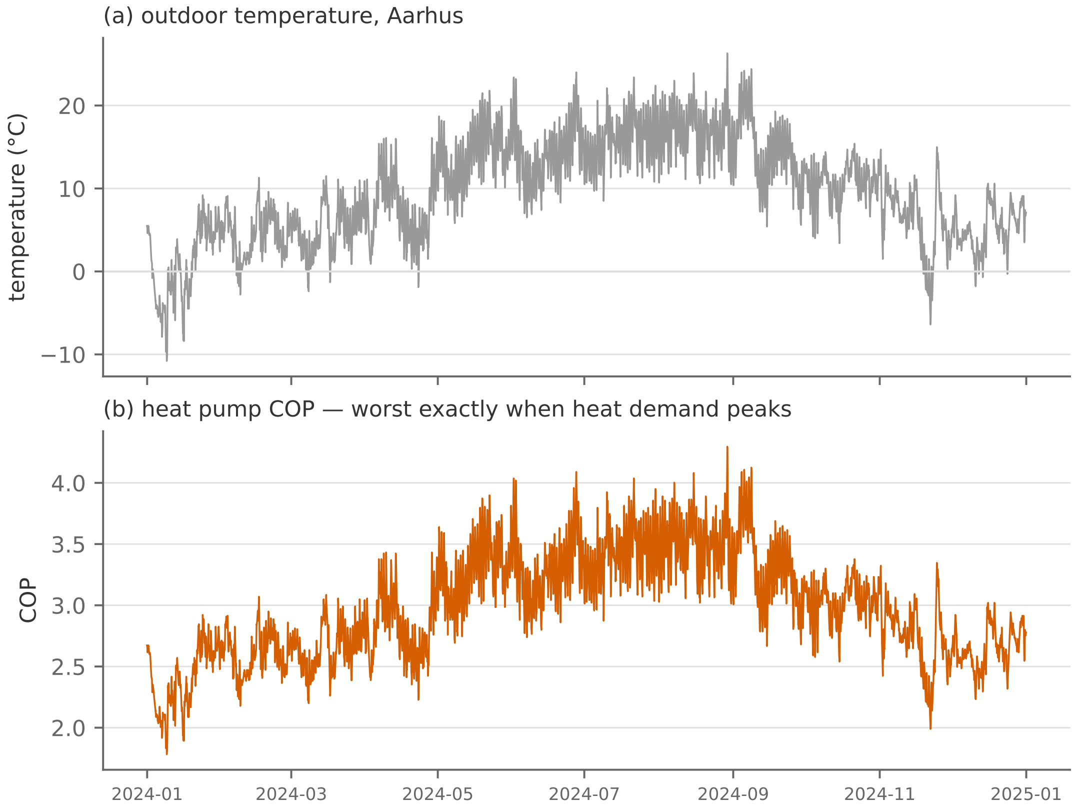

Figure 5.1: Outdoor temperature (a) and the heat pump COP it implies (b), hourly over 2024.

solve

$$
\begin{aligned}
\min \quad & \sum_h \sum_g c_g \, q_{g,h} \; + \; \sum_h c^B q^B_h & & & \text{(26a)}\\
\text{s.t.} \quad & \sum_g q_{g,h} - q^E_h \; = \; D_h & (\lambda^E_h) & \quad \forall h & \text{(26b)}\\
& q^B_h + COP_h \, q^E_h \; = \; D^H_h & (\lambda^H_h) & \quad \forall h & \text{(26c)}\\
& 0 \leq q_{g,h} \leq \gamma_{g,h} \, \bar{q}_g & (\underline{\mu}_{g,h}, \overline{\mu}_{g,h}) & \quad \forall g, h & \\
& 0 \leq q^E_h \; \leq \; \bar{q}^{HP} & (\underline{\mu}^{HP}_h, \overline{\mu}^{HP}_h) & \quad \forall h & \text{(26d)}\\
& 0 \leq q^B_h \; \leq \; \bar{q}^B & (\underline{\mu}^B_h, \overline{\mu}^B_h) & \quad \forall h & \text{(26e)}
\end{aligned}
$$

This is the problem of Section 3 with a second market bolted onto it. Two features are worth dwelling on before we solve it. First, the heat pump appears with a *minus* sign in the electricity balance (26b) and a *plus* sign in the heat balance (26c): It is a consumer in one market and a producer in the other, which is what **sector coupling** means operationally. Second, notice where the COP sits: multiplying q^E_h in eq. (26c). It is a *conversion factor between two markets*, and because it is time-varying, the exchange rate between electricity and heat moves hour by hour.

Note also that the heat pump does *not* need a cost of electricity in its own objective term, although it is tempting to put one there. It is already accounted for: The pump's electricity consumption enters eq. (26b), so it is priced at λ^E_h automatically by the market clearing constraint.

Adding a cost term as well would charge for the same electricity twice.

Duality: the heat price and the fuel-switching margin. The Lagrangian follows the same additive pattern as Section 4's, and it is worth noticing that it branches from the same place: The heat sector is built on Section 3's problem, not on the battery, so it is $\mathcal{L}^{\text{vre}}$ that reappears here, with the pump's withdrawal $-q_h^E$ from market clearing splitting off as its own term and $\lambda_h \equiv \lambda_h^E$,

$$\begin{aligned}\mathcal{L}^{\text{heat}} = \mathcal{L}^{\text{vre}} - \sum_h \lambda_h^E q_h^E - \sum_h c^B q_h^B + \sum_h \lambda_h^H \left(q_h^B + COP_h\, q_h^E - D_h^H\right) \\ + \sum_h \overline{\mu}_h^{HP} \left(\bar{q}^{HP} - q_h^E\right) + \sum_h \underline{\mu}_h^{HP} q_h^E + \sum_h \overline{\mu}_h^B \left(\bar{q}^B - q_h^B\right) + \sum_h \underline{\mu}_h^B q_h^B. \end{aligned} \tag{27}$$

The generators' conditions are untouched, for the same reason as in Section 4: $\mathcal{L}^{\text{vre}}$ enters whole. The two new variables give

$$q_h^E: \quad -\lambda_h^E + COP_h\, \lambda_h^H + \underline{\mu}_h^{HP} - \overline{\mu}_h^{HP} = 0, \tag{28a}$$

$$q_h^B: \quad -c^B + \lambda_h^H + \underline{\mu}_h^B - \overline{\mu}_h^B = 0, \tag{28b}$$

with complementary slackness on each bound. Equation (28b) is the merit order of Section 2 on the heat market: Whenever the boiler runs below capacity, $\lambda_h^H = c^B$. Equation (28a) is the coupling, and we read it as usual by asking where q_h^E can sit:

- Interior: the pump is marginal in the heat market. Both bounds are slack and $\lambda_h^H = \lambda_h^E / COP_h$: The heat price is the electricity price converted through the hour's COP.
- Upper bound: the pump runs flat out. Then the boiler is marginal, $\lambda_h^H = c^B$, and $\overline{\mu}_h^{HP} = COP_h\, c^B - \lambda_h^E > 0$ is the pump's scarcity rent: One more MW of pump would displace boiler heat worth $COP_h\, c^B$ at an electricity cost of λ_h^E.
- Lower bound: the pump is off. Then $\underline{\mu}_h^{HP} = \lambda_h^E - COP_h\, \lambda_h^H > 0$: Electricity is too dear, in the hour's exchange rate, to beat the boiler.

The dispatch rule is therefore a comparison of λ_h^E / COP_h with c^B. Because COP_h moves with the temperature, the same electricity price can put the pump on one side of the boiler in a mild hour and on the other side in a cold one. Sector coupling thus means that the two prices are linked by an arbitrage condition, eq. (28a), exactly as two zones will be linked by an interconnector in Section 6.

5.2 The coupled dispatch

Figure 5.2 shows how the heat load is served in the coldest week of the year, and the electricity price the heat pump faces.

The dispatch follows the rule derived above: The pump serves heat whenever λ_h^E / COP_h is below the boiler's cost, and the boiler covers what the pump's capacity cannot, or takes over entirely when power is dear. Over the year the pump serves 59% of the heat.

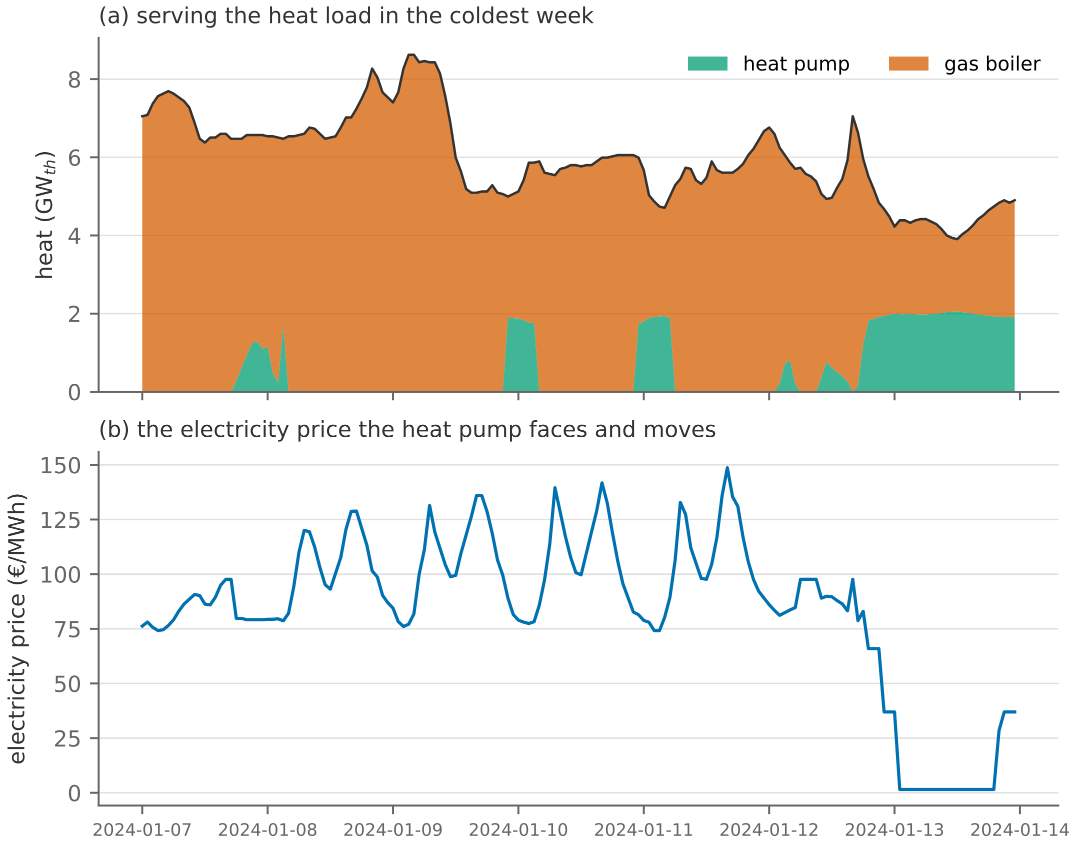

Figure 5.2: The coldest week of the year: how the heat load is served (a) and the electricity price the heat pump faces (b).

This is demand response. It is worth naming what we have just built. The heat pump is the first *price-elastic electricity consumer* in this note. Every load we have met until now was a fixed number D_h that had to be served (Section 2.4 showed what it takes to relax even that, and it took the value of lost load to do it). The heat pump is different: It looks at λ_h^E and switches off when power is expensive, because it has an outside option, the boiler. That is precisely what **demand-side flexibility** means, and it is the second item on the menu of responses to intermittency that Section 4 set out.

It is worth briefly noting that demand flexibility comes in many shapes and forms, and they differ quite a lot in *how* flexible they actually are. Our current model with district heating, the heat pump can be very flexible: It can be turned on/off at any given hour, in particular because the backup boiler is always there to kick in if needed. Other types of "flexible" demand may in fact be much less flexible. Recent Danish evidence makes the point sharply: Andersen and Dietrich (2026) match household smart meter records to the hourly price and find that most households do not respond to it at all, while nearly a third cut consumption significantly when the price rises. Averaged over everyone, a DKK 1/kWh increase in the hourly price — some 130 €/MWh, a very large move — lowers demand by 2.6%, an implied price elasticity of about −0.06. What separates the responsive third from the rest is the instructive part. Of every household characteristic they measure — income, education, age, total consumption — the largest single effect is owning an electric vehicle: It adds some 1.4 percentage points to a household's response per DKK/kWh, against that sample average of 2.6, and it survives controlling for all the others.

That is the same mechanism this subsection has just derived, met in the data. A household's lighting and cooking have no outside option and no store, so a price signal has nothing to work on; a car battery has both, exactly as our heat pump has the boiler behind it. Flexibility is not really a property of consumers, then, but of what they are consuming for: It comes from having somewhere else to put the demand, or some later hour to put it in. This is worth carrying into the sections ahead, where "flexible demand" will appear as a modelling assumption rather than an estimate. We return to it in more detail later in the course.

5.3 The rollout experiment

Let us now use the model to ask the policy question: What happens to emissions and prices as more and more heat is electrified? Figure 5.3 sweeps the heat pump's capacity from zero to 2 GW, counting emissions across *both* sectors.

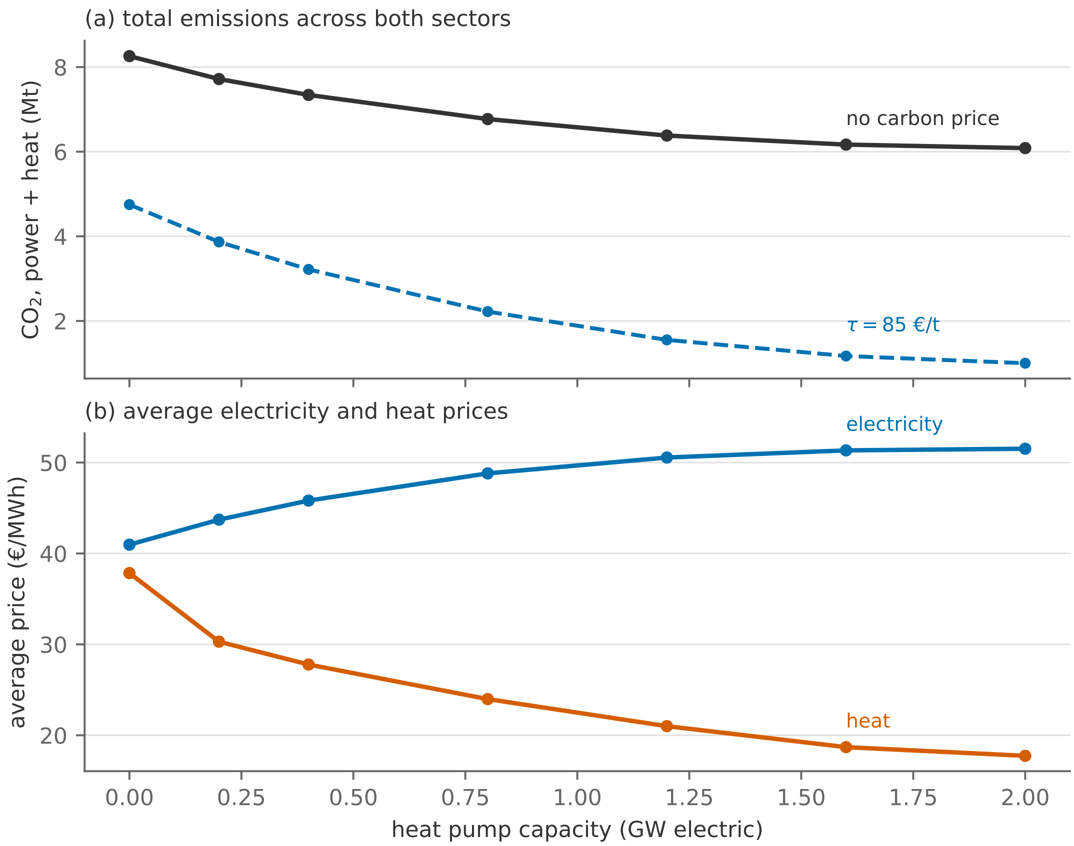

Figure 5.3: Heat pump capacity swept from zero to 2 GW electric: (a) total CO_2 across power and heat, without and with the carbon price of 85 €/t from Section 2; (b) average electricity and heat prices.

The figure shows the following:

i. Without a carbon price, emissions fall from 8.26 to 6.09 Mt over the sweep, a decrease of 26%.

ii. If we instead combine the rollout of the heat pumps with a carbon tax around the level of the EU-ETS, we reach a reduction of 53.2% after 800 MW of pumps.

iii. Beyond 2.0 GW the taxed line turns back *up*: Further rollout raises emissions.

iv. The average electricity price rises (from 41.0 to 51.5 €/MWh), while the heat price falls toward the pump's cost as pump capacity stops being scarce.

Emission reductions in the model. The main mechanism of the rollout is the following: A MWh of heat moved from the heat boiler to the pump avoids the boiler's gas, and costs $1/COP$ MWh of electricity. While this clearly reduces emissions from the heat boiler, the full emission reduction depends on how emission intensive the *marginal* generator in the electricity market is; in the current model system, this is coal 16% of hours. At a winter COP of around two, the emissions avoided in the boiler and the emissions added at the coal plant nearly cancel. Heat pumps are only as clean as the marginal electricity behind them: The right emissions number for an electrification decision is the marginal intensity, not the grid average, and the two can differ by a factor of several. This is also why the two curves in panel (a) behave as they do. With the carbon price pushing coal off the margin, the same pumps do far more good; electrification pays off *jointly* with clean marginal power.

Carbon leakage. Another perspective that our calculations above do not catch is that in 35% of hours, the marginal unit supplying electricity is not a Danish plant at all, but the import plant. We do not count those emissions in our scheme (following the territorial convention that counts what is emitted inside the country). As we rely on more imported electricity, however, foreign emissions are likely to increase as well; we refer to this as carbon leakage. This is an interesting topic that is not unique to this experiment. We will not elaborate much more on this point here, except to note that this is *the* main reason why coordinated carbon policies e.g. like the EU-ETS are more efficient at reducing global emissions than national policies.

Complementarity of heat pumps and VRE. It would be easy to leave this section believing that heat pumps and wind are natural allies: The pump consumes when power is cheap, cheap hours are windy hours, so electrification should lift wind's capture price. Ruhnau et al. (2020) test exactly that proposition in a full European model and find the effect on wind's market value to be *almost nil*. They give three reasons, and the third is a lesson about this whole family of models:

i. Heat demand varies *seasonally*, while the storage available to shift it (hot water tanks, building inertia) is measured in hours. This is the same time-scale mismatch that defeated the battery in Section 4.4.

ii. In a model where investment responds, adding flexible heat pumps makes the optimiser build *less* of the other flexibility it would otherwise have chosen (pumped hydro, interconnection), so the net gain in wind's value is largely competed away.

iii. *Linearity itself.* In a linear model, moving a modest amount of demand into a cheap hour does not move that hour's price at all, because the price is pinned at the marginal technology's cost until the shift is large enough to change *which* technology is marginal. The step-shaped supply curve of Figure 3.4 is exactly this: Prices jump between plateaus rather than responding smoothly. So a model can simultaneously report that heat pumps consume in cheap hours and that wind gains nothing. That is not a contradiction but a property of the step function.

5.4 What is left out: combined heat and power

Danish district heating is historically built on **combined heat and power** (CHP): plants that generate electricity and capture the waste heat instead of throwing it away. This type of coupling creates some interesting dynamics: Recall from Figure 3.8, the local CHP plants kept producing electricity through hours of storm-level wind and prices at or below zero, and Section 3.6 attributed this to a commitment to deliver district heat in a cold month. That is the joint-output constraint described below at work, seen from the electricity side: A plant whose heat is needed produces electricity whether or not the electricity price covers its cost.

A regular condensing power plant converts perhaps 25–35% of its fuel into electricity and rejects the rest as warm water. A *back-pressure* plant produces electricity and heat in a roughly fixed ratio $\nu = q^E/q^H$, so with η^E its electrical efficiency, its total conversion efficiency is

$$\frac{q_h^E + q_h^H}{\text{fuel in}} = \left(1 + \frac{1}{\nu}\right)\eta^E, \tag{29}$$

which lands in the range 85–95%.

Modelling it would a fixed proportion between two outputs exactly like eq. (26c) but with ν in place of the COP, and one genuinely new economic result. A back-pressure plant's value depends on *both* prices at once: It is worth running when $\lambda_h^E + \lambda_h^H/\nu$ exceeds its marginal cost. Where a heat pump *correlates* the two prices (it buys power to make heat, so it pushes power prices up and heat prices down at the same time), CHP *decorrelates* them: A back-pressure plant is profitable on high electricity prices even when heat is cheap, and vice versa. Two coupling technologies, opposite signs on the same statistic, which is a good reminder that "sector coupling" is not one thing with one effect.

Which way does it bias the answer? *A constant coefficient of performance, and a fixed fleet.*

- *Had we used one year-round COP of 3.0 instead of the hourly curve.* The pump runs hardest when it is coldest, and cold hours are low-COP hours, so the hours that carry the most weight are systematically the hours where the true COP is below the mean. A constant COP therefore **overstates** the pump's efficiency where it matters and **overstates** its value: It understates the electricity drawn in the tightest hours, the peak load electrification adds, and the backup capacity the system needs.
- *The power fleet is held fixed.* This stacks the deck against electrification, and **overstates** its

emissions and its price effect: A planner adding 2 GW of heat pumps would likely also add wind to serve them. What the fleet *should* look like once investment responds is exactly the question of Section 7.

From duals to markets.

Constraint	Dual	What it measures	Market counterpart
electricity clearing (26b)	λ_h^E	power system marginal cost	the day-ahead spot price
heat clearing (26c)	λ_h^H	marginal cost of a MWh of heat	the marginal-cost basis of a district heating tariff
heat pump capacity (26d)	$\overline{\mu}_h^{HP}$	value of one more MW of pump	the scarcity premium in cold snaps
derived: λ_h^E / COP_h vs boiler cost		the fuel-switching margin	heat-pump-versus-gas dispatch parity, the heat sector's "spark spread"

The three questions.

Adding heat pumps to the system. . .	
. . . does what to the two tails?	Fills in the *cheap* tail only. The pump adds demand when power is cheap and withdraws it when power is dear, so surplus hours shrink while scarcity hours are untouched: It cannot *supply* anything on a cold calm evening. The average power price rises (41.0 → 51.5 €/MWh).
. . . does what to the value of wind?	Raises it in principle, by putting demand into windy hours. But see Ruhnau et al. (2020) above: In a model where other flexibility investment responds, and where prices move in steps, the realised effect may be small.
. . . does what to system cost?	Lowers the cost of *heat* substantially, raises the cost of power, and cuts total emissions by only 26% against this fleet, because the marginal electricity is often coal. Electrification pays off jointly with clean marginal power, not before it.

One forward pointer closes the section. Flexible heat is storage in disguise: Buildings and district heating networks hold heat cheaply for hours, so an electrified heat sector is not only new load but potentially the cheapest flexibility on the system, the thermal analogue of Section 4. We have left it out here deliberately, to teach the coupling mechanism with a single technology and a single new equation. The next section leaves the single node behind and asks what trade with the neighbours can do.

6 Transmission and trade

Denmark's electricity system cannot be understood as an island: On a windy day DK1 exports most of what it produces, and on a calm one it lives off Norwegian water and German gas. This section puts the single node of Sections 3 to 5 into its geography, and adds the third item on the flexibility menu of Section 4: trade, which moves energy in space rather than in time. Trade needs wires, and wires have physics that no previous section required. So the section first describes the market's geography and the grid that carries it, at the level an economist needs, and then extends the model with zones and the lines between them. From the model we derive what a congested line is worth, confront the prices and rents with the real market once more, and ask what a bigger cable is worth.

> **Reading alongside this section.** This section requires no supplementary reading. Cretì and Fontini (2019, ch. 14–16) cover the ground of Section 6.2 in more detail: the basic principles of transmission, meshed networks and congestion, and transmission pricing in practice. Gonzales et al. (2023) is a good answer to the question "do economists actually use these models?". They study two Chilean grid expansions connecting a solar-abundant desert region to demand centres, using a cost-based dispatch model with ramping and hydro storage (essentially this note's model with the extensions of Section 4) plus endogenous renewable investment, and find solar generation up 180%, generation costs down 8% and emissions down 5%. Bjørndal and Jörnsten (2001) is the Nordic origin of the zonal-versus-nodal question of Section 6.1. On who should pay for a cable, Joskow and Tirole (2005) is the reference on merchant transmission and Hogan (2018) a short primer on benefits and cost allocation; Section 6.7 draws on both.

6.1 The geography: bidding zones

There are different ways of arranging wholesale electricity markets. In Europe, generators and consumers are collected in **bidding zones**: an area within which the market pretends there is no grid at all. Any generator in the zone can serve any consumer in it, every trade inside the zone clears at one price, and only *between* zones is transmission capacity rationed. Most zones are countries, but a few countries are split, because a bottleneck inside them is too large to ignore: Norway has five zones, Sweden four, Italy seven, and Denmark two, DK1 for Jutland and Funen and DK2 for Zealand. Figure 6.1 shows the map, with the twelve zones this section models highlighted.

This zonal design can be computationally and politically convenient, but it comes at a cost: congestion *inside* a zone has not gone away; it has only been hidden from the price. When the market's outcome would overload a line inside a zone, the system operator must fix it afterwards by **redispatch**: paying a generator on the wrong side of the bottleneck to produce less and another on the right side to produce more, at cost, after the market has cleared. German redispatch costs run into billions of euros a year. That cost is real, it is absorbed by network tariffs. The alternative is **nodal pricing**, the design of most American markets, in which every node of the network gets its own price computed from a model that includes the full grid. Internal congestion is then priced

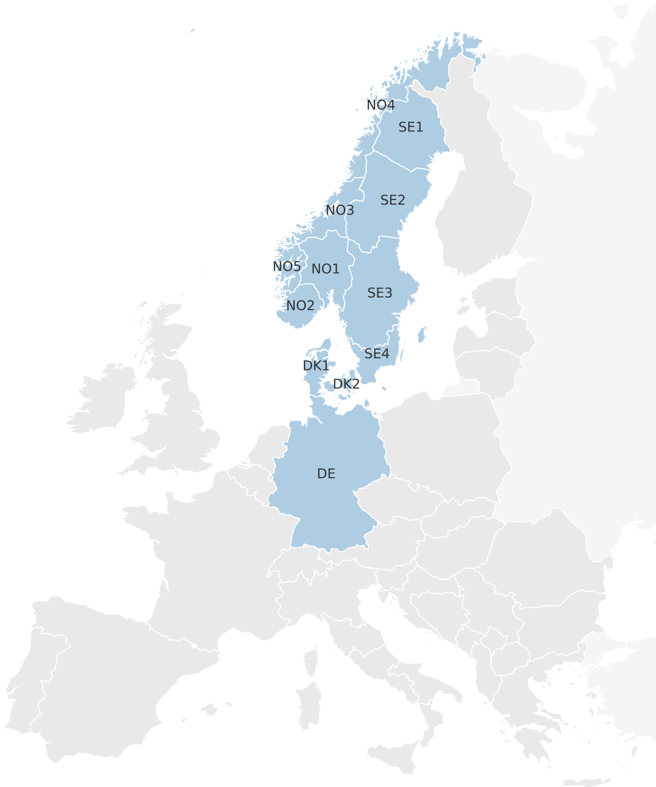

Figure 6.1: The bidding zones of Europe's day-ahead market, with the twelve zones of this section's network highlighted. Zone borders inside Norway, Sweden and Denmark follow county borders and are approximate.

rather than redispatched: A generator behind a bottleneck sees a low price and responds to it, and a consumer in a constrained area sees a high one. This is the textbook-efficient answer, and it goes back to Schweppe et al. (1988); Bjørndal and Jörnsten (2001) is the Nordic statement of what zones cost relative to it. We return to the comparison once the model is on the table.

The network of this section. As in the previous models, the following sections build on actual data on networks, power plant fleets, historical weather patterns etc.. To do this, we rely on PyPSA-Eur, the open European model used across research and industry; Appendix C documents this further for the interested reader. We focus on the Nordic part of the European electricity network. The network consists of the 12 **bidding zones** of Denmark and its neighbours: DK1, DK2, Germany, SE1–SE4 and NO1–NO5. This is far as we are prepared to go in a lecture note in terms of model size, but even this is a simplification of the highly integrated European grid.

It is worth noting that the ten stylised plants of Table 1 do not come with us, as this is not detailed enough for our broadened model scope. The fleet is now 67 generators and 14 hydro reservoirs, inventoried in Appendix D, which adds technologies the small table never had (lignite, nuclear, run-of-river) and gives every plant its own efficiency instead of one representative value per type. What does not change, however, is how we interpret and solve the model in general: Every thermal generator here burns fuel at the prices of Table 1, so a combined-cycle plant costs the same to run in this section as it did in Section 2. Nothing in the vocabulary changes either; the shadow variables are read here exactly as they were read there.

6.2 Transmission grids, for economists

The wires between the zones are the new ingredient, and they deserve a few pages before the model, because the model treats two kinds of wire differently and the reason is physical. We keep to what the model needs; Cretì and Fontini (2019, ch. 14–15) go a bit further.

What a line is to the model. The transmission grid is the high-voltage part of the electricity network, 220 and 400 kV in continental Europe, which carries power from large plants to the areas that consume it and from one zone to the next; below it, distribution grids at lower voltages bring the power to the individual consumer, and they play no role in this note.

For our purposes a transmission line ℓ is characterized by four numbers. First, a *capacity* $\bar{f}_\ell$ in MW: A conductor heats as current flows through it, and its *thermal rating* is the flow it can carry without sagging or damage. The market rarely gets to use all of it. What it trades on is the **net transfer capacity** (NTC), a commercial figure the system operators publish for every border, deliberately below the thermal limit to leave a security margin for the failure of any single element and, on AC borders, for flows the market did not schedule; the paragraphs on AC flow below explain where those come from. On the Jutland–Germany border, for example, the market only sees 2.5 GW where the wires could physically carry several times that. Second, a *loss* share ϕ_ℓ: Resistance turns part of the flow into heat, and the loss grows with the square of the current, so a heavily loaded line loses proportionally more than a lightly loaded one. The simulations charge every DC crossing a flat 3% and approximate the quadratic AC loss piecewise (Appendix D). Third, an *investment cost*, Table 3, which the model of this section does not need because the grid is given, but which Section 6.6 and Section 8 will. Fourth, a property with no number: whether an operator can set the flow on the line directly. That depends on whether the line carries alternating or direct current, which is where the physics enters.

Table 3: What a wire costs: investment costs of transmission technologies in the technology-data compilation the note prices everything from (Appendix C), all with a 40-year lifetime. A DC line pays for its converter stations once and for its cable per kilometre.

Technology	Investment	Unit
AC overhead line	442	€/MW per km
DC overhead line	442	€/MW per km
DC submarine cable	1008	€/MW per km
DC converter station pair	166	thousand €/MW

Alternating and direct current. Almost all of the grid runs on **alternating current** (AC): The voltage swings back and forth fifty times a second, and every generator connected to the grid spins in lockstep with that rhythm. An area in which this holds is a **synchronous area**, and it is one coupled machine: A disturbance anywhere in it is felt everywhere in it within a second, and the frequency is one number for the whole area. Continental Europe, from Portugal to Poland, is one synchronous area; the Nordic countries are another. The two Danish zones sit in different ones. DK1 is synchronised with Germany and the continent, DK2 with Sweden and Norway, and no AC line joins them. **Direct current** (DC) does not alternate. A DC line takes AC power in at one end through a *converter station*, sends it as direct current, and converts it back at the other end. The converters are power electronics, and they set the flow: A DC line carries exactly what its operator tells it to. Because the two ends need not be synchronised, DC is also the only way to join two synchronous areas, which is why every cable between DK1 and the Nordic zones, and the Great Belt link between the two Danish zones, is a DC cable. Figure 6.2 draws the network with the distinction made: 15 AC corridors, 7 DC corridors, and the two synchronous areas.

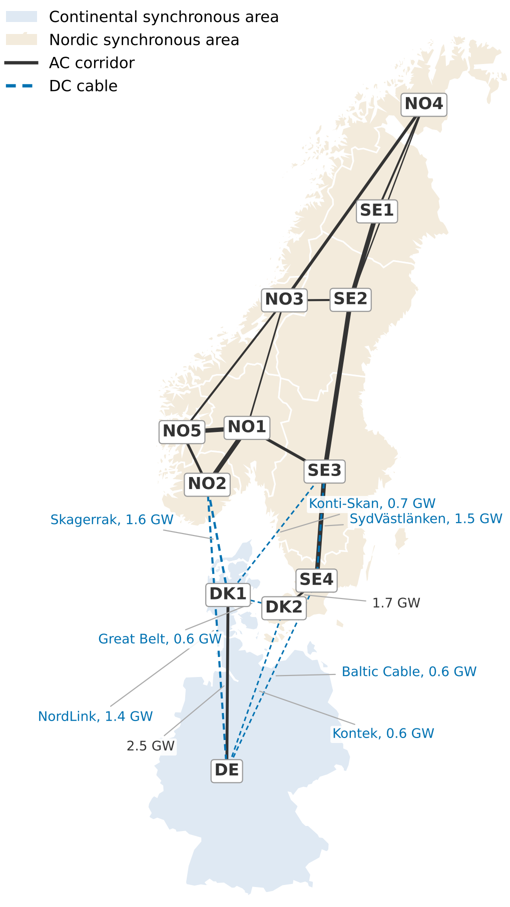

Figure 6.2: The network of this section: twelve zones in two synchronous areas, joined by AC corridors (solid) and DC cables (dashed), with line width scaled to the capacity the market trades on. The named cables and the two AC borders facing Denmark are labelled with their capacity.

Three differences that matter. First, *cost*. An AC line needs only the wire, while a DC line pays for a converter station at each end, on the order of 166 thousand euros per MW for the pair, before a single kilometre is built (Table 3). Over short distances AC therefore wins. Over long ones DC does. The compilation behind the table prices the two overhead wires alike, and in practice a DC line is only somewhat cheaper per kilometre, two conductors where AC needs three; what decides the long-distance case is the lower losses and, above all, the sea. Under water DC is the only option for anything beyond a few tens of kilometres: A long AC cable spends its capacity charging its own insulation rather than delivering power. This is why every long sea crossing in the North Sea and the Baltic is a DC cable. For a cable of the Skagerrak's class, 360 km under water, the figures of Table 3 give roughly 529 M€ per GW; the European network planners' own candidates for the next cable on that corridor are priced at 850–1150 M€ per GW (Appendix C). Second, *losses*: lower per kilometre on DC, but the converters add their own share at each end, which is what the flat 3% per crossing stands for. Third, *control*, the difference that matters most for the model. A DC flow is set by its operator; an AC flow follows from everything else that is happening on the network. We take the two cases in turn.

Controllable flow: the DC line. A DC line from zone a to zone b is the simplest object in this section. Its operator picks a flow $f_{\ell,h}$ between zero and the capacity,

$$0 \;\leq\; f_{\ell,h} \;\leq\; \bar{f}_{\ell}, \tag{30}$$

zone a gives up $f_{\ell,h}$ MWh and zone b receives $(1-\phi_\ell)\, f_{\ell,h}$, the rest having been lost on the way. To let power go the other way we simply give the same cable a second flow variable in the opposite direction. Nothing else constrains it. This is the transport model of trade that an economist would write down unprompted, and for a DC cable it is exactly right.

Flows the operator cannot set line by line: the AC network. What follows is background. You need not carry the physics with you, only the three consequences the paragraph ends with, and the one constraint the model keeps. On an AC network the operator does not set the flow on each line. Power injected in one zone and withdrawn in another spreads over *every* path between them, in proportions fixed by the lines' electrical properties, and the flow on any one line can be changed only by changing where power is injected and withdrawn. The exact description is a system of nonlinear equations in the voltages and phase angles at every node, which this note, and every market model in practice, avoids by linearising it.[13] In the linearised version the flow on a line is proportional to the difference in phase angle across it, divided by the line's *reactance* x_ℓ, a fixed property of the conductor that grows with its length. Two consequences follow. First, between two zones joined by parallel paths the flow splits in inverse proportion to the paths' reactances: The path with half the reactance carries twice the flow. Second, around every loop the reactance-weighted flows sum to zero,

$$\sum_{\ell \in C} \sigma_{C\ell}\, x_\ell\, f_\ell \;=\; 0 \qquad \text{for every loop } C, \tag{31}$$

[13]For a line between a and b the exact active power flow is $f_\ell = V_a V_b \sin(\theta_a - \theta_b)/x_\ell$, with V the voltage and θ the phase angle at each end. Under normal operation the voltages sit near their nominal values and the angle differences are small, so $f_\ell \approx (\theta_a - \theta_b)/x_\ell$ in convenient units. Engineers call this the "DC power flow", an unfortunate name: It is an approximation of AC physics and has nothing to do with DC cables. Stott et al. (2009) review how good it is.

with $\sigma_{C\ell} = \pm 1$ recording whether line ℓ is traversed with or against the direction chosen for the loop. This is *Kirchhoff's voltage law*, and eq. (31) is the one line of physics the model carries. A *loop* here is any set of lines along which power can leave a zone and return to it by a different route. NO1 to NO5 to NO2 and back to NO1 is a loop, and so is the larger ring through Sweden. The single corridor between Germany and DK1 is not, since there is no second route back, and neither is a DC cable, whose flow is set by its converters. In this section's network every loop lies inside the Nordic synchronous area, 5 independent ones in all, and eq. (31) holds on each.

Figure 6.3 shows the consequence with three zones and three identical lines. Send 300 MW from A to C. The direct line has reactance x and the detour through B has $2x$, so 200 MW goes direct and 100 MW through B, and B carries 100 MW of a trade it was no part of. Three lessons carry over to the real network. First, the 100 MW through B is a **loop flow**, capacity used by a trade nobody sold on that line, and the reason the NTC on an AC border sits below the wires' rating. Second, the transfer capacity of a meshed grid is not the sum of its lines: Rate every line in the figure at 250 MW and the direct line fills first, at a transfer well short of the 500 MW of wire touching C. Third, loading one AC line changes the flow on every other line of the loop, which is why an AC system is so much less flexible to operate than a DC one, and why Europe's grid is adding DC cables. The famous continental case is trade between Germany and Austria that loops through Poland and the Czech Republic. Since 2024 the Nordic day-ahead market has rationed its borders by *flow-based* coupling, which puts eq. (31) into the market algorithm itself rather than hiding it behind fixed NTCs.

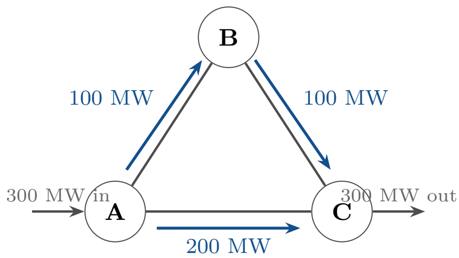

Figure 6.3: The linearised power flow on a loop of three identical lines: A transfer of 300 MW from A to C splits two to one between the direct line and the detour through B.

6.3 The model

Each zone z has its own market clearing constraint with its own price $\lambda_{z,h}$, and the corridors move power between the zones. We index the corridors by direction: $\mathcal{L}$ contains every corridor twice, once for each way power can flow, and for $\ell \in \mathcal{L}$ we write a_ℓ for the zone the flow leaves and b_ℓ for the zone it enters. Every flow is then non-negative, the losses are charged where the power arrives, and a corridor's capacity $\bar{f}_\ell$ is its NTC. Written out, with geography in blue, the planner chooses generation $q_{g,h}$ and flows $f_{\ell,h}$ to solve

$$\min \quad \sum_h \sum_z \sum_{g \in \mathcal{G}_z} c_g \, q_{g,h} \tag{32a}$$

$$\text{s.t.} \quad \sum_{g\in\mathcal{G}_z} q_{g,h} + \sum_{\ell:\, b_\ell = z} (1-\phi_\ell)\, f_{\ell,h} \;=\; D_{z,h} + \sum_{\ell:\, a_\ell = z} f_{\ell,h} \qquad (\lambda_{z,h}) \qquad \forall z, h \tag{32b}$$

$$0 \le q_{g,h} \le \gamma_{g,h}\, \bar{q}_g \qquad (\underline{\mu}_{g,h}, \overline{\mu}_{g,h}) \qquad \forall g, h$$

$$0 \;\le\; f_{\ell,h} \;\le\; \bar{f}_\ell \qquad (\underline{\omega}_{\ell,h},\; \overline{\omega}_{\ell,h}) \qquad \forall \ell, h \tag{32c}$$

$$\textstyle\sum_{\ell\in C} \sigma_{C\ell}\, x_\ell\, f_{\ell,h} \;=\; 0 \qquad (\zeta_{C,h}) \qquad \forall \text{ AC loops } C \tag{32d}$$

Equation (32b) is Section 3's market clearing, one copy per zone, with the zone's imports net of losses on the supply side and its exports on the demand side. Equation (32c) is eq. (30), the capacity limit on every directed corridor. Equation (32d) is eq. (31) for every loop of the AC network, with C now running over the directed corridors along the loop; on a DC cable there is no such constraint, and on the AC corridor between Germany and DK1, which closes no loop, it is vacuous. Nodal pricing, for the record, is this same problem written with a node in place of a zone and eq. (32d) enforced on every loop of the full grid.

One housekeeping note. The problem above is the one the text reasons about, and it stays that minimal on purpose. The *simulations* behind the figures of this and the following sections carry two further constraints on top of it, in order to get closer to more realistic simulations: ramping limits on the slow thermal plants, outages on the thermal fleets.[14] Appendix Appendix D elaborates on the model ingredients.

Duality: the price spread. The Lagrangian extends as it has in every section, with one wrinkle: The market clearing constraint is not merely extended here but *replicated*, one copy per zone. So the starting point is $\mathcal{L}^{\text{vre}}$ read zone by zone — write $\mathcal{L}^{\text{vre}}_z$ for (14) restricted to zone z's generators, load and price $\lambda_{z,h}$ — and the wires add their terms on top,

$$\mathcal{L}^{\text{net}} = \sum_z \mathcal{L}^{\text{vre}}_z + \sum_h \sum_z \lambda_{z,h} \Big[\sum_{\ell:\, b_\ell = z} (1-\phi_\ell)\, f_{\ell,h} - \sum_{\ell:\, a_\ell = z} f_{\ell,h} \Big] + \sum_h \sum_C \zeta_{C,h} \sum_{\ell\in C} \sigma_{C\ell}\, x_\ell\, f_{\ell,h}$$
$$+ \sum_h \sum_\ell \overline{\omega}_{\ell,h} \big(\bar{f}_\ell - f_{\ell,h}\big) + \sum_h \sum_\ell \underline{\omega}_{\ell,h}\, f_{\ell,h}. \tag{33}$$

Now consider a DC cable ℓ from zone a to zone b. Flow carries no cost in the objective and closes no loop, so it appears only in the blue terms of the first line, and differentiating $\mathcal{L}^{\text{net}}$ with respect to $f_{\ell,h}$ gives

$$(1-\phi_\ell)\, \lambda_{b,h} - \lambda_{a,h} + \underline{\omega}_{\ell,h} - \overline{\omega}_{\ell,h} \;=\; 0, \tag{34}$$

with complementary slackness on the two bounds. We read it the usual way:

- <u>Lower bound: nothing is sent from a to b.</u> Then $\underline{\omega}_{\ell,h} = \lambda_{a,h} - (1-\phi_\ell)\lambda_{b,h} \ge 0$: What a MWh fetches at the far end, net of what is lost on the way, does not cover its price at home. Power may well be flowing the other way.
- <u>Interior: the cable is used, below capacity.</u> Both multipliers are zero and $(1-\phi_\ell)\lambda_{b,h} = \lambda_{a,h}$.

[14] In addition, the model approximates the AC losses piecewise rather than with the flat share ϕ_ℓ, because this is the standard approach in the PyPSA open source framework anyway.

Arbitrage across a slack cable does not equalise the two prices; it brings them within the *loss wedge*, $\lambda_{b,h} - \lambda_{a,h} = \phi_\ell\, \lambda_{b,h}$. On a lossless cable the wedge is zero and the prices coincide.

- Upper bound: congested from a to b. Then $\overline{\omega}_{\ell,h} = (1-\phi_\ell)\lambda_{b,h} - \lambda_{a,h} > 0$. The prices decouple, and the spread net of losses is the shadow price of the cable's capacity: the value of one more MW of wire in that hour.

On an AC corridor eq. (34) gains a term $\sum_C \zeta_{C,h}\, \sigma_{C\ell}\, x_\ell$ for every loop the corridor lies on, so two zonal prices can differ across a slack AC line when a loop constraint binds somewhere else on the ring: the price of loop flow. The reading is otherwise the same.

Result 5 (Prices across a corridor)**.** *In a zonal market, power flows toward the dearer zone, and only when the spread covers the losses. Across a slack corridor the two zonal prices differ by the loss wedge $\phi_\ell\, \lambda_{b,h}$ and by nothing else; across a congested corridor the spread net of losses, $(1-\phi_\ell)\lambda_{b,h} - \lambda_{a,h}$, is the shadow price of one more MW of capacity, exactly the $\overline{\mu}_g$ of Section 2 attached to a wire instead of a plant.*

Zonal pricing is thus the planner's market clearing decentralised by geography. Because the zones here are the real bidding zones, each $\lambda_{z,h}$ has a direct empirical counterpart in the Nord Pool and EPEX day-ahead prices for the same areas.

6.4 Prices and rents, in the model and in the market

Let us now solve the year and read the outcome through Result 5: first the zonal prices, then the rents the corridors between them collect, and then both against what the market actually did in 2024. Two choices about the instance should be stated first. Every solve in this section charges the 2024 ETS average of 65 €/t on every emitter, because the section is held against the 2024 market and that year's plants paid it; Section 7 charges a higher reference price because it asks about 2030, not about a year that happened. And the network is calibrated to its year in two respects before it is solved: Norwegian reservoir inflow is scaled up from the normal-year level every section uses so that the model's hydro output equals the 140 TWh Norway's plants actually produced in a record year, and the internal Nordic corridors are derated from their thermal ratings to the transfer capacities the market trades on. Appendix C documents both, and what each does to the prices; the sections that follow, which look at 2050 rather than at 2024, keep the normal-year level.

Zonal prices. Figure 6.4 shows the year's price duration curves for the five zones that face Denmark.

The figure shows the following:

i. The Nordic zones are almost perfectly flat. Reservoir hydro is the intertemporal arbitrage of Section 4 at national scale, and arbitrage flattens the price it trades on. The plateau *is* the water value, the shadow price of stored water, and it pins prices whenever hydro is marginal, which in the north is most of the year (southern Norway averages 59.2 €/MWh, central Sweden 57.7).

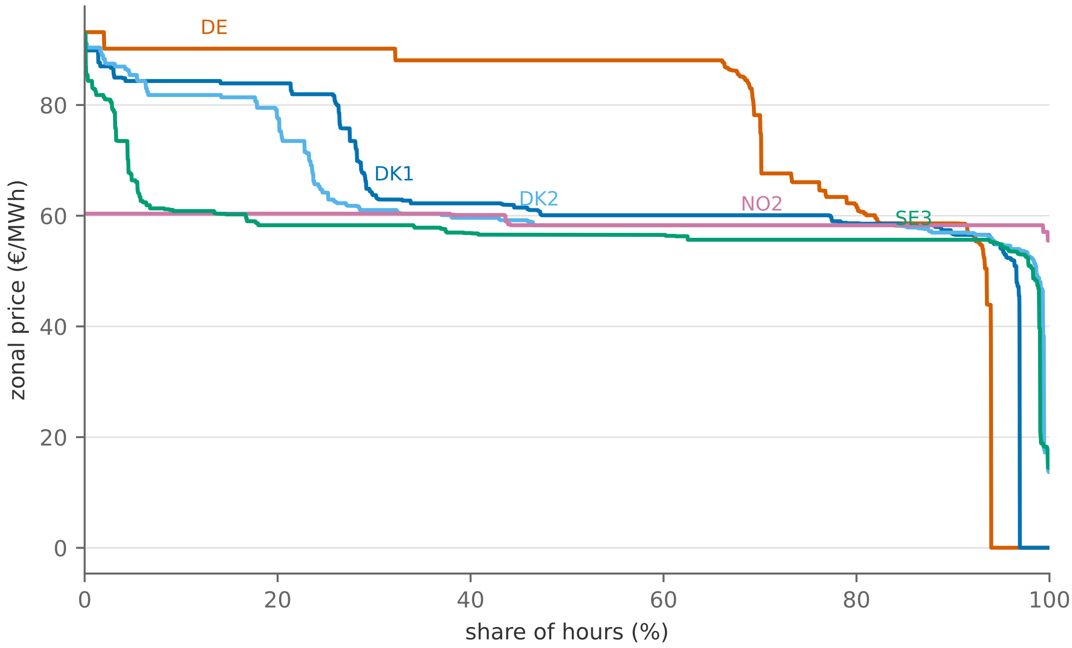

Figure 6.4: Price duration curves for the five zones that face Denmark (the remaining Nordic zones track their neighbours closely).

ii. Germany, thermal at the margin, swings: gas plus carbon in the tight third of the year, coal and lignite plus carbon below that, and zero in wind-surplus hours. It averages 76.9, well *above* the hydro plateau: With the carbon price charged, the thermal fleet is the dear side of every cable to the North.

iii. The Danish zones are the instructive case. Glued to the German price when the cables to the North are full and pulled down to the Nordic plateau when they have room, they end the year at 64.8 (DK1) and 63.7 (DK2), between their two neighbours.

The mechanism is what matters. Denmark sits between a dear thermal neighbour and a cheap hydro neighbour, and its price is the German one in the hours when the Nordic cables are congested and close to the Nordic one in the rest; the German zeros reach it too, through the slack border. A zone between two markets does not average them; it takes a different draw from each tail. The geographical fact of the Danish power market stands: It is a small system whose price is set abroad more often than at home.

Congestion rent. A congested interconnector buys at the cheap end and sells at the dear end. The quantity the system operators report as their *bottleneck income*, and the one Figure 6.5 plots, is the flow times the price spread, summed over the year,

$$R_\ell \;=\; \sum_h f_{\ell,h}\,(\lambda_{b,h} - \lambda_{a,h})\,. \tag{35}$$

By eq. (34) and complementary slackness this splits into two parts,

$$R_\ell = \underbrace{\sum_h \overline{\omega}_{\ell,h}\, \bar{f}_\ell}_{\text{scarcity rent}} + \underbrace{\sum_h \phi_\ell\, \lambda_{b,h}\, f_{\ell,h}}_{\text{value of the losses}}, \tag{36}$$

since the bracket in eq. (35) equals $\overline{\omega}_{\ell,h} + \phi_\ell \lambda_{b,h}$ whenever the cable flows. The first term is the scarcity rent $\overline{\mu}_g \bar{q}_g$ of Section 2 for a wire: the shadow value of the capacity constraint, collected in cash by whoever owns the cable. In Europe that is the TSOs, who must by regulation spend it on more capacity or lower tariffs (European Parliament and Council 2019). The second term is not scarcity at all: On a lossy line the two end prices differ even when the line is slack, and that spread pays for the energy burned in transit. Every term is non-negative, because power flows toward the expensive zone. Figure 6.5 tallies the year by corridor.

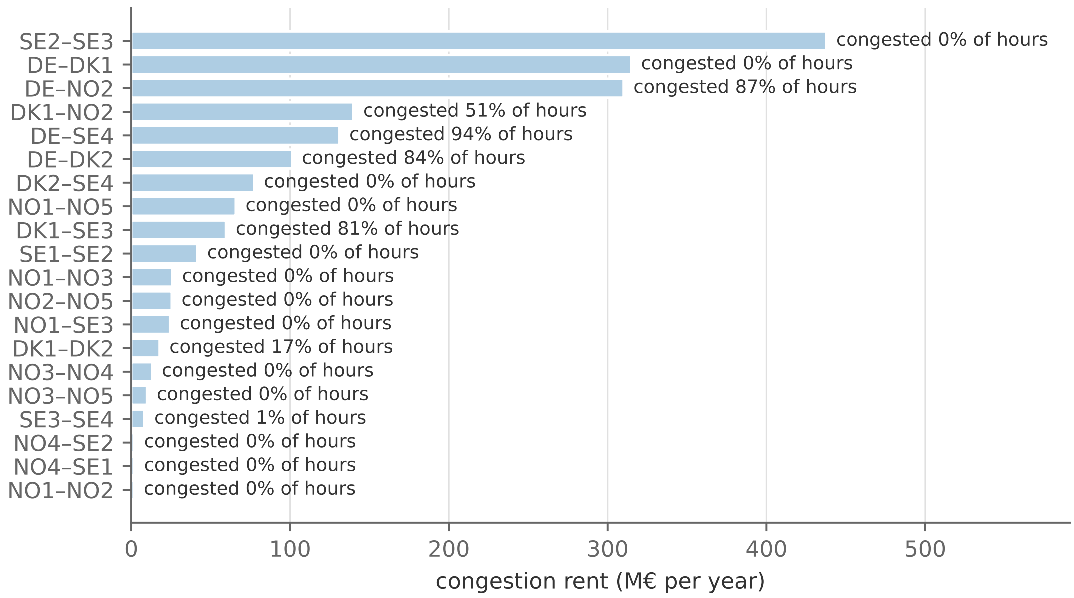

Figure 6.5: Annual congestion rent by interconnector, computed as flow times spread (35), with the share of hours each corridor runs at its limit.

The figure shows that the yearly congestion rent totals 1809 M€. This is partly due to congested cables with significant price spreads (in particular From Germany to the Nordics), but also because of non-congested and heavily traded corridors' value of losses (e.g. internally in Sweden). It is worth noting that, in reality, what we here represent as income from "non-congested" lines is not always collected as such. For instance, the two largest bars , the corridor between central and southern Sweden and the Jutland–Germany border, earn their whole "rent" this way. In the real market this happens only on the DC cables: The Nordic day-ahead coupling schedules the Great Belt, Skagerrak and Konti-Skan with an explicit loss factor, so a flow on them is scheduled only when the spread covers the loss, and the TSOs' bottleneck income on them includes that wedge. An AC border between two zones is treated as lossless by the coupling; its two prices are equal whenever the capacity is slack, and the TSOs buy the physical losses separately and recover them through tariffs. The two largest bars are both AC corridors that never congest, so in the market they would show no spread and no income at all. The Jutland–Germany corridor never reaches its limit and loses about three percent of what it carries, yet the two zonal prices differ by a dozen euros in most hours. Two features of the network do that. The network measures every corridor between the centres of the zones it joins, which makes this border several hundred kilometres long where the real tie is a fraction of that; and the piecewise approximation of the quadratic loss, with two tangents, puts the *marginal* loss rate, which is what enters the flow condition and hence the spread, at up to twice the average rate. The bar credits a slack corridor with roughly four times the income the Danish TSO reported on it; the validation below returns to it.

The model against the market. Because the zones are the real bidding zones and the year is 2024, every number above has a published counterpart: the day-ahead area prices, and the congestion rent actually collected on the Danish borders, computable from public data as hourly exchange flow times the hourly price spread, exactly eq. (35). Figure 6.6 makes both comparisons. The prices are compared for the six zones Energi Data Service publishes, the two Danish zones and the four that border them.

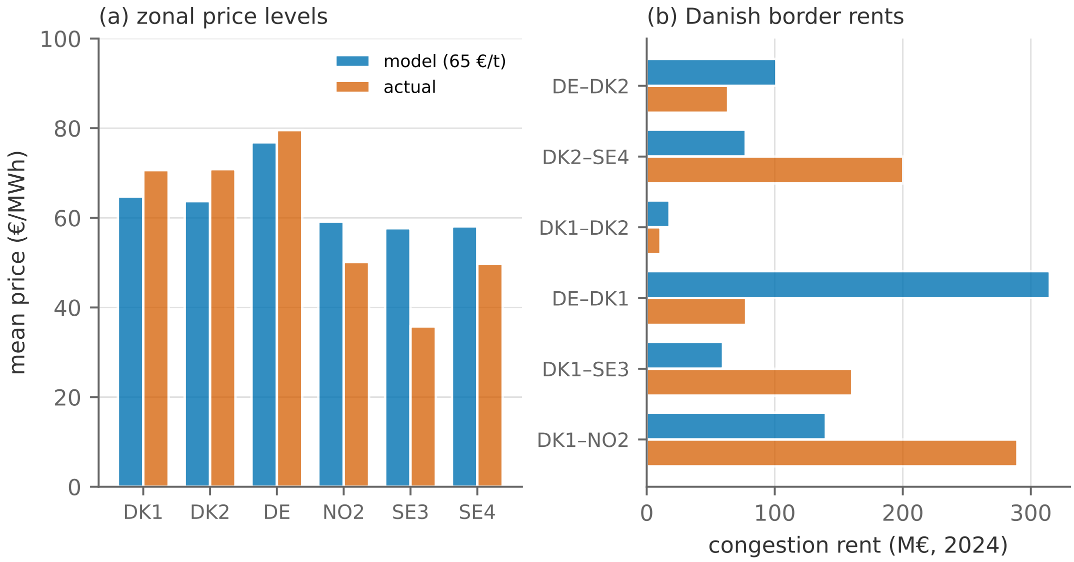

Figure 6.6: Model against market, 2024. (a): mean zonal prices in the model and realised, for the six zones Energi Data Service publishes. (b): congestion rent on the Danish borders, the model against rents computed from Energi Data Service's hourly exchange flows and the zones' price spreads.

Panel (a) shows that average prices fare relatively well, but with a noticeable difference in Norway and Sweden where the model prices are too high. As mentioned earlier, these differences can likely be attributed to the fact that the model is still simplified in various ways. Some of the relevant items include: (i) We do not include British and Dutch interconnectors, (ii) no "must-run" generators and unit commitment, (iii) an exaggerated loss wedge on the Jutland–Germany corridor, (iv) and perfect foresight.

Panel (b) is the harder test, since rents live on the hourly *differences* between prices. Summed over the Danish borders the model collects 711 M€ against the 802 M€ of bottleneck income the

TSOs actually reported: the right order of magnitude, but with the composition wrong. The model collects far too much on the Jutland–Germany border and too little on the Nordic cables (Skagerrak: 140 M€ modelled against 290 M€ realised). The likely culprits are the same as above, but not something we will dwell more on for now.

What is still missing. Of the three boundary errors named in Section 3's box, this section removes two: Denmark can now sell, and the neighbours' prices are computed rather than read from the market. The third, a network inside each zone, is what zonal pricing leaves out by construction, and it is time to return to the targeting principle of Section 3.5. We asked there whether network congestion caused by renewable siting is a genuine externality. Under nodal pricing it largely is not: The congestion is priced, and the generator faces it. Under zonal pricing it largely is: A wind farm sited behind an internal bottleneck imposes redispatch costs on everyone else and sees a price that does not reflect them. The market failure, in other words, is not created by intermittency; it is created by the pricing geography. That is a much more useful diagnosis than "renewables impose system costs", and it points at an instrument, pricing the location, rather than at a subsidy. Our model is zonal, so it inherits this blind spot. Every congestion rent computed here is a *cross-border* rent. Internal Danish, German and Norwegian congestion is invisible by construction, and the model therefore understates the true cost of moving power around. The bias box collects it with the two assumptions the validation just exposed.

Which way does it bias the answer? *Zonal prices, fixed transfer capacities, and perfect foresight over the water.*

- *No network inside a zone.* Internal congestion and redispatch are invisible, so the model **understates** the cost of moving power and **overstates** the value of variable generation sited behind internal bottlenecks.
- *Fixed NTCs instead of flow-based coupling.* The right corridors congest, at partly the wrong hours, so the model **understates** the rents collected on the Nordic cables.
- *Losses priced on AC borders, over corridors measured between zone centres, with two tangents.* A property of how this network was built, not of zonal models. The market couples AC borders without a loss wedge; the model charges one, over a long line and at a marginal rate up to twice the true one, so it **overstates** the price wedge across the Jutland–Germany border, **understates** DK1's price, and **overstates** the "rent" on a corridor that never congests.
- *Perfect foresight over inflow.* Real reservoir owners hedge both ways: In a wet year they produce ahead of the risk of spilling and the price falls, in a dry one they sell at normal water values until the shortage is visible and then face a spike. The planner never spills and saves from the first week, so the model **overstates** Nordic prices in a wet year and, in a dry one, **overstates** them early and **understates** the spike.

6.5 Geographic smoothing

Interconnection does more than arbitrage average price levels; it aggregates weather. Figure 6.7 shows the two Danish zones' onshore wind profiles and the duration curves of each zone and of their average.

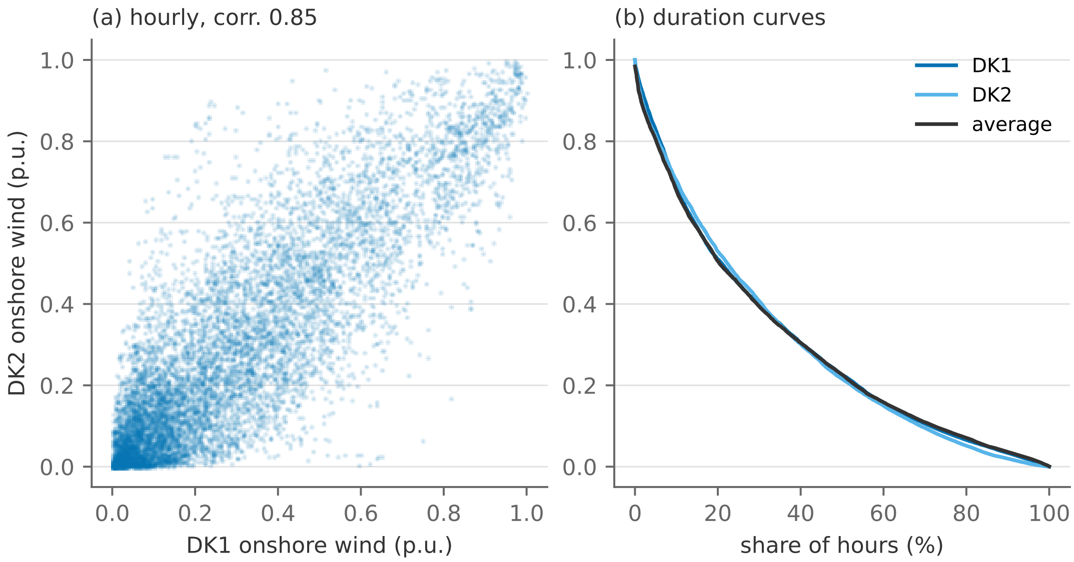

Figure 6.7: DK1 and DK2 onshore wind, hourly 2024: (a) the two profiles against each other, (b) the duration curves of each zone and of the averaged profile.

The figure shows a *negative* result: At a correlation of 0.85 (three hundred kilometres is simply not far in meteorological terms) averaging the two zones flattens the duration curve only barely. Denmark cannot diversify its wind against itself. The value of aggregation is real but it lives at *distance*: Correlations fall roughly exponentially with separation, Iberian and Nordic wind are close to uncorrelated, and the same figure drawn for DK1 against Spain would show the averaged curve visibly flatter at both ends. That is the statistical argument for a continental grid rather than national self-sufficiency. It is also the reason that the smoothing this network can offer runs through its *price* channel (Norwegian reservoirs absorbing Danish surpluses) rather than through wind diversification among near-identical neighbours.

6.6 The value of another gigawatt

Let us now ask the planner's question directly: What is a bigger cable worth? Figure 6.8 re-solves the year with the Skagerrak cables at every size from zero to three times today's 1.6 GW, and reports two things: the annual system cost saving relative to no cables, and the congestion rent the corridor itself earns. Both are gross of what the cable costs to build. The figure is the benefit side of a cost-benefit analysis, to be set against the annualised investment of Table 3; the model of this section takes the grid as given and does not do that arithmetic.

The figure shows the following:

i. The *social* value, system cost saved relative to no cables, rises concavely, from 237 M€ a year

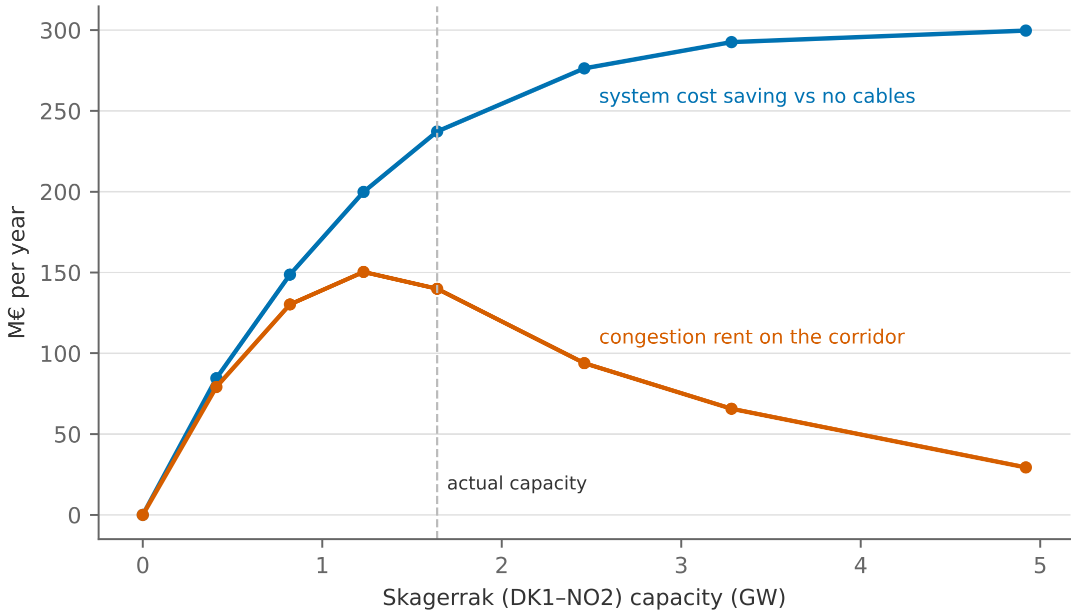

Figure 6.8: The Skagerrak corridor swept from zero to three times its actual capacity: system cost saving relative to no cables, and the congestion rent the corridor earns.

at actual capacity to 300 M€ at triple: Each gigawatt closes more of the spread the next would have monetised.

ii. The *merchant* value, the rent the corridor itself collects, traces a hump. It peaks a little below today's size, stands at 140 M€ at the capacity actually built, and then collapses, to 29 M€ at triple capacity, because a big enough corridor competes away the very spread it earns. It does not fall to zero at any size the figure shows, since reservoirs are finite and no cable can export water that has not rained. Where the peak sits is a property of this instance, its calibrated water and its fuel prices, not a finding.

iii. A merchant investor, paid only the rent, would stop at the hump, while every gigawatt beyond it still creates social value.

Beyond the peak, more capacity is strictly *against* the cable owner's interest even as it benefits the system. This is the standing argument for regulated rather than merchant transmission investment, made formally by Joskow and Tirole (2005), and a live issue every time a TSO's bottleneck income depends on the bottleneck not being fixed. We leave the annuity arithmetic to Section 8, where the model builds corridors itself.

6.7 Who wins, who loses, and who pays

So far throughout this note, we have generally looked at system efficiency. However, a distinct and very important topic is how policies and infrastructure investments affect different stakeholders and consumers. This is particularly clear in a model with the transmission network, where we do not only distinguish between consumers and producers, but also between different countries/zones.

To understand this more carefully, let us again deploy the model and analyse who wins, loses, and who pays what. In general, connecting a low-price zone to a high-price zone generally means that prices tend to converge; Table 4 lists who gains and who loses from that.

Table 4: Who gains and who loses when a cable joins a low-price zone to a high-price zone.

	Low-price (exporting) zone	**High-price (importing) zone**
Price moves	up	down
Consumers	**lose**	**win**
Producers	**win**	**lose**
Cable owner	collects the congestion rent (35)	
Net	positive: the sum is the system cost saving in Figure 6.8	

The total effect is unambiguously positive. That is the gains-from-trade result, and it is why the model builds the cable. But two of the four cells are negative, and they are concentrated. Figure 6.9 puts the model's own numbers on the cells, country by country, for two projects: the Skagerrak corridor as built, against no corridor at all, and a second corridor of the same size on top of today's. Consumers gain what their bill falls by, since load is fixed; producers gain what their revenue net of variable cost rises by, hydro included; and each corridor's rent is split equally between the two countries it joins, which is how the Nordic cables are in fact owned.

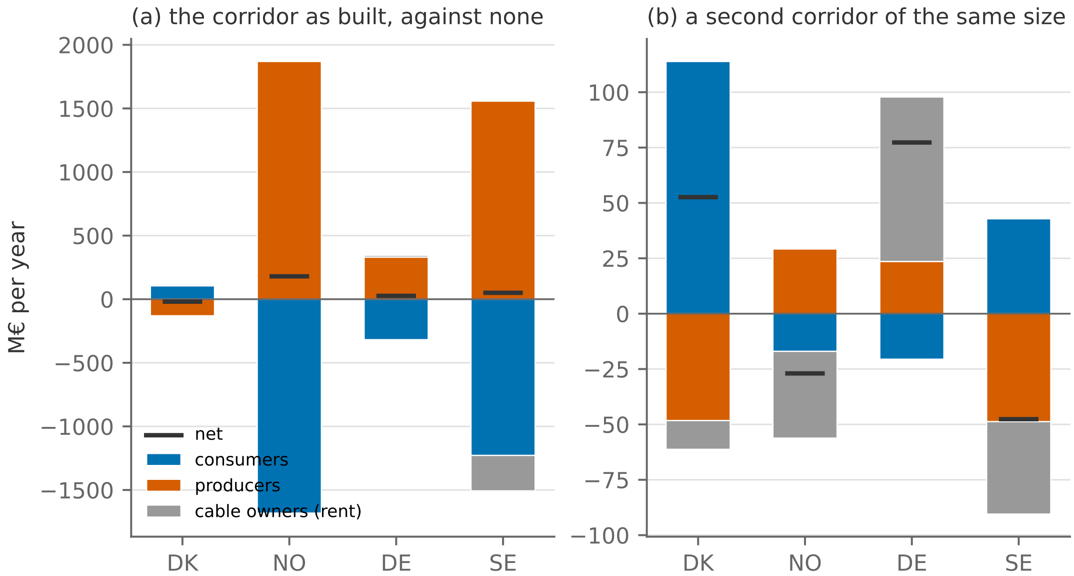

Figure 6.9: Who wins and who loses from the Skagerrak corridor, by country and by group: the change in consumer surplus, producer profit and the country's half of the congestion rents, (a) for the corridor as built against none, (b) for a second corridor of the same size on top of today's. The dash marks each country's net.

Panel (a) shows the following:

i. The cells of Table 4 are there, and they are large. In this wet 2024 Norway is the cheap side: Without the corridor, southern Norway's water is trapped and its price collapses, so the corridor lifts the Norwegian price toward the continent's. Norwegian consumers pay

1682 M€ more and Norwegian producers earn 1869 M€ more, and Sweden, wired to Norway by AC, moves with it, consumers paying 1228 M€ more and producers earning 1557 M€ more. These transfers are an order of magnitude larger than the system cost saving of 237 M€ that the planner sees.

ii. On the importing side Danish consumers gain 106 M€ and Danish producers, selling into a DK1 price that cheap water now holds down, lose 130 M€. Denmark as a country ends a small net loser, by 19 M€, while owning half the cable; Germany gains a little.

iii. Norway takes most of the surplus, a net 180 M€, through its producers. Sweden, which owns none of the cable, still nets 50 M€, because its producers gain more than its cable owners lose: The new corridor takes over export duty from Sweden's own corridors, whose owners lose 279 M€. A third country moves as much as the two parties to the deal.

Panel (b), a second corridor of the same size, is the case that matters for the next investment decision, and it looks different. The system gains 55 M€. Germany gains 77 M€, most of it as income on the Jutland border, whose spread widens as cheap water lowers DK1; Denmark gains 53 M€, nearly all of it to consumers. Norway is now a net *loser*, by 27 M€, because its half of the rent falls on the far side of the hump in Figure 6.8 while its consumers and producers barely move, and Sweden loses 48 M€ the same way. The country that must host and half-own the cable would, on these numbers, have to be paid to build it, and the two countries that gain own none of it. Two reading notes. The rent shares here are what the cable owners actually collect, the value delivered at the receiving end minus the value taken at the sending end, so they are net of losses, unlike the flow-times-spread bars of Figure 6.5; with that convention the country nets add up exactly to the system cost saving. And the signs are those of one instance, a wet year in which the Nordics are the cheap side, as they were in 2021 and 2022 when the debate about the cables began.

Sharing the bill across a border. The figure states the problem a cross-border project has to solve before it is built. Its benefit, the system cost saving of Figure 6.8, is spread over consumers, producers and cable owners in two countries and in their neighbours, in shares the market decides; its cost, Table 3, has to be put on somebody's tariff. There are four ways this is done, and the model has something to say about each.

i. *Split the cost by ownership.* The Nordic convention, and the simplest: Skagerrak is owned half by Energinet and half by Statnett, NordLink half by Statnett and half by the German TSO, and each pays its half and keeps its half of the rent. Simple to agree, but unrelated to who benefits: In Figure 6.9 the two halves of the rent are small and alike while the two countries' nets differ by two hundred million euros.

ii. *Let the beneficiaries pay.* The principle written into both American and European rules. In the United States, FERC's Order 1000 requires the cost of a regional line to be allocated "roughly commensurate with" the benefits it brings (FERC 2011). In the European Union, a cross-border project designated a *project of common interest* goes through a **cross-border cost allocation**: The promoters compute each country's net benefit with the European network planners' scenarios and cost-benefit method, the regulators decide who pays what, a country whose net benefit is negative can be compensated, and if the regulators cannot agree the European agency ACER decides (European Parliament and Council 2022; ACER 2023). The benefit that is allocated is exactly the sum of the bars in Figure 6.9, including the bars of the third countries, which is what makes the exercise contentious: Hogan (2018) explains why the benefits are hard to measure and easy to dispute.

iii. *Let the cable pay for itself.* A merchant line keeps its rent and nothing else. Figure 6.8 says what that buys: a corridor sized to the hump, short of what the system would want, and Joskow and Tirole (2005) show that a merchant investor also has reasons to build the line in the wrong place and to keep it scarce. The European rules accordingly treat merchant interconnectors as an exemption to be applied for, not the default.

iv. *Bargain over the surplus.* Once the benefits are on the table, how to divide a joint surplus is a question cooperative game theory has answers to. Kristiansen et al. (2018) apply them to a North Sea offshore grid, and show that a split that keeps every country better off than it would be alone, such as the Shapley value, can differ substantially from splitting by ownership or by generation share.

Whatever the rule, the congestion rent is the money on the table. In Europe it may by regulation be used only for keeping the corridor available or for more capacity, or, failing that, for lower network tariffs (European Parliament and Council 2019), so Figure 6.5 is also a map of where the next cable could be financed from. And the Norwegian case stands: Norwegian households facing higher bills because their hydro can now be sold to Germany is not a hypothetical grievance but a recurring political controversy, and the model says the grievance is *correct*. The consumers of an exporting country genuinely are worse off. What the model adds is that the country as a whole is better off, because its producers gain more than its consumers lose, which is an argument for compensating transfers, not for refusing to build.

From duals to markets.

Constraint	Dual	What it measures	Market counterpart
zonal clearing (32b)	$\lambda_{z,h}$	the zone's system marginal cost	the Nord Pool / EPEX area price
corridor capacity (NTC) (32c)	$\overline{\omega}_{\ell,h} = (1 - \phi_\ell)\lambda_{b,h} - \lambda_{a,h}$	value of one more MW of corridor	the TSOs' *bottleneck income* (35), net of losses
AC loop flow (32d)	$\zeta_{C,h}$	the physics constraint on routing	loop flows; the redispatch a TSO must buy
reservoir energy (Section 4's θ_h)	water value	value of a stored MWh of water	Nordic producers' water-value bids

The three questions.

Adding interconnection to the system...	
...does what to the two tails?	Compresses both, and by more than storage does: Imports cover the scarcity tail, exports drain the surplus tail. Note the asymmetry with storage: Trade moves energy in *space*, so it works on week-long weather systems that a 4-hour battery cannot touch, provided the neighbour's weather differs.
...does what to the value of wind?	Raises it substantially, but through the *price* channel (Norwegian reservoirs absorbing Danish surpluses) rather than through weather diversification: DK1 and DK2 wind correlate at 0.85, so Denmark cannot diversify against itself.
...does what to system cost?	Lowers it by 237 M€ a year on the Skagerrak corridor alone, with concave returns. In Section 7 it turns out to be load-bearing: Cut off from trade, Denmark needs a much larger and dearer fleet to serve the same load.

To sum up: Trade adds a subscript, not a new theory. Zonal prices, rents as capacity shadow prices net of losses, and decreasing returns to relaxing a constraint are by now familiar from every previous section. What is new is the physics, one linear constraint per AC loop, and the politics, a distribution the objective function cannot see. What this section cannot answer is what the fleet behind the prices should be. That is where the note turns next.

7 Investment

Every model so far has taken the fleet as given and asked how to run it. The transition's real question runs the other way: What should be *built*? This section makes capacity a choice variable. Each technology's installed MW becomes part of the optimisation, charged to the objective at an annualised capital cost. The change to the model is small; the change to the economics is a whole layer: Prices must now cover not only fuel but capital, and the scarcity rents that Sections 2 to 6 treated as a distributional curiosity become the system's investment signal. It is also where the closing remark of Section 2 is finally answered: whether solar and wind are cost-effective once the whole system, and not only the marginal cost of running it, is priced.

We proceed in five steps. First the annuity arithmetic that puts capital and fuel on one clock, and the screening curve it produces. Second the model, and the result that ties this section to every previous one: long-run zero profit. Third the reference system, which we use to check that result in the numbers and to see how the peakers get paid. Fourth the sweep over the carbon price. Fifth, other worlds, including the one Section 2 called a cap.

> **Reading alongside this section.** This section is where the note's threads tie together, so the reading is mostly backward. Result 6 below is the formal version of the cost-recovery question Borenstein (2012) discusses informally, and Joskow (2011) is the standing warning against the levelised-cost reasoning that screening curves invite. Butters et al. (2025) show what happens when battery investment is determined in market equilibrium rather than by a planner, the closest published analogue to what this section computes. Dechezleprêtre et al. (2023) is worth rereading here for how far the real EU ETS is from the frictionless carbon price of this section. Appendix B carries the annuity arithmetic, and Appendix D lists what the model does not represent, which matters more in this section than in any other, because a capacity expansion answer is only as good as its constraint set.

7.1 Annualised capital and screening curves

A MW of capacity costing I euros with lifetime n is charged to the annual objective as $(a(r,n)+f)\,I$: the annuity of eq. (49) plus fixed operation and maintenance, derived in Appendix B. At the note's $r = 7\%$ the candidates' annual fixed costs run from 29 k€/MW (solar) through 45 (OCGT) and 100 (CCGT) and 94 (onshore wind) to 168 k€/MW (offshore wind). Investment costs are 2050 projections from an open technology-cost database, and the fuel prices are the forward ones for the same horizon rather than the present-day levels of Table 1; Appendix C says exactly which, and Section 9.2 asks what happens if the projections are wrong. Figure 7.1 plots each candidate's total annual cost per MW against how many hours a year it runs.

The classic reading of the figure is the following: OCGT's low intercept and steep slope win the peaking duties, CCGT the mid-range, and the renewables, flat lines since their fuel is free, undercut everything if one ignores that they cannot *choose* their hours. That proviso is exactly what the two-dimensional picture cannot hold, and why the section's actual tool is the LP: The full model prices each technology's hours at the hours it actually delivers. The markers make the proviso concrete:

i. Solar's line is the lowest on the chart at every duty, and solar lands at 962 full-load hours, because that is all the Danish sun offers. Offshore wind, several times dearer per MW-year, lands at 4,532 hours and is built anyway. Neither technology chose its position; the weather

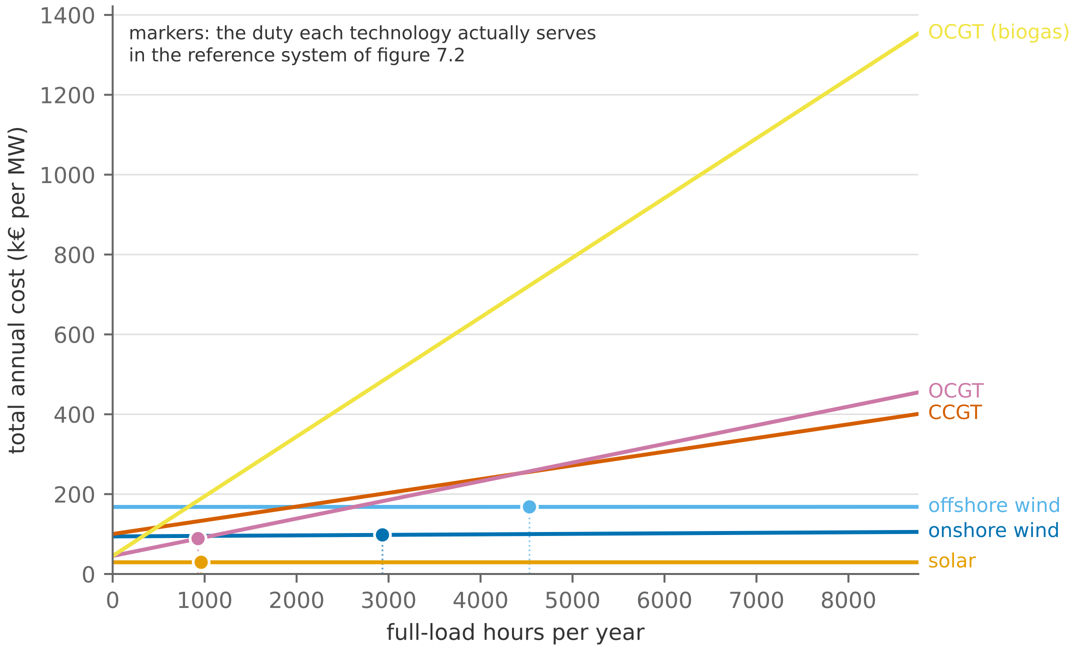

Figure 7.1: Screening curves: total annual cost per MW against running hours. Fixed cost is the intercept, marginal cost the slope. The markers show where each technology ends up in the reference system of Section 7.3.

did, and the screening chart has no axis on which to say so.

ii. The peaker runs 928 hours, a small fraction of the year. Nothing on the screening chart says a technology should be built to run that little; what puts it there is the value expression derived next, in which a handful of scarce hours can pay for a whole machine.

The screening curve tells you what a duty costs; only the LP tells you which duties exist, and who is left to serve them.

What makes a backup plant worth building. The algebra answers a natural question: Why does a system full of cheap renewables still want to build gas turbines? Differentiate this section's Lagrangian $\mathcal{L}^{\text{inv}}$ — the accumulated object of Section 6 plus the terms investment adds, written out at (39) below — with respect to a technology's capacity $\bar{q}_g$. Capacity enters the objective through its annualised fixed cost, and it enters the constraint set through the hourly capacity limits, whose multipliers we have been reading as scarcity rents since Section 2. So

$$\frac{\partial \mathcal{L}^{\text{inv}}}{\partial \bar{q}_g} = -\underbrace{\left(a(r, n_g) + f_g\right) I_g}_{\text{annual fixed cost}} + \sum_h \gamma_{g,h}\, \overline{\mu}_{g,h}, \qquad \overline{\mu}_{g,h} = \begin{cases} \lambda_h - c_g, & q_{g,h} = \gamma_{g,h}\bar{q}_g \\ 0, & \text{otherwise.} \end{cases} \tag{37}$$

This is an envelope argument: The dispatch variables are already at their optimum, so their derivatives drop out and only the direct effects survive (Appendix A.4). Now read eq. (37) for a peaking plant that runs, say, 100 hours a year, and notice how each term behaves:

- *Marginal cost barely matters.* The indicator inside $\overline{\mu}_{g,h}$ is zero in *almost every hour*, because the plant is almost never running. A technology's fuel cost is weighted by the number of hours it is used, and for backup capacity that number is tiny. Two peakers differing by 20 €/MWh in fuel cost differ by almost nothing in annual value.

- *Fixed cost matters enormously.* It enters with weight one, every year, regardless of utilisation. For a plant that runs 100 hours, the fixed cost is essentially the entire cost.
- *The value comes from a handful of extreme hours.* The revenue term is $\sum_h(\lambda_h - c_g)$ over the hours the plant is at capacity, precisely the scarcity hours, where Section 2.4 showed λ_h can reach a consumer's willingness to pay. A peaker is not selling energy; it is selling *availability*, and it is paid for it in a few hours a year at large margins.
- *Flexibility matters, in the same expression.* With ramping (Result 4), a plant that cannot get to full output quickly fails to be at capacity in exactly the hours when λ_h spikes, so the indicator switches off where the money is. Inflexibility does not show up as a cost; it shows up as forgone revenue.

So the ideal backup technology has low capital cost, high flexibility, and we frankly do not care what its fuel costs. That is an open-cycle gas turbine, the technology with the *worst* efficiency and the *highest* marginal cost on Table 1. It is also, for the same reason read in reverse, the argument against nuclear as a partner for variable renewables: High capital cost and low flexibility are precisely the wrong combination for a plant whose job is to fill residual gaps.

7.2 The greenfield problem

The instance takes the network of Section 6, wipes the two Danish zones clean, and lets the model rebuild them from candidates: solar, onshore wind, offshore wind, CCGT, OCGT, the same open-cycle machine fired with biomethane at 149 €/MWh marginal cost and zero fossil CO_2, nuclear at 2050 costs, and batteries at 2, 4 and 8 hours of duration (capacity and energy are priced separately, so duration is itself a choice, the lesson of Section 4 turned into an investment margin). Each candidate carries an annualised capital cost; solar and both winds are capped by buildable potential (the land constraint), at 16.0, 7.5 and 30.0 GW across the two Danish zones.

The neighbours are not wiped clean, but they are not frozen either. Each keeps the fleet it has and may add to it from the same candidate menu up to its published potential. The question is what *Denmark* should build, given a neighbourhood that is itself adapting to the horizon's demand. That last clause matters, because demand is not today's: Every zone carries a projected 2050 load, which for Denmark is the electrification the transition assumes and is also, candidly, what gives the question content. At today's load the interconnectors alone could serve Denmark, and the model, asked what to build, would answer "nothing". A 2050 load on a frozen 2024 fleet, on the other hand, would put the neighbours at the value of lost load for most of the year and every Danish export price with them.[15]

[15] The section's figures solve on 1095 chronological segments of the year rather than on every eighth hour, and on the zonal weather archive's 2024 rather than the network's own profiles; Appendices C.5 and C.6 say why and record the full-year check.

Carbon enters as a *price*: Every emitter in every zone pays τ euros per tonne, as in Section 2.5, and the section sweeps τ from zero to 300 €/t. The reference case charges 85 €/t, the level of recent EU ETS prices that Section 2 also used. The other instrument of Section 2, a budget in tonnes, appears in Section 7.5 as a scenario, and Section 8 makes it the main event.

Demand, finally, is no longer must-serve. Section 2.4 introduced the value of lost load as the honest way to close a planning problem; here it is implemented, as a small step demand curve. Every zone carries a last-resort block priced at 23.4 thousand €/MWh, a Danish survey-based VoLL averaged across sectors,[16] and two cheaper blocks: flexible industry and power-to-heat (5% of hourly load, willing to pay 100 €/MWh) and interruptible heavy industry (10%, 3,000 €/MWh). Formally each block is a "generator" at the load's node whose dispatch is *curtailed demand*, so the LP stays linear and the zonal price is still the market-clearing dual. Economically the objective is now surplus maximisation rather than cost minimisation, since serving every MWh is no longer an axiom but a choice the model can decline at the block's willingness to pay. In scarcity hours the price settles at the marginal block's valuation: the step demand curve of Section 2.4, now with more than one step. A finer staircase, a consumer index by sector with survey-based valuations per block, is the natural extension, and Brown, Neumann, et al. (2025) show in precisely this model class that even a little demand elasticity transforms the price distribution a renewables-heavy system generates.

Written out, the change from Section 6 is remarkably small: the capacities move from the data to the variables, and they acquire a price. The planner chooses generation $q_{g,h}$, flows $f_{\ell,h}$, storage operation, and capacities $\bar{q}_g$ to solve

$$
\begin{aligned}
\min \quad & \sum_h \sum_g (c_g + \tau e_g)\, q_{g,h} \;+\; \sum_g \big(a(r, n_g) + f_g\big)\, I_g\, \bar{q}_g && && \text{(38a)} \\
\text{s.t.} \quad & \sum_{g \in \mathcal{G}_z} q_{g,h} + \sum_\ell A_{z\ell} f_{\ell,h} \;=\; D_{z,h} && (\lambda_{z,h}) \quad \forall z, h && \\
& q_{g,h} - \gamma_{g,h}\, \bar{q}_g \;\leq\; 0 && (\overline{\mu}_{g,h}) \quad \forall g, h && \text{(38b)} \\
& 0 \leq \bar{q}_g \;\leq\; \bar{Q}_g && (\underline{\beta}_g, \overline{\beta}_g) \quad \forall g && \text{(38c)} \\
& \text{plus storage eqs. (19d) to (19g) and network (32c);} \quad q_{g,h} \geq 0.
\end{aligned}
$$

Two things are new. Equation (38c) is the *buildable potential*, there is only so much Danish land and seabed, and its multiplier $\overline{\beta}_g$ is a land rent, the value of one more permitted MW. The subtle one is eq. (38b). Compare it with the version in Section 3, $q_{g,h} \leq \gamma_{g,h}\bar{q}_g$. Algebraically nothing has changed. But $\bar{q}_g$ was a *parameter* there and is a *variable* here, which changes the constraint's character: What was a simple bound on one variable is now a genuine inequality linking two of them. That is why $\bar{q}_g$ acquires an optimality condition of its own, the one we wrote in eq. (37).

Duality: the investment condition. The Lagrangian is the one this note has been accumulat-

[16] Surveyed VoLL measures the cost of an *unannounced* outage; as a bid price for anticipated, voluntary curtailment it overstates willingness to pay, perhaps by nearly half. The level matters little here, since any four-digit backstop produces the same solutions, but the caveat matters when such numbers are used as market price caps.

ing since Section 2 — the network of Section 6, the storage terms of Section 4, the carbon charge of Section 2.5 — plus what investment adds,

$$\mathcal{L}^{\text{inv}} = \mathcal{L}^{\text{net}} + \big(\text{storage terms of } \mathcal{L}^{\text{sto}}\big) - \sum_h \sum_g \tau e_g \, q_{g,h}$$
$$- \sum_g \big(a(r,n_g) + f_g\big) I_g \, \bar{q}_g + \sum_g \underline{\beta}_g \, \bar{q}_g + \sum_g \overline{\beta}_g \big(\bar{Q}_g - \bar{q}_g\big). \tag{39}$$

The subtlety is not in the new terms but in an old one: The capacity terms $\sum_h \overline{\mu}_{g,h}(\gamma_{g,h}\bar{q}_g - q_{g,h})$ inherited from $\mathcal{L}^{\text{net}}$ now multiply a *variable* $\bar{q}_g$ where they multiplied a parameter. That is why $\bar{q}_g$ gets its own optimality condition,

$$-\,(a(r,n_g) + f_g)\, I_g + \sum_h \gamma_{g,h}\, \overline{\mu}_{g,h} + \underline{\beta}_g - \overline{\beta}_g \;=\; 0, \tag{40}$$

which is eq. (37) with the two bound multipliers added. We read it the usual way:

- Interior: built, below the cap. Both multipliers vanish and the rents equal the fixed cost.
- Lower bound: not built. $\underline{\beta}_g > 0$: The rents a MW would earn fall short of its fixed cost.
- Upper bound: at the potential cap. $\overline{\beta}_g > 0$ collects the excess of rents over fixed cost, the land rent.

Result 6 (Long-run zero profit)**.** *At the optimum of the capacity expansion problem, every technology that is built and not blocked by its potential cap earns exactly its annualised fixed cost in availability-weighted scarcity rents:* $\sum_h \gamma_{g,h}\, \overline{\mu}_{g,h} = (a(r,n_g) + f_g)\, I_g$ *for every* g *with* $0 < \bar{q}_g^* < \bar{Q}_g$*. A technology at its cap earns its fixed cost* plus *the land rent,* $\sum_h \gamma_{g,h}\, \overline{\mu}_{g,h} = (a(r,n_g) + f_g)\, I_g + \overline{\beta}_g$*; and no unbuilt technology could earn more than its fixed cost. Equivalently: In the price system the model generates, entry is exactly worthwhile for what is built and not worthwhile for what is not, and where entry is rationed by land, the excess return is capitalised into the site.*

This is the logic of Section 2.3 applied one level up. There, the *dispatch* condition priced energy; here, the *investment* condition prices capacity. Rents are no longer a transfer; they are the mechanism by which prices recover capital, the modern form of the peak-load pricing results of Boiteux (1960). The LP imposes the condition by optimality; a competitive market with free entry imposes it by arbitrage; and this equivalence is the welfare theorem extended to the capital stock. Every market design debate about whether "energy-only" prices can finance a capital-intensive system is a debate about whether the premises of Result 6, scarcity prices allowed to spike and investors able to collect them, hold in practice (Section 9 returns to this as *missing money*).

7.3 The reference system

At the reference price of 85 €/t the model builds 8.3 GW of solar, 7.5 GW of onshore wind (its entire land allowance), 6.6 GW of offshore wind, 0.8 GW of open-cycle gas turbines and 2.7 GW of batteries, every megawatt of them at the 8-hour duration. It builds no combined-cycle plant

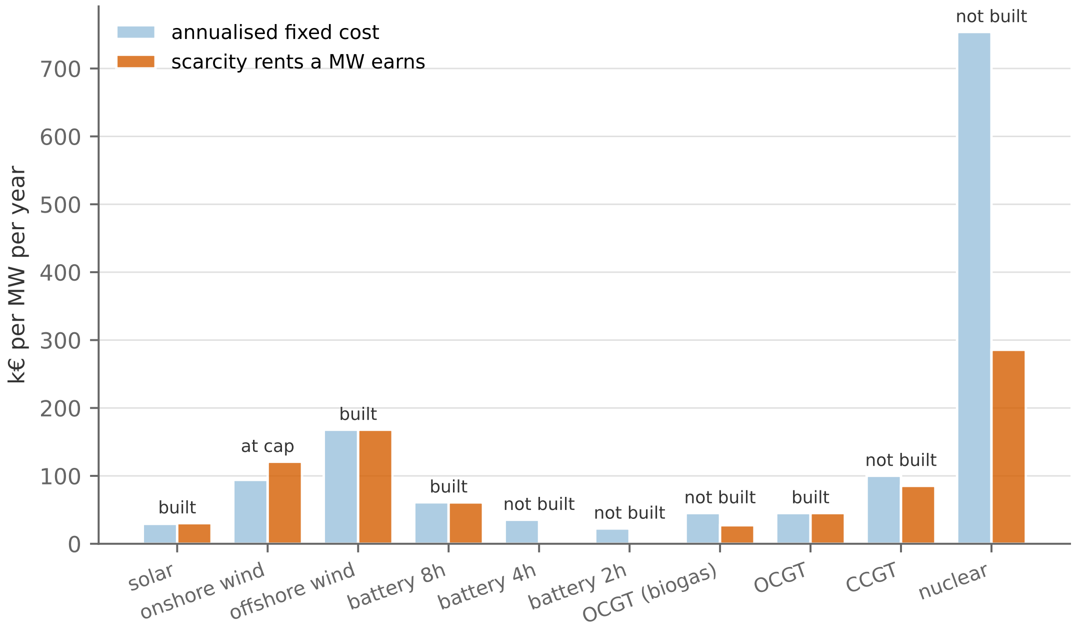

Figure 7.2: Long-run zero profit in the reference system: the annualised fixed cost of a MW of each candidate against the scarcity rents a MW earns in the solved prices.

and no nuclear. Denmark generates 60.8 TWh against a load of 48.6 TWh, so it is a net exporter, and its territorial emissions are 0.36 Mt. We check three things in this system before moving the carbon price.

Zero profit, in the numbers. Result 6 is a theorem about the optimum, and the solved case lets us verify it. Figure 7.2 shows, for each candidate, what one MW would earn in scarcity rents in the reference system's prices, $\sum_h \gamma_{g,h} \max(\lambda_h - c_g - \tau e_g, 0)$, against what it costs to hold for a year.

The figure shows the following:

i. Offshore wind, the gas peaker and the 8-hour battery are built below their caps and earn exactly their fixed cost (100%, 100% and 100% of it).

ii. Onshore wind is at its land cap and earns 128% of its fixed cost. The excess, 27 k€ per MW and year, is the land rent $\overline{\beta}_g$: what a permit for one more MW of onshore wind would be worth.

iii. Nothing that is not built could pay for itself. CCGT would earn 85% of its fixed cost, the biomethane peaker 60%, nuclear 38%, and the 2- and 4-hour batteries nothing at all, since the price spreads that would pay for them are already arbitraged away by the longer ones.

Built and uncapped means rents equal to cost, to the euro.

Levelised cost against capture price. Section 3 warned that a technology's levelised cost says little about its value, and defined the capture price and value factor to say the rest. Table 5 evaluates both at the optimum for every technology the model built.

Table 5: Levelised cost and capture price in the reference system, per built technology. The LCOE excludes the carbon price; for the peaker the carbon cost of 39 €/MWh has to be added to compare with its capture price.

Technology	Capacity GW	Full-load hours h/yr	LCOE €/MWh	Capture price €/MWh	Value factor	
solar	8.3	962	30.5	31.5	0.59	interior
onshore wind	7.5	2,934	33.4	42.5	0.79	at its cap
offshore wind	6.6	4,532	37.1	37.1	0.69	interior
OCGT	0.8	928	95.5	134.6	2.51	interior

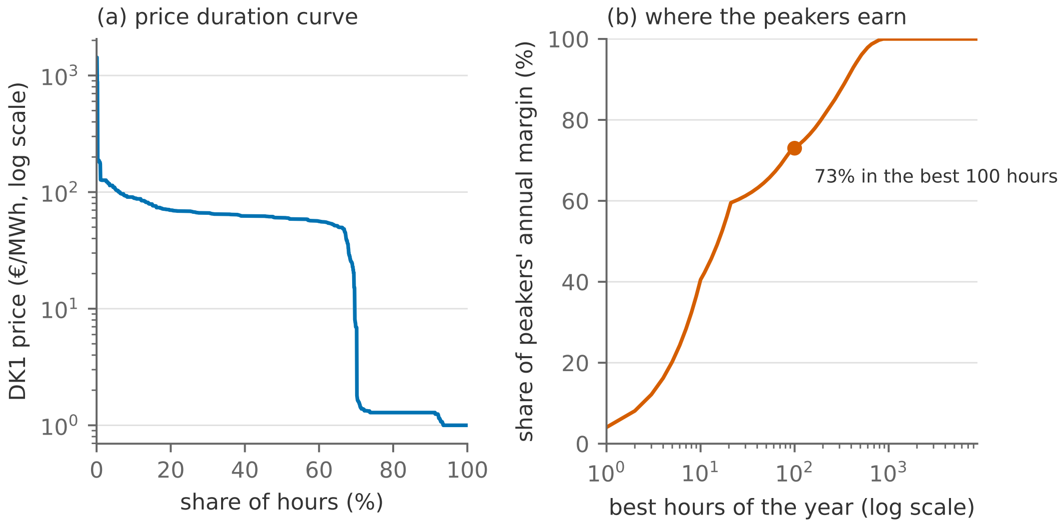

Figure 7.3: How the peakers get paid in the reference system: (a) the DK1 price duration curve, on a log axis; (b) the cumulative share of the peaking fleet's annual operating margin earned in its best hours.

The table shows that (i) for the technologies built below their caps, LCOE and capture price coincide, which is Result 6 restated per MWh: A technology is built exactly until the average price it captures equals its levelised cost; (ii) for onshore wind, at its cap, the capture price exceeds the LCOE by the land rent per MWh; (iii) the value factors are well below one for all three variable technologies (solar 0.59, onshore 0.79, offshore 0.69), so each is built up to the point where its *own* capture price, not the base price of 53.7 €/MWh, covers its cost; and (iv) the peaker has a value factor of 2.5, because it sells only into the expensive hours. Read the first and third points together: Solar's LCOE of 30.5 €/MWh is the lowest on the table, and the model still stops building it at 8.3 GW, because the price solar sells at is 31.5 €/MWh and falling with every MW added. That is the cannibalisation of Section 3 in its investment form, and it is why comparing technologies by LCOE alone, as Joskow (2011) warned, gets the answer wrong.

How the peakers get paid. Section 2 promised to show how capacity that exists to be available rather than to generate earns its keep. Figure 7.3 shows the DK1 price duration curve of the reference system, and the cumulative share of the peaking fleet's annual operating margin earned in its best hours.

The figure shows that (i) the price exceeds the flexible block's willingness to pay of 100 €/MWh

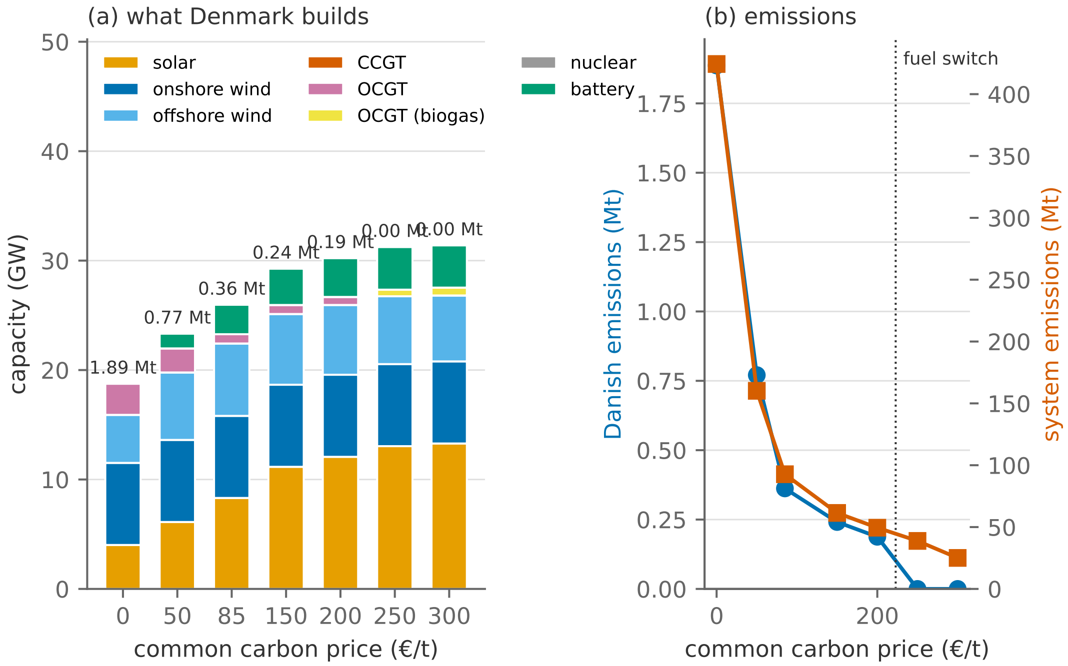

Figure 7.4: The common carbon price swept from zero to 300 €/t: (a) the optimal Danish capacity mix, with Danish territorial emissions above each bar; (b) Danish and twelve-zone emissions against the price, with the fuel-switching price of the peakers marked.

in 553 hours of the year, and peaks at 1,431 €/MWh, so this system never reaches the industrial block, let alone the value of lost load; (ii) the peakers earn 73% of their annual margin in their best 100 hours and 96% in their best 500. This is eq. (37) in the data: An energy-only market finances its backup capacity through a small number of expensive hours, and it must, because that is the arithmetic of paying for a machine that runs 928 hours a year. Section 9 returns to what happens when those hours are not allowed to occur.

7.4 As the carbon price rises

Figure 7.4 sweeps the common carbon price from zero to 300 €/t and shows what Denmark builds and what everyone emits.

The figure shows the following:

i. Even without a carbon price the Danish system is mostly renewable: 4.0 GW of solar, onshore wind at its 7.5 GW land limit, 4.4 GW of offshore, and 2.8 GW of gas turbines that exist for firmness rather than bulk energy. At 2050 capital costs, carbon policy is not what makes renewables enter; it is what pushes out the remaining fossil *energy*.

ii. Danish emissions fall from 1.89 Mt at zero to 0.36 Mt at the reference price and to zero by 300 €/t. Panel (b) is the marginal abatement cost curve of Section 2, read from the tax side: Each price buys the abatement that costs less than it.

iii. The peakers do not disappear as the price rises; they *switch fuel*. The fossil turbine is still built at 200 €/t and gone at 250 €/t, where the biomethane machine takes its place. The hand calculation of Section 2 says exactly where the switch should sit: The two are the same machine, so the abatement cost is the fuel-price premium divided by the gas turbine's emission rate, $(149 - 47)/0.46 = 223$ €/t, and the sweep brackets it.

iv. The rest of the fleet responds smoothly. Solar grows from 4.0 to 13.3 GW and batteries from 0.0 to 3.9 GW, every megawatt of them at 8 hours; offshore wind settles near 6.6 GW early and stays there. No 2- or 4-hour battery and no nuclear is built at any price.

v. The twelve-zone emissions fall from 424 to 25 Mt over the same sweep, most of it German coal and lignite giving way to German solar. Denmark's own tonnes are a rounding error in that total, which is worth remembering when a Danish target is discussed as if it were a climate policy.

Result 7 (The fuel-switching plateau)**.** *A candidate that abates at a constant cost per tonne puts a horizontal segment into the marginal abatement curve at that cost. Under a carbon price, emissions drop by the width of the segment as the price crosses it: Here the peakers' energy switches from gas to biomethane at* $\tau = (c_{clean} - c_{dirty})/e_{dirty}$*. Under a budget, the budget's shadow price sits on the segment for as long as the budget can be met by switching, and leaves it once the switchable energy is exhausted. It is a* plateau*, not a ceiling: Its width is the quantity of energy that can be switched, and a budget that demands more abatement than that moves the price to the next margin.*

The tempting misreading is "the backstop caps the carbon price". The sentence is nearly true and very tidy, and it is wrong because of the word *caps*: A backstop bounds the price only over the range in which it is still available, and a backstop with a finite resource behind it (a limited peaking fleet here, a limited biomethane supply in Section 8) is a plateau of finite width. The misreading has the shape of many modelling errors in this area: a correct piece of arithmetic about a *margin*, promoted to a claim about the *system*, with the quantity behind the margin never checked. When you meet a model whose carbon price is suspiciously stable, the question to ask is not what sets it but how much of that thing there is.

Figure 7.5 decomposes the reference system's 2.4 bn€ annual bill by technology, into annuitised capital and operating cost.

A decarbonised system is a *capital-intensive* system, paying for machines rather than molecules. That shift has consequences beyond the model. It raises the stakes of the discount rate (see the scenarios below), moves the political economy from fuel prices to financing costs, and makes the sector's costs fixed and its marginal costs near zero, which is precisely the world in which the missing-money questions of Section 9 bite hardest.

7.5 Other worlds

Figure 7.6 re-asks the reference question under different assumptions: Denmark cut off from its neighbours, a doubled gas price, capital at 2% instead of 7%, and finally the instrument Section 2

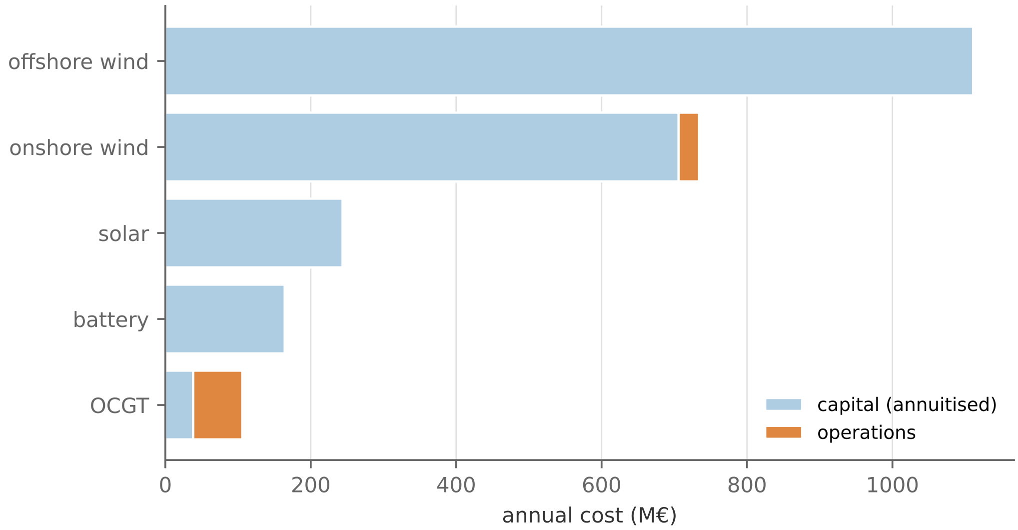

Figure 7.5: The annual bill of the reference system: annuitised capital and operating cost by technology.

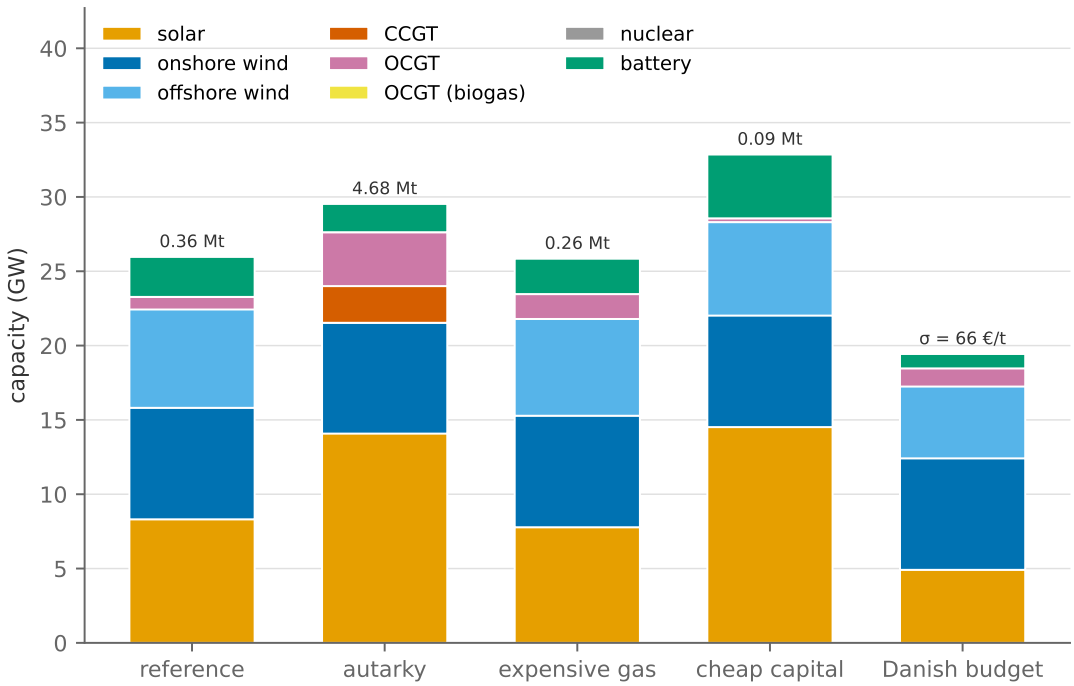

Figure 7.6: The reference question under different assumptions: autarky (no interconnectors), doubled gas price, capital at 2% instead of 7%, all at the reference carbon price; and a Danish CO_2 budget at −90% of Denmark's unpriced emissions, with no carbon price on anyone. Danish emissions above each bar, the budget's shadow price above the last.

called a cap, a budget on Danish territorial emissions with no carbon price anywhere.

Autarky. Denmark cut off prices security of supply. The figure shows a system that has to serve the same load alone, and it answers the question "what did the interconnectors do?" by rebuilding what they replaced. Offshore wind disappears entirely (there is nobody to sell it to), solar goes to 14.1 GW, and in place of imports the system builds 2.5 GW of combined-cycle plant and 3.6 GW of peakers, all fossil: At 85 €/t the fuel switch is far away, so the dark calm week is covered by gas. Danish emissions rise to 4.68 Mt, fifteen times the reference. Net of trade, autarky costs Denmark 2.55 bn€ a year against 2.35 with trade, 9% more to serve the same load. What trade buys is a cleaner system at a lower bill, and a smaller one: The reference fleet is larger in offshore wind and smaller in everything firm. One measurement lesson comes with it. The gross Danish bill includes the capital of an offshore fleet that is partly built to export, so comparing gross bills would compare an exporter's cost base with a self-supplier's. The comparison has to be made net of trade revenue, and a table answers the question its denominator asks.

Expensive gas separates the peakers' two roles. With gas at twice the price the turbines run less (Danish emissions fall from 0.36 to 0.26 Mt), but the peaking capacity *rises*, from 0.8 to 1.7 GW: The neighbours' gas is dearer too, so the hours in which Denmark would import from them are dearer, and holding one's own turbine against them pays. Capacity is bought for availability, and availability is worth more when everyone's energy is dear.

Cheap capital tilts the mix toward the capital-heavy technologies, the annuity arithmetic of Appendix B made flesh: At 2% the same offshore turbine costs a third less per year, and the optimal system buys more solar (14.5 GW against 8.3) and more batteries (4.3 GW against 2.7), for a bill of 1.6 bn€ against 2.4.

A Danish budget. The last scenario replaces the common price with the instrument of Section 2.6: a budget on Danish territorial emissions at −90% of what Denmark emits with no carbon price anywhere, and nothing on the neighbours. Its shadow price comes out at $\sigma = 66$ €/t, a fraction of the fuel-switching price, and the mix barely moves from the unpriced one. The reason is the boundary. The budget covers *Danish generation only*, while imports cross the border carbon-free as far as the accounting is concerned, the standard territorial convention. So the cheapest way to meet the budget is not to switch fuel but to generate less and import more, and with the neighbours free to build unpriced gas, that is what the model does: Danish territorial emissions fall by 1.70 Mt, but emissions summed over all twelve zones fall by only 1.46 Mt. 14% of the Danish cut is **carbon leakage**, tonnes that moved rather than disappeared. This is why the section swept a price rather than a budget: A territorial budget in an open economy is a constraint on accounting, not on emissions, and its shadow price says what it costs to move tonnes across a border, not what it costs to abate them. Section 8 puts the budget over the whole neighbourhood, where there is nowhere to leak to.

Which way does it bias the answer? *Perfect foresight over the investment horizon.*

The model chooses capacity knowing every hour of weather and every price it will face. This **understates** the cost of the transition, because a real investor hedges against a future they cannot see and therefore builds a portfolio more diversified, and more expensive, than the optimum for any single realised future. And it **overstates** how confidently the mix can be specified: With the future known, the LP picks a corner (Appendix A.1), so it never builds a diversified portfolio for diversification's own sake, even though diversification is exactly what uncertainty rewards. Both effects are measured in Section 9.1.

From duals to markets.

Constraint	Dual	What it measures	Market counterpart
zonal clearing	$\lambda_{z,h}$	marginal cost incl. scarcity	energy-only market prices
investment optimality (40)	—	rents = annualised fixed cost	free entry; cost recovery without subsidy
buildable potential (38c)	$\overline{\beta}_g$	land rent: value of one more permitted MW	the price of a site permit or seabed lease
a Danish CO_2 budget (scenario)	σ	the cost of moving a tonne across the border	a national target's implicit price, leakage included

To sum up: Making capacity a variable adds one optimality condition per technology, and that condition is the long-run zero-profit result that ties the rents of every previous section to cost recovery. The reference system verifies it to the euro. A common carbon price then moves the mix smoothly and the peakers' fuel in one step, at the price the hand calculation predicts; a territorial budget mostly moves tonnes across the border. All of it was computed for one year of weather and one set of cost projections. Section 8 widens the boundary and turns the price into a budget; Section 9 then asks how much of this section's precision survives the weather.

8 Expansion at European scale

Section 7 fixed a carbon *price*, charged it on every zone, and read off what Denmark builds and how much everyone emits. This section makes the move that Section 2 made from the tax to the cap: It fixes the *quantity*, one CO_2 budget for all twelve zones, and reads off the price. At the same time it widens the question from what Denmark should build to what the whole neighbourhood should build. Every zone may add to the fleet it has, the interconnectors between the zones become a decision rather than data, vehicles may charge when the system wants them to, and a hydrogen sector arrives. It is built as a *ladder* of three models, each the previous one plus one ingredient, so that what each ingredient does is visible rather than asserted.

> **Reading alongside this section.** This section tests claims from two papers. Victoria et al. (2019) is the canonical statement of the division of labour between storage technologies in a decarbonising European system, batteries for the daily cycle and hydrogen for the multi-day one, and its central claim, that storage only starts to matter below about a fifth of 1990 emissions, is tested here rather than cited. Neumann et al. (2023) is the hydrogen half of the same research programme and the source of the extension this section most obviously points at. Both are built with the same tool, on

> the same data, at a scale about fifteen times this one; the point of reading them beside this section is to see what changes with scope and what does not.

8.1 The objectives

Four questions, in the order the ladder answers them:

i. *What does a common budget do that a common price did not?* Section 7 showed that a territorial budget in an open economy is met by importing: Its shadow price carries little information. A budget over all twelve zones has nowhere to leak to, and its dual is the carbon price the whole system would need.

ii. *Can flexible demand stand in for storage?* Victoria et al. (2019) argue that electric vehicles, charged when the system wants them to, can largely displace stationary batteries. Rung (ii) carves a shiftable charging block out of the load and measures it.

iii. *Does the system electrolyse its hydrogen, and having built electrolysers, does it store hydrogen and burn it back?* That is the division of labour Victoria et al. (2019) describe, and rung (iii) is where this note can test it, because it is the first model here with a hydrogen sector at all.

iv. *Which of the lessons of Section 7 survive?* This is the one that matters for how you read the rest of the note.

8.2 The ladder

Table 6: The ladder: what each rung adds and what it tests.

Rung	What it adds	What it tests
(i) basic	All 12 zones extendable beside their existing fleets, one system-wide CO_2 budget, buyable interconnectors	Where the carbon price settles when the budget is continental
(ii) + flexible charging	The vehicle charging block's *timing* becomes a choice	Whether flexible demand displaces stationary storage, and at what price
(iii) + hydrogen	An industrial hydrogen demand and every route to meeting it: reforming, imports, electrolysis, storage, and a turbine to burn it back	Whether hydrogen is made electrolytically, and whether it is used as electricity storage

Three modelling decisions in the ladder could have gone the other way.

It is brownfield. Every existing plant is kept and none may be retired. The capital is sunk, so a plant may simply not run; wiping twelve zones clean, as Section 7 does for Denmark, would ask the model to re-invent Norwegian hydro from scratch. This is the honest setting for a question about the next twenty years, and it is also why this section cannot be read as a greenfield answer: Some of what the system does is inherited.

Nobody is special. The electrification, the flexible demand blocks and the value of lost load apply to all 12 zones, as they did in Section 7. In a European model, leaving eleven zones inflexible while Denmark is flexible would produce a Danish result and disguise it as a European one.

Rungs (i) and (ii) share a budget; rung (iii) does not. Rungs (i) and (ii) are the same system, differing only in whether charging may move, so one budget in tonnes covers both and their carbon prices compare directly.[17] Rung (iii) carries an industrial hydrogen sector the others do not, and one absolute target across all three would ask it for a far deeper cut than its neighbours. So it runs on its own baseline, 514 Mt against 424. Its carbon price therefore compares along its own path and not with the other two rungs, and this section never puts the two side by side without saying so.

The model. Almost nothing in eq. (38) changes. The capacity variables and their potentials now run over every candidate in every zone, the budget is summed over all zones, and the corridor ratings join the capacity variables:

$$\begin{aligned}
&0 \le \bar{q}_g \;\le\; \bar{Q}_g && (\underline{\beta}_g, \overline{\beta}_g) && \forall g \text{ in every zone} \\
&-\left(\bar{f}_\ell^0 + \bar{f}_\ell\right) \;\le\; f_{\ell,h} \;\le\; \bar{f}_\ell^0 + \bar{f}_\ell && (\underline{\omega}_{\ell,h}, \overline{\omega}_{\ell,h}) && \forall \ell, h && \text{(41a)} \\
&0 \le \bar{f}_\ell \;\le\; \bar{F}_\ell && (\underline{\phi}_\ell, \overline{\phi}_\ell) && \forall \ell && \text{(41b)} \\
&\sum_h \sum_z \sum_{g \in \mathcal{G}_z} e_g\, q_{g,h} \;\le\; E && (\sigma) && && \text{(41c)}
\end{aligned}$$

where $\bar{f}_\ell^0$ is the existing rating of corridor ℓ, $\bar{f}_\ell$ the capacity bought, priced in the objective at its own annualised cost $(a(r, n_\ell) + f_\ell) I_\ell \bar{f}_\ell$, and $\bar{F}_\ell$ the candidate ceiling. The common carbon price of Section 7 is gone; the budget (41c) takes its place, and its dual σ is the endogenous carbon price of Section 2.6, now at continental scale. Rung (iii) adds a hydrogen market per zone, with the electrolyser, the reformer, the store, the turbine and the import corridors as conversions into and out of it.

Duality: the value of a wire. The Lagrangian changes as little as the model does. Section 7's carbon charge is *removed* and the budget put in its place — the one section of the note where a term leaves rather than arrives — and the corridor variables join those of the generators:

$$\begin{aligned}
\mathcal{L}^{\text{exp}} = \mathcal{L}^{\text{inv}} &+ \sum_h \sum_g \tau e_g\, q_{g,h} + \sigma\Big(E - \sum_h \sum_z \sum_{g \in \mathcal{G}_z} e_g\, q_{g,h}\Big) \\
&- \sum_\ell (a(r, n_\ell) + f_\ell) I_\ell\, \bar{f}_\ell + \sum_\ell \underline{\phi}_\ell\, \bar{f}_\ell + \sum_\ell \overline{\phi}_\ell (\bar{F}_\ell - \bar{f}_\ell), \qquad (42)
\end{aligned}$$

where the second term cancels the tax out of $\mathcal{L}^{\text{inv}}$, and the line capacity terms $\overline{\omega}_{\ell,h}, \underline{\omega}_{\ell,h}$ inherited from $\mathcal{L}^{\text{net}}$ now multiply the variable rating $\bar{f}_\ell^0 + \bar{f}_\ell$ — the same parameter-turned-variable move

[17] Giving each its own relative target would make flexibility look harmful: The flexible system's unconstrained emissions are lower (399 Mt against 424), so the same *fraction* would be a tighter target in tonnes, and its carbon price would come out higher for a purely mechanical reason.

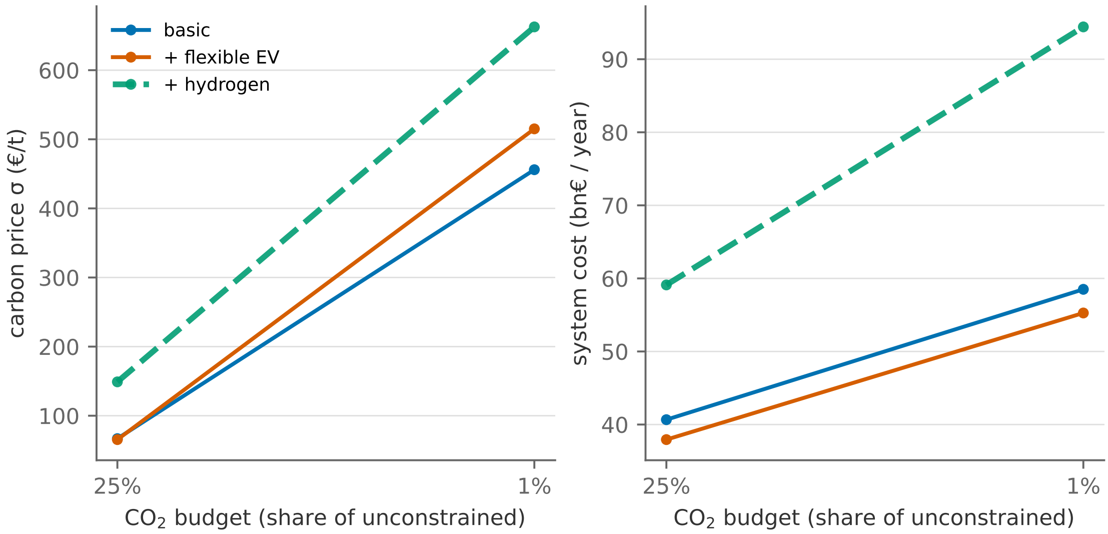

Figure 8.1: The carbon price (a) and the annual system cost (b) along each rung, as the system-wide CO_2 budget tightens. The hydrogen rung is dashed because it is a different system on its own baseline.

that Section 7 made for $\bar{q}_g$, applied to a wire.

The new optimality condition is therefore the one for $\bar{f}_\ell$. The corridor's rating enters the objective through its annualised cost and both sides of eq. (41a), whose multipliers Section 6 read as the hourly price spread. So

$$-\left(a(r, n_\ell) + f_\ell\right) I_\ell + \sum_h \left(\overline{\omega}_{\ell,h} + \underline{\omega}_{\ell,h}\right) + \underline{\phi}_\ell - \overline{\phi}_\ell \;=\; 0, \tag{43}$$

which is eq. (40) for a wire: Built corridors earn exactly their annualised cost in congestion rent per MW, corridors that are not built would earn less, and a corridor at its candidate ceiling earns more, with $\overline{\phi}_\ell$ collecting the excess. Section 6 asked what a scarce wire earns; here the model may respond to that rent by building the wire, and Result 6 applies to a corridor exactly as it applies to a turbine.

8.3 The carbon price of a continental budget

Figure 8.1 shows the carbon price and the annual system cost along each rung as the budget tightens.

The figure shows the following:

i. Unconstrained, the twelve zones emit 424 Mt. Cut that by 75% and the shadow price of the budget is 67 €/t. Push to −99% and it is 456 €/t.

ii. The cheap end is cheap because Europe's cheapest abatement at this depth is displacing German lignite and coal with German solar, which is far cheaper per tonne than anything on the Danish menu. Result 7 holds here as it did there: σ sits on the cheapest margin that has not run out, and at −75% that margin is coal.

iii. The dear end is dear because the last tonnes sit where no cheap margin reaches them. At -99% the price is well past the fuel-switching cost of 223 €/t, the peakers have switched to biomethane (47.0 GW of it), and the resource ceiling on biomethane is still slack (its shadow price is 0 €/MWh), so what sets σ is the storage and wire needed to displace the remaining gas energy.

Now hold this beside Section 7. There the same twelve zones faced a common carbon *price*; at the reference price of 85 €/t the system emitted 93 Mt. Result 2 says that a budget set at the emissions a price produced returns that price, and the reader can check the sense of it on Figure 8.1: A budget of 93 Mt sits between the unconstrained point and the -75% point, where the ladder's price is below 67 €/t. The two sections are not two runs of one model (this one is brownfield everywhere, buys wires and uses published potentials for Denmark), so the correspondence is one of magnitude, not to the euro. Section 7 told the system a price and read off tonnes; this section tells it tonnes and reads off a price, and the two agree on which is cheap and which is dear.

The contrast with the *territorial* budget of Section 7.5 is sharper. There a Danish budget at -90% came out at $\sigma = 66$ €/t, with 14% of the Danish cut reappearing abroad, because imports count as carbon-free and the neighbours could build to supply them. Here there is no border to push the tonnes across.

Result 8 (Scope changes the answer, not just its precision)**.** *A carbon price is a property of a constraint set, not of an energy system. A territorial budget in an open economy constrains accounting rather than emissions: Its dual prices the cheapest way to move tonnes across the boundary, not the cheapest way to abate them. A budget over the whole trading area removes the accounting escape and prices the marginal tonne, at a level set by whichever abatement margin the budget reaches last. Widening the boundary lengthens the menu of margins, which lowers the price at a loose target, and removes the escape routes, which raises it at a tight one.*

8.4 What each ingredient builds

Figure 8.2 shows what each rung builds at the reference budget, in two panels.

The figure shows the following:

i. Panel (a) is the scale: hundreds of GW of solar in every rung. The *unconstrained* system already builds 203 GW of solar, so carbon policy here is not what makes renewables enter; it is what pushes out the remaining fossil *energy*, as in Section 7.

ii. Panel (b) shows the differences between the rungs. Rung (ii) changes nothing about the fleet except when vehicles charge, and the battery build falls from 63.9 to 29.5 GW at -75% (54% of it gone) and from 102.6 to 69.7 GW at the reference, a further 32%. The bill falls with it, by 7% at -75% (40.7 to 37.9 bn€).

iii. The carbon price hardly moves: 67 against 65 €/t at -75%, 456 against 515 at the reference.

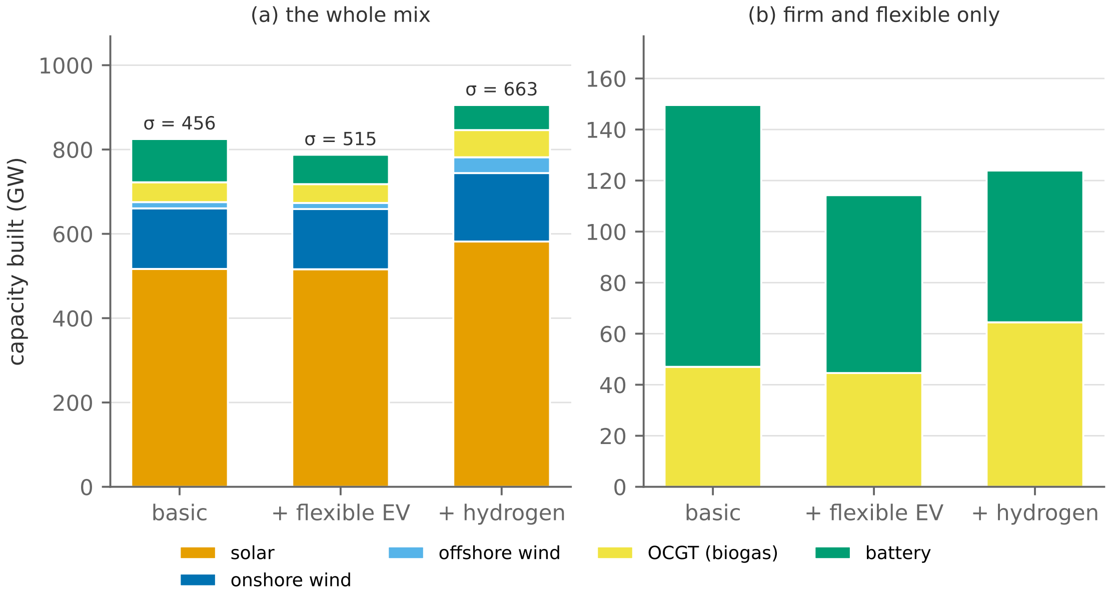

Figure 8.2: What each rung builds at the reference budget (−99%), with each rung's own carbon price above its bar. Panel (b) drops the variable renewables.

The second point is Victoria et al. (2019)'s claim confirmed at our scale rather than cited, and the mechanism is the one Section 4 made obvious: A battery and a shiftable load are the same product. Both move energy from an hour of surplus to an hour of scarcity; one of them has to be paid for, and the other is a fleet of vehicles that had to be plugged in anyway. The third point, as a rule: Flexibility changes *how much capital* the system needs to exploit its cheapest abatement margin; it does not change which margin that is. When a policy intervention moves quantities and leaves σ alone, it has made the target cheaper to hit without making it easier to hit.

Figure 8.3 shows where the reference system builds.

The figure shows that 77% of everything built goes into Germany: 458 GW of solar and 118 GW of onshore wind. This is not an artefact: Germany is most of the demand, most of the land and most of the coal in this footprint, so most of the transition happens there. Norway builds almost nothing, because it already has the flexibility everyone else is trying to buy.

Transmission needs a caveat. Offered 30 GW of candidate interconnection, the basic rung buys 19.9 GW at the reference budget and far less when the budget is loose, so the wires are bought *because* of the carbon target, as the cheapest way to let German solar serve a Danish evening. But the existing grid this expansion sits on is the network of Section 6, at its 2024 ratings, carrying a 2050 demand. A model given a modern load and an old grid and then asked where it is congested is measuring its own assumption, which is exactly why the corridors are a decision here rather than data.

8.5 Hydrogen: what it is for, and what it is not for

Section 7 had no hydrogen at all. Rung (iii) adds the whole sector: an industrial hydrogen demand of 429 TWh across the four countries, and four ways to meet it. Reform it from natural gas, import

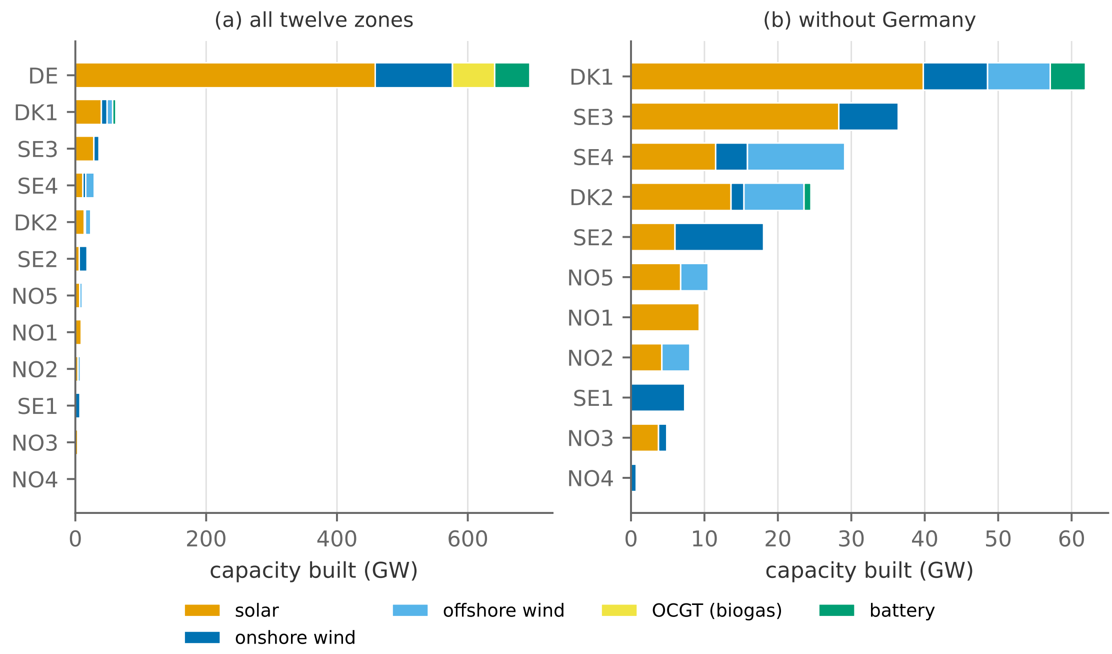

Figure 8.3: Where the reference system chooses to build. Panel (b) drops Germany so the rest can be read.

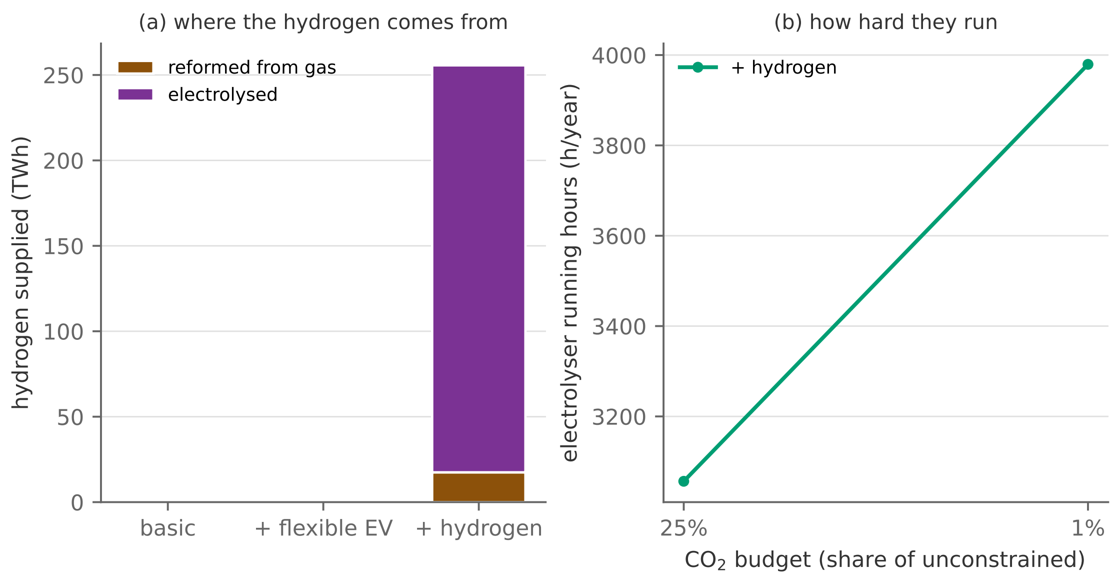

Figure 8.4: (a) where the hydrogen comes from at the reference budget. (b) the electrolysers' running hours as the budget tightens.

it, electrolyse it, or take it out of a store filled earlier.[18] The store may also feed a turbine that burns hydrogen back to electricity, which is the configuration Victoria et al. (2019) find doing the long-duration work in a decarbonised Europe. Figure 8.4 shows where the hydrogen comes from at the reference budget, and how many hours a year the electrolysers run as the budget tightens.

The result has two halves:

[18]Import corridors exist only into Germany, and as a step supply curve rather than an unlimited offer. That matters more than it sounds: An unlimited offer at a flat price is a backstop, and by Result 7 a backstop with no quantity limit behind it would take the entire market and set the price.

i. *Hydrogen as an industrial fuel: yes, decisively.* Unconstrained, all of it is reformed from natural gas (47.9 GW of reformers and no electrolysis whatever). Tighten the budget and the electrolytic route takes over: 0.1 GW of electrolysers at −75% and 85.6 GW at the reference, by which point 93% of the hydrogen this system makes for itself is electrolysed and the reformer fleet has fallen to 7.6 GW.[19] Hydrogen storage follows: 19.31 TWh of it, almost all in salt caverns, in the zones that have salt to store it in (DE, DK1, DK2, SE1, SE2, SE3, SE4).[20]

ii. *Hydrogen as electricity storage: no, at every budget.* The turbine is built at 0.0 GW in every solve on the ladder, and 0.0 TWh of hydrogen is burnt back to electricity. The store that fills up is buffering an electrolyser against a *flat industrial demand*, not holding energy for the dark week.

Panel (b) says why the two halves differ. An electrolyser is capital: Its cost per MWh of hydrogen is its annuity divided by how many hours it runs. With an industrial customer it reaches 3,056 hours a year at −75% and 3,979 at the reference, and at that utilisation the hydrogen is cheap enough to beat reforming plus the carbon price. Now imagine the same electrolyser with *only* the turbine to sell to. The turbine's whole economic function is to cover the deep backup hours, of which a system like this has a few hundred; an electrolyser feeding nothing else could therefore run at most a couple of thousand hours, and its capital per MWh delivered would be several times what the alternatives cost. The decisive term is the electrolyser's utilisation, not the store, the turbine or the carbon price.

Result 9 (Round-trip storage cannot buy its own utilisation)**.** *A conversion chain whose only customer is a peaking plant inherits that plant's running hours. Since the capital cost per unit delivered is inversely proportional to utilisation, such a chain is expensive precisely because it is only needed rarely, and no carbon price fixes this, because the carbon price raises the value of the output and the cost of the input together. What makes the chain viable is a* second *customer that runs continuously, which pays for the plant and leaves the peaking duty as an opportunistic use of capital already bought.*

The claim this section can make is therefore narrow, as Table 7 sets out.

The difference from Victoria et al. (2019) and Neumann et al. (2023) is exactly this. Their electrolysers serve industry, transport and synthetic fuels across a whole continent, so they run thousands of hours and the turbine is an opportunistic second use of a plant paid for elsewhere. Ours has an industrial customer too, which is why our electrolysers get built, but the second use never arrives, because at this footprint the deep backup duty is small and there is a cheaper way to cover it.

[19] The remaining 173 TWh is imported, at the corridors' full capacity, so imports are rationed by geography, not chosen on price.

[20] The store is the number in this note most exposed to how the year is aggregated, and it has been checked at full resolution: At all 8760 hours it is 18.93 TWh against 19.31 on the 1095 segments the figures use, the carbon price 663 against 663 €/t, and the turbine 0.0 GW in both. Sampling every eighth hour instead gives 33.72 TWh, 1.8 times the full-year store, because eight-hour steps hide the intra-day flexibility that makes most of it unnecessary (Appendix C.6).

Table 7: What the model shows, and what it does not.

What the model shows	What it does *not* show
Hydrogen as *electricity storage* (electrolyse, store, burn back) does not pay in this system at any budget	That hydrogen does not pay. It plainly does: The same model builds 85.6 GW of electrolysers for industry
The binding term is the electrolyser's utilisation, not the store or the turbine	That the store and turbine are cheap. They are not; they are simply not what decides

Which way does it bias the answer? *No hydrogen network, a small footprint, and a biogenic backstop.*

All three make hydrogen look worse here than it is, and all three point the same way.

- *Hydrogen cannot move between zones.* Each zone's hydrogen sector is self-contained, so Norway, with cheap power and no domestic hydrogen demand, gets no hydrogen sector at all and cannot export. Neumann et al. (2023) find a European hydrogen network worth up to 26 bn€ a year; its absence **understates** the electrolytic route.
- *The footprint is too small for seasonal storage to shine.* Four countries with a large hydro reservoir in the middle can ride through a calm week on trade and water; a whole continent has correlated calm over a larger area and a thinner buffer. The model **understates** the value of long-duration storage.
- *A cheap biogenic backstop is on the menu.* The biomethane peaker covers the deep backup hours instead of hydrogen, and its resource ceiling never binds. The model **understates** hydrogen storage relative to a system without biomass, which is the mirror image of what Victoria et al. (2019) note about their own work.

So the honest summary is not that hydrogen storage loses, but that it loses *to whatever firm low-carbon option is on the menu at this scale.* Testing that claim needs a bigger footprint, not a tighter budget.

From duals to markets.

Constraint	Dual	What it measures	Market counterpart
system-wide CO_2 budget (41c)	σ	marginal abatement cost across twelve zones	a multi-country cap's allowance price
corridor candidate ceiling (41b)	$\overline{\phi}_\ell$	value of one more MW of *permitted* transmission	the option value of a corridor a TSO is not allowed to build
corridor investment optimality (43)	—	congestion rent = annualised cost of wire	a merchant cable's break-even; regulated cost recovery
hydrogen clearing, per zone	$\lambda^{H_2}_{z,h}$	marginal cost of a MWh of hydrogen	a hydrogen contract price, absent a hydrogen market
biomethane resource ceiling	shadow price	value of one more MWh of biogenic fuel	the premium on a scarce fuel; zero here, since the ceiling is slack

8.6 What survives from Section 7

Survives. Result 6, every built candidate earning exactly its annualised fixed cost in rents, holds at twelve zones as it did at two, because it is a property of the optimisation and not of the instance. Result 7 survives too, and does more work here than it did there: σ sits on the cheapest margin that has not run out, and the loose-budget and tight-budget results of Figure 8.1 are that sentence with different margins substituted in. The capital intensity of a decarbonised system survives. And the duration lesson survives: The batteries built here are long ones, as they were in Denmark.

Does not survive. The identity of the marginal abatement technology does not: In Denmark it was a fuel switch on a peaker, here it is German solar displacing coal at loose budgets and storage displacing gas energy at tight ones. The territorial budget's price does not survive either, and that was the point of computing it. And the land caps of Section 7 are not these: This section uses published per-zone potentials for all twelve zones, which for Danish onshore wind are roughly twice the stylised limit of Section 7. The two sections are therefore not two runs of one model, and the comparisons above are between systems, not between scenarios.

To sum up: Neither section computed the cost of decarbonising Europe, and neither could. What they did was to make one number, the carbon price, traceable to one margin, twice, at two scales, with different answers both times. That traceability is what these models are for. The next section asks how much of a single run's precision to believe; this section has asked the prior question, which is whether the run is answering the question you asked.

9 Uncertainty and the limits of the model

A capacity expansion model returns its answer to four significant digits. This section is about how many of those digits deserve to be believed. We proceed in three steps. First quantitatively, by re-solving the reference case of Section 7 across 10 weather years and then across 3 technology-cost scenarios. Second by returning to the tax-versus-cap question of Section 2, which is a question about uncertainty and can only be answered once there is some. Third structurally, by listing what the linear planner cannot represent at all.

> **Reading alongside this section.** This section requires no supplementary reading. Bushnell and Smith (2025) is its theme done properly: how to be honest about uncertainty when a model is used to evaluate real climate policy, applied to the US Inflation Reduction Act. Reichenberg et al. (2023) is the natural companion to Section 9.1: It recasts the cannibalisation of Section 3 as *revenue risk* for an investor, which is what the weather spread below amounts to seen from the private side. Weitzman (1974) is the source of Section 9.3. Appendix D is the systematic version of Section 9.5: every ingredient the models contain, every one they do not, and the direction of each omission's error.

9.1 The weather-year sweep

Everything in Section 7 conditioned on a single weather year. Figure 9.1 re-solves the identical problem, same network, same costs, same demand and the same carbon price of 85 €/t, on each of the 10 weather years 2015–2024. Only the weather moves, and it moves *everywhere at once*: All twelve zones are re-run on the same reanalysis year, so when Denmark is becalmed, Germany and southern Norway are whatever they actually were in that hour of that year. Wind is strongly correlated over a few hundred kilometres and weakly over a thousand, so a sweep that varied Danish weather while holding the neighbours fixed would quietly assume that imports are available in every Danish scarcity hour, and would understate exactly the risk the exercise is supposed to measure: correlated calm across a whole synchronous area. Appendix C.5 records where the profiles come from and what the construction still simplifies. The 2024 column is the reference system of Section 7 itself, solved on the same archive year with the same construction.

Figures 9.1 and 9.2 show the following:

i. Onshore wind sits at its land limit of 7.5 GW in every year, cheap enough to max out whatever the weather does. Solar ranges from 7.6 to 11.3 GW and offshore wind from 6.4 to 7.3 GW.

ii. The flexible layer moves most: batteries from 1.5 to 3.9 GW and the peakers from 0.0 to 2.1 GW. Storage and standby capacity are close substitutes near the optimum, and which one a year prefers depends on how its calm spells happen to fall.

iii. The Danish bill varies by 19.2% across the years. The cost surface is *flat* near the optimum, so very different fleets are almost equally good.

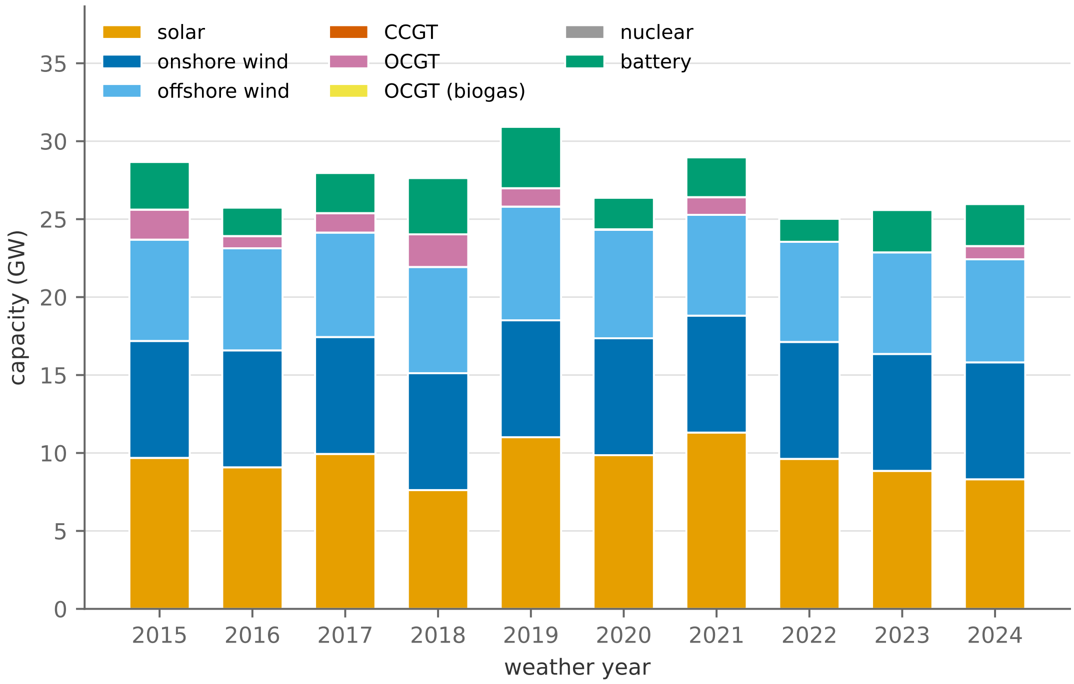

Figure 9.1: The optimal Danish mix, re-solved per weather year with everything else held fixed.

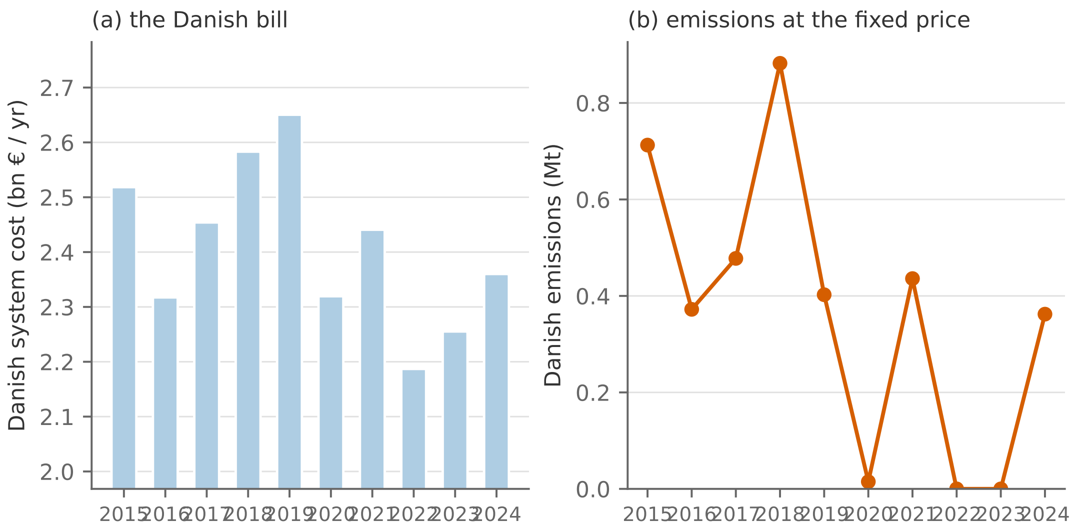

Figure 9.2: The Danish bill (a) and Danish emissions at the fixed carbon price (b) across weather years.

iv. Danish emissions at the fixed carbon price range from 0.00 to 0.88 Mt: Under a price, the tonnes are what the weather makes them, which is the first half of Section 9.3.

Figure 9.3 shows why the years differ. It is not in their averages, which are close, but in their tails: how long the calm spells last, and how much wind arrives in hours that already have enough. It is tails, not averages, that size capacity.

Three conclusions follow. First, a single-year capacity expansion result carries *spurious precision*: Reporting "the optimal offshore capacity is 6.4 GW" from one run misstates what the model knows, which is a range. Second, this is *weather* uncertainty only; Section 9.2 asks the same

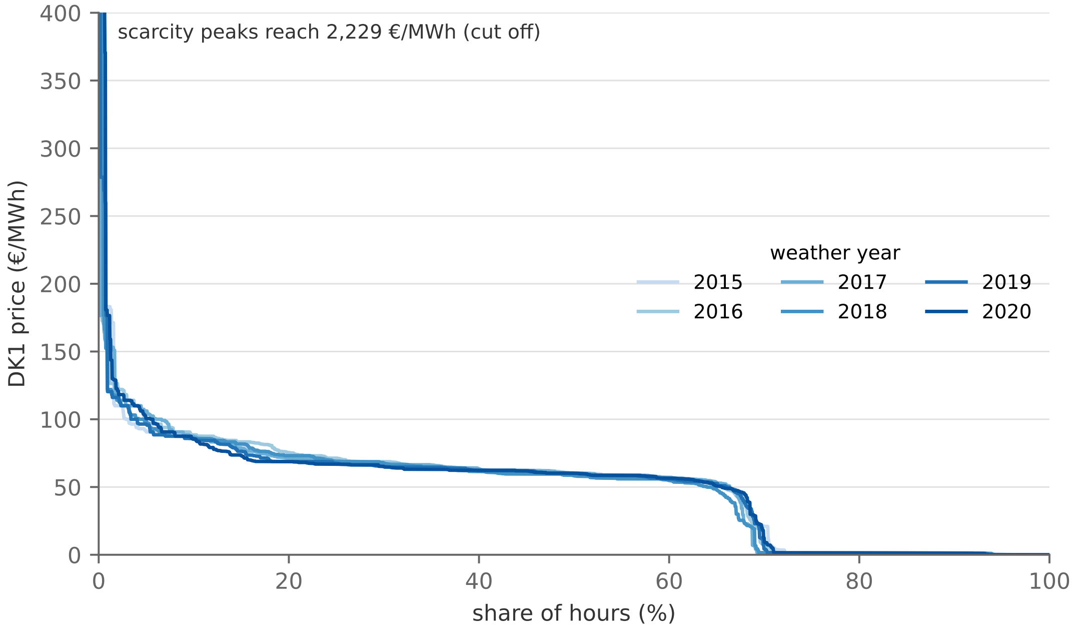

Figure 9.3: DK1 price duration curves of the optimal systems, by weather year.

question of technology costs, and fuel prices, demand growth and policy add spread of their own again (the scenarios of Figure 7.6 sample that space crudely). The honest statement compounds them. Third, the right response is not despair but design. The *robust* findings (the system is overwhelmingly renewable in every year, onshore wind is at its limit in every year, batteries appear in every year, and gas appears only as standby capacity, in some years not at all) deserve confidence, while the *fragile* ones (the exact split between solar and offshore wind, and between batteries and peakers) deserve hedging, staging and option-value language, none of which a deterministic LP will volunteer. Flat optima are good news: If several fleets cost nearly the same, the planner is free to let other criteria, ports, supply chains and public acceptance, break the tie.

9.2 What the machines will cost

Weather is the uncertainty this kind of model is usually made to confess to, and it is the easiest to sweep, because a weather year is a thing that happened and can simply be substituted. But look back at Section 7 and ask what else was asserted rather than derived. The annuitised capital costs, 29 k€/MW for solar and 168 for offshore wind, are *projections*. Every capacity in Figure 7.4 rests on them at least as heavily as on the weather. So we can ask the same question of them. The database publishes a projection for each of 2025, 2050 and the years between, and the obvious move is to re-solve on each of them and call the spread cost uncertainty. It is not. A vintage is a projection of what a machine will cost in a *stated year*; sweeping across vintages asks "what if it were 2035?" when Section 7 has already fixed the horizon at 2050. Moving along the vintages is moving *forward in time*, not sideways into an equally likely world.

What we want instead holds the horizon and varies the projection. The database's own path from 2025 to 2050 is the only statement about the future we have, so we build the scenarios out

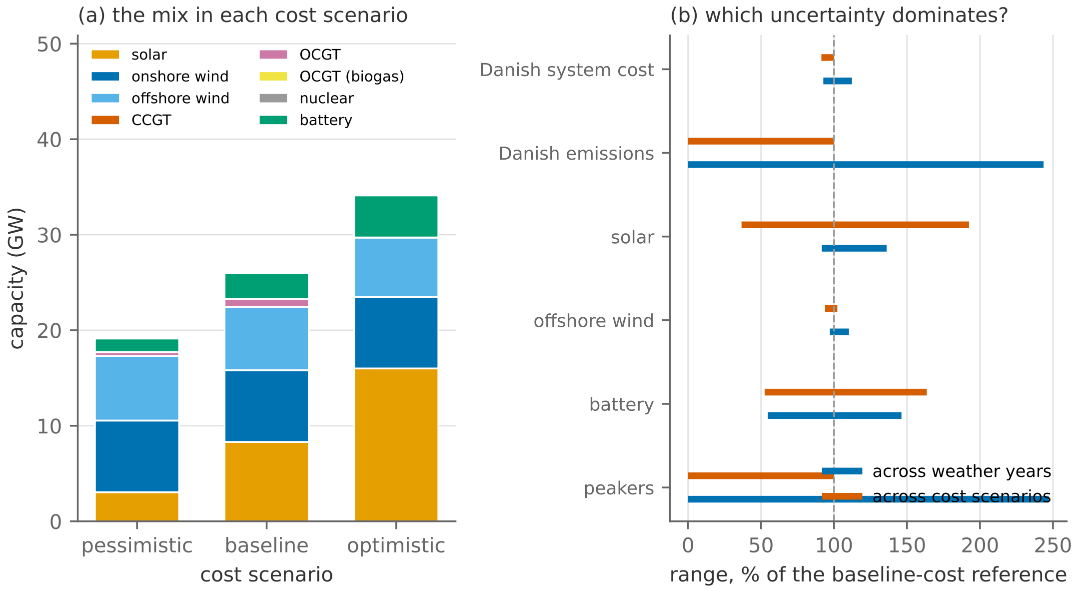

Figure 9.4: (a) the optimal mix in each cost scenario. (b) the two uncertainties side by side: for each output, the range it takes across weather years against the range it takes across cost scenarios, both as a percentage of the reference system.

of it. For each technology, write Δ for the fall the database projects between those two years, and set

$$C^{\text{pess}} = C_{2050} + \tfrac{1}{2}\Delta, \qquad C^{\text{opt}} = C_{2050} - \tfrac{1}{2}\Delta.$$

The *pessimistic* world realises only 50% of the projected decline by 2050; the *optimistic* one overshoots it by half again. This is applied uniformly to every technology's capital cost, so nothing is cherry-picked; a machine the database expects to stay flat, such as nuclear, has $\Delta = 0$ and does not move at all. For the four that matter most here Table 8 lists the three tables' values in €/kW.

Table 8: Investment costs under the three cost scenarios, in €/kW, for the four technologies that matter most here.

	pessimistic	baseline	optimistic
Solar PV	383	293	202
Onshore wind	1080	1019	959
Offshore wind	1763	1643	1522
Battery storage	104	80	56

Figure 9.4 re-solves the reference problem on each of the 3 scenarios, holding the network, the weather, the demand and the carbon price fixed. The middle scenario is not a copy of the cost table of Section 7 but the table itself, so the centre of this sweep reproduces the reference system exactly.

The figure shows the following:

i. Panel (a) is what cheap machines do, and it is orderly in a way the weather sweep was not, because it is a *substitution along a cost gradient* rather than a response to a different year.

Batteries grow from 1.4 to 4.4 GW as cells cheapen, solar from 3.0 to 16.0 GW, and offshore wind from 6.2 to 6.8 GW.

ii. The Danish bill falls from 2.4 to 2.2 bn€, and emissions at the fixed carbon price fall with it, from 0.36 to 0.00 Mt: Cheap capital does abatement that the price alone would not have bought.

iii. In panel (b), the bill is the least sensitive thing on either axis (19.2% across weather years, 9.0% across cost scenarios), so the flat optimum of Section 9.1 is not a weather artefact; the cost surface is simply flat near the answer, in both directions. Note that this is the *Danish* bill. Quote the solver's objective instead, 51.3 bn€ across twelve zones, and either axis appears to move the system by a fraction of a per cent, which says nothing about Denmark and everything about the denominator.

iv. The capacities are far more sensitive than the bill, and here the cost axis moves them further than the weather does: The peakers move by 247 percentage points of the reference system across weather years and 100 across cost scenarios, the battery by 92 and 111. A cheaper battery displaces the peaker, and a dearer one restores it, so the size of the peaking fleet is at least as much a statement about what its competitors cost as about how bad the calm week can get.

Which way does it bias the answer? *A cost projection read as a probability.*

The two ranges in Figure 9.4(b) are not the same kind of object. The weather years are ten draws from a process that will keep drawing, from a distribution that is itself drifting as the climate warms. The cost scenarios are not a sample of anything: They are two points we *constructed*, and their separation is a number we chose, 50% of the database's projected decline in each direction. The database publishes one number per technology per year, not a distribution, so everything we do not know about whether that *path* is right is missing from the panel entirely, and the direction of past error is known: Projections of solar and wind costs have been revised down repeatedly. The cost range therefore **understates** cost uncertainty, and the note never prints it as a confidence interval. Bushnell and Smith (2025) is the best treatment of what to do about that when a model of this family is used to evaluate real policy.

9.3 Tax versus cap under uncertainty

Result 2 stated that a tax and a cap are equivalent, and that the equivalence breaks under uncertainty: A tax fixes the marginal cost of abatement and lets emissions fluctuate, while a cap fixes emissions and lets the carbon price fluctuate, and which is better depends on the relative slopes of marginal abatement cost and marginal damage (Weitzman 1974). We now have some uncertainty to run that argument on. The experiment lives on the European model of Section 8 rather than on the Danish one of Section 7, because a cap has to bind for its price to mean anything, and Section 7.5 showed that a territorial budget in an open economy barely does. The basic rung of

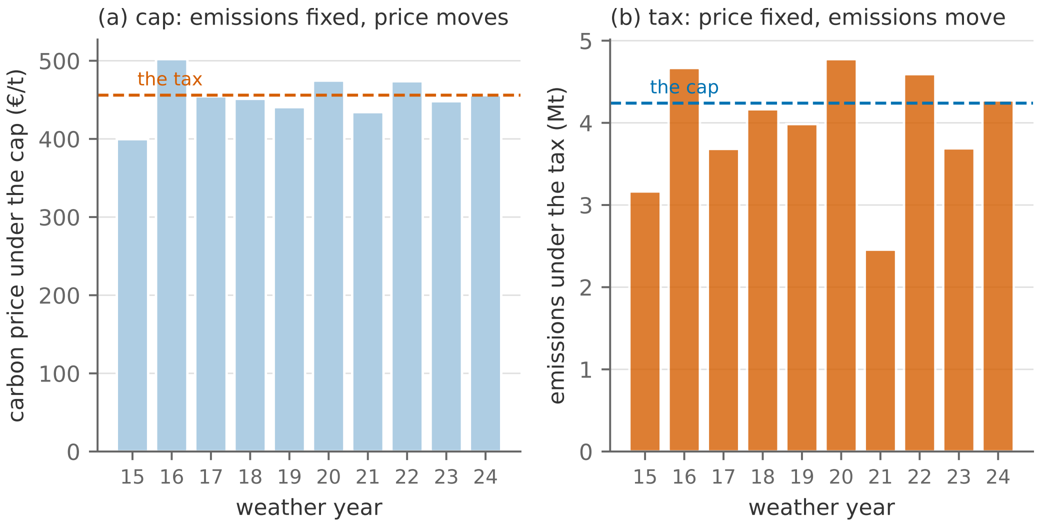

Figure 9.5: Tax versus cap across weather years, on the European model: (a) under the cap, the carbon price the budget requires; (b) under the tax, the emissions that result.

the ladder is re-solved on each of the 10 weather years twice: under its reference budget of 4.2 Mt, and under a carbon tax of 456 €/t, which is the budget's shadow price in the reference year 2024. In that year the two coincide, by Result 2. Figure 9.5 shows what happens in the others.

The figure shows that (i) under the cap, emissions are fixed by construction and the carbon price ranges from 400 to 502 €/t, a ratio of 1.3 between the extreme years; (ii) under the tax, the price is fixed and emissions range from 2.5 to 4.8 Mt, overshooting the budget by up to 13% in a bad year and undershooting it by up to 42% in a good one. This is Weitzman's trade-off in the numbers: The cap delivers the tonnes and makes the price a lottery; the tax delivers the price and makes the tonnes a lottery. Which lottery to prefer depends on what is being protected. If what matters is a cumulative carbon budget, the cap's fluctuating price is the cost of certainty about the quantity. If what matters is that investors can plan against a known price, and the marginal damage of a tonne is roughly the same in a windy year and a calm one, the tax's fluctuating tonnes are the cheaper way to get roughly the same abatement. That is the argument for a carbon price with a floor and a ceiling, which is what most real emissions trading systems have grown into.

9.4 What perfect foresight is worth

The LP does not merely know the future's statistics; it knows the future. Every price a storage unit arbitraged in Section 4 was, in effect, known a year in advance. Figure 9.6 prices that clairvoyance: the battery of Section 4 operated by a perfectly foresighted planner against the same battery operated myopically, one day at a time (366 daily windows), each midnight's charge handed to a tomorrow the operator never anticipates.

For a 4-hour battery the gap is 8.20 M€ a year: real, but modest, because daily arbitrage needs to see only a day ahead, and day-ahead prices are forecast well. The lesson should scale with the storage duration. For a hydro reservoir or a hydrogen store deciding whether to hold energy

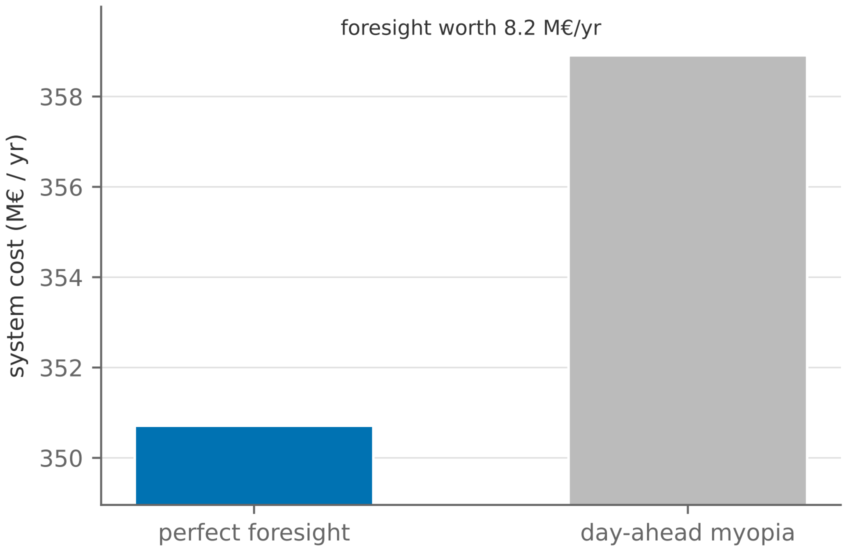

Figure 9.6: System cost with the battery under perfect foresight and under day-at-a-time myopia.

for months, foresight is plausibly a far larger share of the value (we assert this from the horizon argument rather than compute it), and a perfect-foresight LP then materially overstates what an actual operator can extract. As a rule of thumb: Trust the LP's operational numbers where the relevant horizon is short and forecastable, and discount them where it is long.

9.5 Structural limits

Weather spread can be quantified inside the model. The deeper limits cannot, because they are things the model's mathematics excludes by construction. Four matter most, and each maps to a specific way a policy conclusion can go wrong.

Non-convexity. Real plants have start-up costs, minimum loads and indivisibilities; real investments come in lumps. Convexity is what bought us strong duality, prices that support the optimum (Appendix A), and with it gone there are hours in which no uniform price supports the efficient dispatch at all: Markets run side-payments ("uplift") instead, and the crisp identity between dual and price frays. The LP's smooth answers understate the value of flexibility and overstate how finely capacity can be tuned.

Perfect foresight. As above, and at investment horizons the problem compounds, since the model builds for weather and prices it has already seen. Real investment under uncertainty carries option value: Waiting has a price, and deterministic LPs set it to zero.

Market power. The welfare theorems that let us read the planner's shadow variables as market prices assume price-taking. Concentrated ownership of pivotal capacity, a hydro cascade in a dry year or the last gas plant in a scarcity hour, breaks the equivalence, and the model's prices become a lower bound on what a strategic player could extract. A model of this kind can tell you what the competitive benchmark is; it cannot tell you whether the market will deliver it.

Missing money. The note has now assembled every piece of this argument. Start from Section 2.4: In an hour when capacity is short, the price rises to what the marginal consumer will pay, and a peaking plant earns that price less its fuel on every MWh. Next, Result 6: At the long-run optimum those rents, summed over the year, exactly cover the plant's annualised capital cost. Finally Figure 7.3: For the peakers of the reference system, 73% of that annual revenue arrives in the best 100 hours. An energy-only market finances its backup capacity through a small number of expensive hours, and it must, because that is the arithmetic of paying for a machine that runs a few hundred hours a year.

Now ask what happens in the world. Regulators impose price caps. Politicians intervene when prices spike, and spikes coincide with cold snaps, which is when intervening is most popular. Investors, anticipating both, discount rents they doubt they will be allowed to collect. Every one of these breaks a premise of Result 6, and the result is the **missing money** problem: Capacity that the model says pays for itself is not financeable in practice, and the market under-builds relative to the optimum. This is the economic case for capacity remuneration mechanisms, which this note does not model, and it is why the debate over "energy-only versus capacity markets" is really a debate about whether scarcity pricing is politically survivable. Two things follow. First, the dispatch models of Sections 3 to 6 cannot even display the problem: With load fixed and no value of lost load, they have no scarcity prices to cap. The expansion models can, and the reference system's prices never go above 1,431 €/MWh, since the flexible blocks absorb the scarcity before it reaches anyone's value of lost load. Second, note the direction of the error: A model that reports an adequate, self-financing fleet is presuming a price distribution that no European regulator has yet been willing to allow.

None of this is a reason to discard the tool; every alternative to a transparent optimisation model is a less transparent one. It is a reason to use the model the way this note has tried to: as a machine for producing *mechanisms and ranges*, whose every number comes with the question "which constraint produced you, and would you survive a different year?". A student who asks that question by reflex has learned what these models are for.

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Appendices

A Convex optimisation, duality and KKT: a refresher

The main text assumes you can set up a Lagrangian, write down first-order conditions, and interpret multipliers, including Karush-Kuhn-Tucker conditions for inequality constraints. This appendix provides a refresher on the techniques including a short cookbook for how to approach it. This is not a strictly rigorous presentation with fancy proofs; we focus on the basics and on giving you a tool that can be used. Note that to arrive at one simple cookbook for how to use it, we always write up the Lagrangian for a *maximization problem.* When a model is naturally written as a cost *minimisation*, we maximise minus the costs instead.

A.1 Convex problems and linearity

In this note, all models belong to the class of so-called *linear programs* (LP). While I will spare you a presentation of the different types of optimisation problems, the LP deserves a short mention. First, LPs belong to a greater family of optimisation models called *convex problems* that are famous for being (relatively) easy to solve (both theoretically and numerically). A convex problem maximises a concave function (or minimises a convex one) over a convex set. The convex set roughly means that the space of possible solutions we search over is easy to navigate and the concave function means that gradients/derivatives will tell you in what direction the optimum lies.

The linear models in this note are special cases of the convex problems that are linear in objective, equality, and inequality constraints. A linear function is both concave and convex, so an LP is the friendliest corner of the convex world, and three properties are used repeatedly:

i. *Local is global.* A local optimum of a convex problem is a global optimum. When a solver reports an optimum, there is no better dispatch hiding elsewhere (up to ties).

ii. *First-order conditions are exact.* For a convex problem the KKT conditions below are not merely necessary at an optimum; they are also sufficient.

iii. *Solutions sit on corners.* A linear objective on a polyhedron attains its optimum at a vertex (roughly meaning a point where as many constraints bind as there are decision variables). This is why optimal dispatch is bang-bang (plants at capacity or at zero, one marginal plant in between) and why, in Section 7, an LP never builds a diversified portfolio for diversification's own sake.

A.2 The standard form, and a recipe for the Lagrangian

You will typically meet optimisation problems written in a *standard form* that looks like this:

$$\max_{x_1,\,x_2} \quad f(x_1, x_2) \tag{44a}$$

$$\text{s.t.} \quad g(x_1, x_2) \le k_g \qquad (\lambda_g) \tag{44b}$$

$$h(x_1, x_2) = k_h \qquad (\lambda_h) \tag{44c}$$

$$x_1 \in [\underline{x}_1, \overline{x}_1] \qquad (\underline{\theta}_1,\ \overline{\theta}_1) \tag{44d}$$

$$x_2 \in [\underline{x}_2, \overline{x}_2] \qquad (\underline{\theta}_2,\ \overline{\theta}_2) \tag{44e}$$

where x_1, x_2 are the decision variables, f is the *objective*, g is an *inequality constraint*, h is an *equality constraint*, and $(\underline{x}_i, \overline{x}_i)$ are *domain constraints* (also inequality constraints, really, but so common that the recipe treats them separately). As in the main text, each constraint's multiplier is named in parentheses next to it.

The *Lagrangian* replaces each constraint by a linear penalty term:

$$\begin{aligned}\mathcal{L} = \; & \underbrace{f(x_1, x_2)}_{\text{objective}} + \underbrace{\lambda_h\Big(k_h - h(x_1, x_2)\Big)}_{\text{equality constraint}} + \underbrace{\lambda_g\Big(k_g - g(x_1, x_2)\Big)}_{\text{inequality constraint}} \\ & + \underbrace{\underline{\theta}_1\Big(x_1 - \underline{x}_1\Big) + \overline{\theta}_1\Big(\overline{x}_1 - x_1\Big)}_{\text{domain constraint I}} \; + \; \underbrace{\underline{\theta}_2\Big(x_2 - \underline{x}_2\Big) + \overline{\theta}_2\Big(\overline{x}_2 - x_2\Big)}_{\text{domain constraint II}},\end{aligned} \tag{45}$$

where $\lambda_h, \lambda_g, \underline{\theta}_1, \overline{\theta}_1, \underline{\theta}_2, \overline{\theta}_2$ are called *shadow variables* (equivalently: dual variables, or Lagrange multipliers); we return to their interpretation below. If there were more constraints of any type, the Lagrangian would simply extend additively, exactly as it does for the two domain constraints.

Three remarks on getting the terms right, one per constraint type. This is where sign mistakes are made, so the rules are worth committing to memory:

- *Equality constraints:* λ_h can be positive or negative, and the constraint may be added either way round:

$$\text{Do this: } \lambda_h\left(k_h - h(x_1, x_2)\right), \qquad \text{or this: } \tilde{\lambda}_h\left(h(x_1, x_2) - k_h\right).$$

Both give the same solution except for the sign of the multiplier: $\lambda_h = -\tilde{\lambda}_h$.

- *Inequality constraints:* write the term so that it **is non-negative when the constraint holds**. With the constraint $g(x_1, x_2) \leq k_g$:

$$\text{Do this: } \lambda_g \underbrace{(k_g - g(x_1, x_2))}_{\geq 0}, \qquad \text{and } \underline{\text{not}}\text{: } \tilde{\lambda}_g \underbrace{(g(x_1, x_2) - k_g)}_{\leq 0}.$$

  The reason the direction matters is that the sign of λ_g is itself a condition for optimality: the KKT conditions below require $\lambda_g \geq 0$, and that requirement only means what it should if the term is written this way round. This changes if the problem is a *minimisation* instead of *maximisation* problem.

- *Domain constraints:* $x_1 \in [\underline{x}_1, \overline{x}_1]$ is just the pair of inequality constraints $x_1 \geq \underline{x}_1$ and $x_1 \leq \overline{x}_1$, so the same rule applies: We add both terms such that they are non-negative:

$$\begin{aligned} &\text{Do this:} && \underline{\theta}_1 \overbrace{(x_1 - \underline{x}_1)}^{\geq 0} + \overline{\theta}_1 \overbrace{(\overline{x}_1 - x_1)}^{\geq 0} \\ &\text{and } \underline{\text{not}}\text{:} && \tilde{\underline{\theta}}_1 \underbrace{(\underline{x}_1 - x_1)}_{\leq 0} + \tilde{\overline{\theta}}_1 \underbrace{(x_1 - \overline{x}_1)}_{\leq 0} \end{aligned}$$

A.3 Optimality conditions

Now that we have set up the Lagrangian, identifying an optimum (x_1^*, x_2^*) requires two sets of conditions:

(1) First-order conditions for optimality. All derivatives of $\mathcal{L}$ with respect to the decision variables equal zero at the optimum:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_1} &= \frac{\partial f}{\partial x_1} - \lambda_h \frac{\partial h}{\partial x_1} - \lambda_g \frac{\partial g}{\partial x_1} + \underline{\theta}_1 - \overline{\theta}_1 \quad &=: 0 \\ \frac{\partial \mathcal{L}}{\partial x_2} &= \frac{\partial f}{\partial x_2} - \lambda_h \frac{\partial h}{\partial x_2} - \lambda_g \frac{\partial g}{\partial x_2} + \underline{\theta}_2 - \overline{\theta}_2 \quad &=: 0 \end{aligned}$$

(2) Complementary slackness conditions. For every inequality constraint, including the domain constraints, three things hold at once:

$$\begin{aligned} \lambda_g \Big(k_g - g(x_1, x_2)\Big) &= 0, & k_g - g(x_1, x_2) &\geq 0, & \lambda_g &\geq 0 \\ \underline{\theta}_i \Big(x_i - \underline{x}_i\Big) &= 0, & x_i - \underline{x}_i &\geq 0, & \underline{\theta}_i &\geq 0 \\ \overline{\theta}_i \Big(\overline{x}_i - x_i\Big) &= 0, & \overline{x}_i - x_i &\geq 0, & \overline{\theta}_i &\geq 0, \end{aligned}$$

for $i = 1, 2$. Each row is one inequality constraint, and the three columns are, in order, the complementary-slackness condition proper, *primal feasibility* (the constraint itself holds) and *dual feasibility* (its multiplier is non-negative).

The interpretation of these conditions, and of the shadow variables in them, is generally really important for understanding the models. We read the set as follows:

- For every inequality constraint, at least one of the following two must hold: (i) the shadow value of the constraint is zero (e.g. $\lambda_g = 0$), or (ii) the constraint holds with equality, e.g. $k_g - g(x_1, x_2) = 0$. Both may hold at once; neither may fail.
- If a constraint holds with equality, $k_g - g(x_1, x_2) = 0$, we say that the constraint is *active* (or *binding*).
- If it holds strictly, $k_g - g(x_1, x_2) > 0$, we say that the constraint is *inactive* (or *slack*), and its shadow value is then zero.

Complementary slackness is the condition that carries the most economics per symbol: *only active constraints can have positive prices.* A capacity limit that does not bind is worth nothing at the margin ($\lambda_g = 0$); a capacity limit with a positive price must be exhausted. Scarcity and value are the same fact seen from the primal and the dual side. A useful corollary is that the lower- and upper-bound multipliers of the same variable can never both be positive, since the two bounds naturally cannot both bind at once (unless $\underline{x}_i = \overline{x}_i$).

One caveat worth a sentence: for general nonlinear problems the KKT conditions require a regularity condition on the constraints (a "constraint qualification") to be necessary. For linear constraints (every model in this note) no such condition is needed.

A.4 Interpreting the multipliers: derivatives of the value function

Let f^* be the optimal value of (44). This is, naturally, a function of the parameters that define how *tight* each constraint is: $(k_h, k_g, \underline{x}_i, \overline{x}_i)$. This is useful to interpret what the multipliers measure: **the change in the optimised objective from a marginal relaxation of its own constraint**. Wherever f^* is differentiable, it is straightforward to show that[21]

$$\lambda_h = \frac{\partial f^*}{\partial k_h} \lesseqgtr 0, \qquad \lambda_g = \frac{\partial f^*}{\partial k_g} \geq 0, \qquad -\underline{\theta}_i = \frac{\partial f^*}{\partial \underline{x}_i} \leq 0, \qquad \overline{\theta}_i = \frac{\partial f^*}{\partial \overline{x}_i} \geq 0. \tag{46}$$

This envelope property is the licence for every sentence of the form "the dual of market clearing is the marginal cost of serving one more MWh". Taking the constraint types in turn:

- *Equality constraints:* λ_h is the gain in the objective from increasing k_h marginally, and it can go either way. If f is a utility function and k_h represents income, we'd have $\lambda_h > 0$ measuring specifically the marginal utility of income.[22]
- *Inequality constraints:* the two cases of complementary slackness are exactly the two cases of the interpretation. If the constraint is inactive, a marginal relaxation changes nothing that

[21] You can show this by differentiating $\mathcal{L}$ with respect to the relevant parameter, using the fact that at the optimum the Lagrangian coincides with the objective, $\mathcal{L} = f^*$.

[22] Note that which way round the term was written (Appendix A.2) matters: with the $\tilde{\lambda}_h\,(h - k_h)$ version, the interpretation flips to the gain from *decreasing* k_h.

was binding, so $\lambda_g = 0$. If it is active, relaxing k_g enlarges the feasible set in a direction the optimiser actually wants, and the objective rises at rate $\lambda_g > 0$.

- *Domain constraints:* raising $\overline{x}_1$ hands the optimiser more options, which can never hurt ($\overline{\theta}_1 \geq 0$) and helps only if the bound was the thing stopping it ($\overline{\theta}_1 > 0$ exactly when active). Raising $\underline{x}_1$ removes options, which can never help ($-\underline{\theta}_1 \leq 0$) and hurts only if the optimiser is pushed off its preferred choice.

Two further points matter specifically for the LPs of this note.

Solvers may report the other sign. Once again, whether or not we report the problem as minimisation or maximisation problem matters. If the numerical solvers work on the minimisation form and report duals as derivatives of that objective instead, the signs flip: the dual of a $\leq$-constraint in a cost minimisation comes back non-positive (relaxing a constraint cannot raise minimised cost). The course code flips the sign once, so that what you see matches the non-negative multipliers of the Lagrangian as written here. If you ever wire a solver up yourself and the rents come out negative, it is important to check this convention.

At a kink, the multiplier is not unique. The value function of an LP is piecewise linear in the constraint levels, and at a kink where the optimal basis changes, i.e. where the identity of the marginal plant or the active abatement margin switches, the derivative in eq. (46) does not exist: left and right derivatives differ, and any multiplier between them is valid. This is not a technical nuisance to be waved off; it is visible in the results. Figure 2.4's step function is the derivative of a piecewise-linear value function, flats and jumps included, and on any of its vertical segments the permit price is indeterminate between the segment's endpoints. When a solver reports one number at such a point, it has picked an endpoint of the valid interval, not discovered a unique price.

A.5 Duality, and the dual as the price-setting problem

Every optimisation problem comes with a twin. The problem we started from, (44), is called the *primal*: it chooses quantities (x_1, x_2) and takes the constraint levels (k_g, k_h and the bounds) as given. Its *dual* is a problem in the multipliers instead: it chooses a price for each constraint and takes the quantities as given. The two are built from the same ingredients and they swap roles completely, so that each constraint of the primal supplies one variable of the dual, and each variable of the primal supplies one constraint of the dual. For the models in this note the twins also have the same solution, and that is what lets us move between a planner choosing quantities and a market setting prices without changing the model at all.

Where does the dual come from? From asking how good the primal optimum f^* could possibly be, *without* solving for it. A lower bound is easy: any feasible x we can think of provides one, since the optimum is at least as good as anything feasible. An upper bound is the hard direction, because it has to hold for every feasible point at once, including all the ones we have not looked at. This is what the Lagrangian gives us. Fix the multipliers, and let the planner choose freely, paying the penalties instead of respecting the constraints:

$$D(\lambda, \theta) = \max_{x} \ \mathcal{L}(x, \lambda, \theta). \tag{47}$$

For any x that is actually feasible, every penalty term in $\mathcal{L}$ is non-negative, so $\mathcal{L}(x) \geq f(x)$: the penalised planner can always claim at least what the constrained one achieves. And since D maximises over *all* x, feasible or not, it is at least as large again. Hence $D(\lambda, \theta)$ is an *upper* bound on the optimum of (44) for any sign-feasible multipliers: each set of prices certifies a ceiling on what the constrained planner can achieve, and it does so without anybody having solved (44). This is *weak duality.* Some of those ceilings are useless (set every multiplier to zero, and $D(0, 0)$ is just the value of the problem with all constraints thrown away), so the *dual problem* is the search for the tightest one:

$$\min_{\lambda, \theta} \ D(\lambda, \theta), \qquad \lambda_g, \underline{\theta}_i, \overline{\theta}_i \geq 0. \tag{48}$$

For linear programs the bound closes completely: primal and dual optima are *equal* (*strong duality*), provided the primal is feasible and bounded. Carrying out the inner maximisation for the dispatch LP of Section 2 is a one-line computation, because $\mathcal{L}$ is linear in q: unless the coefficient on every q_g is exactly zero, the maximum is infinite, so the dual's constraints are precisely those zero-coefficient conditions — the first-order conditions, reappearing as the constraints of the price-setting problem. That computation turns (48) into the dual program (6a)–(6b) quoted there. Note also that the prices solving the dual are exactly the multipliers of Appendix A.3: the dual is not a second computation to carry out, but the one we have already done, seen from the other side.

The economic reading of the dual pair: the primal chooses *quantities*, with prices emerging as multipliers; the dual chooses *prices*, subject to a no-arbitrage condition on quantities (no generator can be made to earn more than its cost structure allows), with quantities emerging as multipliers. Strong duality is the statement that the quantity-planner and the price-setter agree on the value of the system — the LP restatement of the welfare theorems that Section 2.3 uses to move between planner and market without changing the model.

A.6 The technique, end to end

Every "derive and interpret" exercise in this course has the same shape, and it is the shape the main text follows silently each time:

i. Write the objective and constraints in the standard form (44) — as a maximisation — and name a multiplier per constraint.

ii. Form the Lagrangian, writing every inequality term so that it is non-negative when its constraint holds (Appendix A.2).

iii. Differentiate with respect to each decision variable and set the derivatives to zero — the first-order conditions for optimality — remembering the multiplier of a non-negativity constraint if the variable has one.

iv. List complementary slackness for every inequality.

v. Solve the resulting case distinction — which constraints bind is the economic content, and there are usually only a few coherent patterns.

vi. Interpret each multiplier as the price of its constraint, using eq. (46).

One clarification from experience: steps 1–4 are mechanical, but step 5 is not, and it is where the training lies. With inequality constraints there is no single formula — solving usually means *guessing* which constraints are active, solving the easy system that remains, and checking the guess against feasibility and the sign conditions. A failed check is not failure; it is information about which guess to try next, and the example below runs exactly that loop. Sometimes the case distinction is too rich to solve analytically at all; the conditions are still worth deriving, because interpreting them — or handing the problem to a numerical solver — is the payoff. Applied to (3a)–(3c), the recipe yields the merit order in half a page, as Section 2.3 shows. The exam expects the technique at exactly this level: set up, derive, interpret — not reprove strong duality.

A.7 A worked example: boots and Teslas

With the technical material in place, let us run the recipe once on a small consumer problem where every number can be checked by hand.
The problem: As an example, consider the "simple" consumer utility maximization problem:

- x_1, x_2 are two consumer goods, e.g. boots and Teslas. Personally, I enjoy all boots just the same - no matter the number, but the more Teslas I get, the less it means to me. So, my utility function is:

$$f(x_1, x_2) = x_1 + 3 \cdot \ln(x_2) .$$

- I have 3€ available to buy boots at 1€ a piece and Teslas at 1/2€ (good price on Teslas, yes). I also have a horrendous credit score, so I cannot establish any debt. So, I have a budget:

$$\underbrace{x_1 + 0.5 \cdot x_2}_{h(x_1,x_2)} = \underbrace{3}_{k_h}$$

- I have promised Marie Kondo never to own more than 5 items, so:

$$\underbrace{x_1 + x_2}_{g(x_1,x_2)} \leq \underbrace{5}_{k_g} .$$

- It is silly to have "negative" amounts of boots, and I only have two feet, so I also face the domain constraint:

$$x_1 \in [0, 2].$$

Figure A.1 draws the three constraints and the set of bundles that satisfies all of them at once. Note what the budget does: because it must be spent *exactly*, the choice is confined to the edge of that set where the budget line runs — the thick stretch between the item cap and the two feet. That is worth holding on to, because it is what makes the answer a corner.

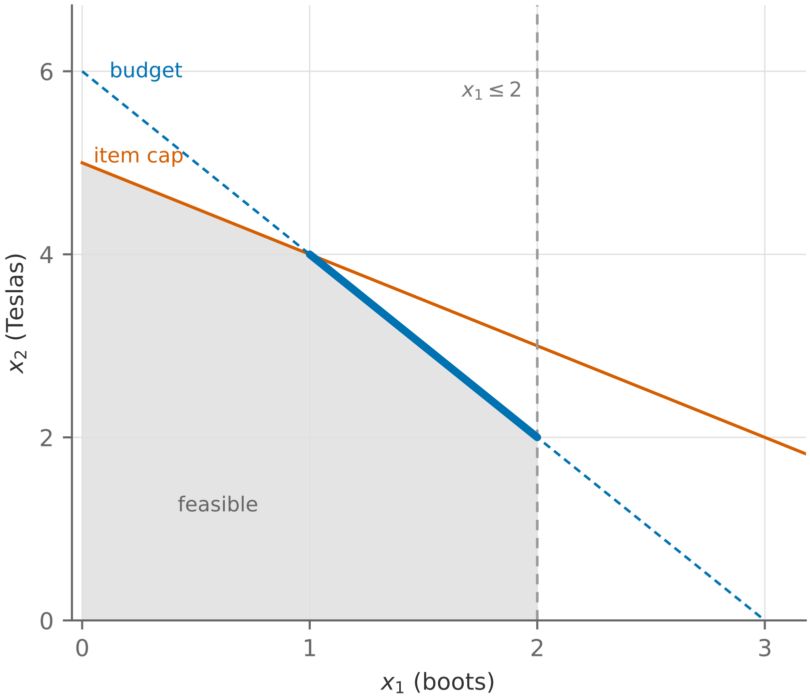

Figure A.1: The constraints of the example, and the feasible set they leave. Shaded is every bundle that is affordable and allowed: at most five items, at most two pairs of boots, nothing negative. Its upper-right edge is built from the two constraints that bind somewhere — the item cap from $(0, 5)$ to $(1, 4)$, then the budget from $(1, 4)$ to $(2, 2)$, where the two feet stop it. Since the budget holds with equality, the bundles actually on offer are that thick stretch of budget line.

<u>The Lagrangian:</u> following the recipe:

$$\mathcal{L} = \underbrace{x_1 + 3 \cdot \ln(x_2)}_{\equiv f(x_1,x_2)} + \lambda_h \Big(\underbrace{3 - (x_1 + 0.5 \cdot x_2)}_{\equiv k_h - h(x_1,x_2)} \Big) + \lambda_g \Big(\underbrace{5 - (x_1 + x_2)}_{\equiv k_g - g(x_1,x_2)} \Big) + \underline{\theta}_1 \Big(\underbrace{x_1 - 0}_{\equiv x_1 - \underline{x}_1} \Big) + \overline{\theta}_1 \Big(\underbrace{2 - x_1}_{\equiv \overline{x}_1 - x_1} \Big).$$

<u>The KKT conditions:</u> the first-order conditions for optimality,

$$\frac{\partial \mathcal{L}}{\partial x_1} = 1 - \lambda_h - \lambda_g + \underline{\theta}_1 - \overline{\theta}_1 := 0$$
$$\frac{\partial \mathcal{L}}{\partial x_2} = \frac{3}{x_2} - 0.5 \cdot \lambda_h - \lambda_g := 0,$$

and the complementary slackness conditions:

$$
\begin{aligned}
\lambda_g \left(5 - (x_1 + x_2)\right) &= 0, & 5 - (x_1 + x_2) &\geq 0, & \lambda_g &\geq 0 \\
\underline{\theta}_1 x_1 &= 0, & x_1 &\geq 0, & \underline{\theta}_1 &\geq 0 \\
\overline{\theta}_1 \left(2 - x_1\right) &= 0, & 2 - x_1 &\geq 0, & \overline{\theta}_1 &\geq 0.
\end{aligned}
$$

Solve and interpret: this is the guess-and-check loop promised in Appendix A.6.

First guess: no constraint binds except the budget. Set $\lambda_g = \underline{\theta}_1 = \overline{\theta}_1 = 0$. The remaining system,

$$
\begin{aligned}
1 - \lambda_h &= 0 \\
\frac{3}{x_2} - 0.5\lambda_h &= 0 \\
x_1 + 0.5 \cdot x_2 &= 3,
\end{aligned}
$$

solves to $\lambda_h = 1$ and $(x_1, x_2) = (0, 6)$: spend everything on Teslas. But check the guess: $x_1 + x_2 = 6 > 5$ violates the Marie Kondo constraint, so this is *not* the optimum — and the failed check tells us which constraint to activate next.

Second guess: the item cap is active. Keep $\underline{\theta}_1 = \overline{\theta}_1 = 0$ but let $\lambda_g > 0$, so complementary slackness forces $x_1 + x_2 = 5$:

$$
\begin{aligned}
1 - \lambda_h - \lambda_g &= 0 \\
\frac{3}{x_2} - 0.5\lambda_h - \lambda_g &= 0 \\
x_1 + 0.5 \cdot x_2 &= 3 \\
x_1 + x_2 &= 5.
\end{aligned}
$$

You may verify that the solution is $\lambda_h = \lambda_g = 0.5$ and $(x_1, x_2) = (1, 4)$. Now every check passes: $x_1 = 1$ lies strictly inside $[0, 2]$ (so the domain multipliers are indeed zero), and both remaining multipliers are non-negative. By sufficiency (Appendix A.1), this *is* the optimum. Figure A.2 puts both candidates on the picture, and the geometry is worth a look, because it is the same geometry every time. Utility rises along the feasible segment in the direction of more Teslas, so the consumer walks up it until something stops her, and what stops her is the item cap. The optimum is therefore the corner where budget meets cap, with both constraints active. The two feet cut the segment off at the other end, but she never wants to go that way, which is exactly why their multipliers are zero.

The multipliers now carry the economics, via eq. (46). $\lambda_g = 0.5$: at the margin, loosening Marie Kondo's rule is worth 0.5 of utility per item. $\lambda_h = 0.5$: a marginal euro is worth 0.5 in utility. Note that the euro was worth 1 in the first guess, before the item cap entered: with the cap binding, extra money is less useful, because the cheap way to spend it — more Teslas — is exactly what the cap forbids. Constraints interact through the multipliers, and reading those interactions is what the last step of the recipe is for.

One caution to close on, and it is Appendix A.4's: a multiplier is a *derivative*. Allowing a sixth item does not raise utility by 0.5, but by $3\ln(3/2) - 1 \approx 0.22$, because the value of loosening the cap falls as the cap loosens: at six items the cap no longer binds at all, so λ_g has reached zero by the time we get there. In the rest of this note the value function is piecewise *linear* rather than curved, so a shadow price does stay valid over a range of the constraint — but only up to the next kink.

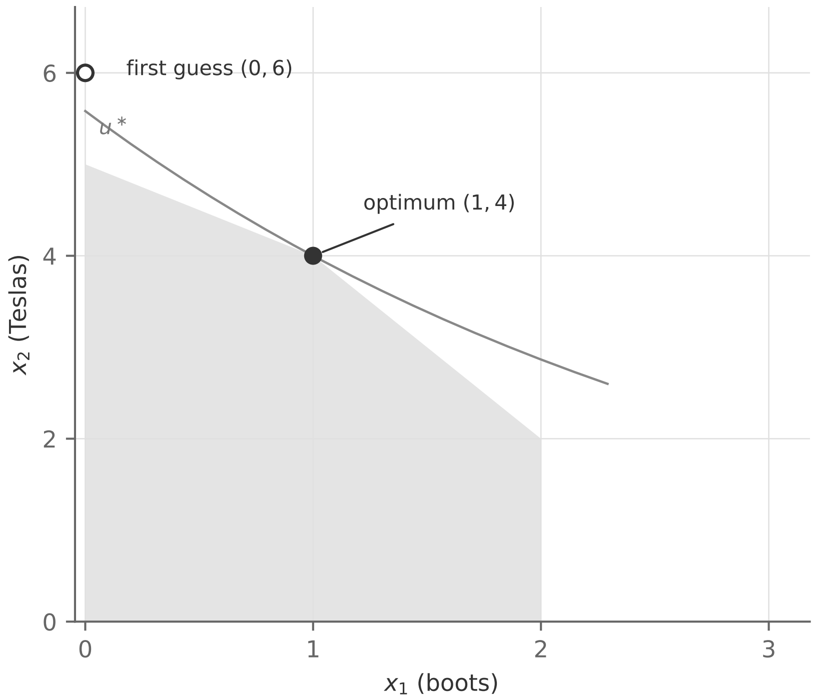

Figure A.2: The two candidates against the same feasible set, with the constraint lines dropped — the shape of the region carries them. The first guess ignores the item cap and lands outside it, which is what the check catches. The optimum is the feasible bundle on the highest indifference curve u^*: the whole feasible set lies below that curve, and everything the consumer would prefer to $(1, 4)$ lies outside it. Note that u^* passes through the corner with a slope between those of the two constraints that meet there — the geometry behind $\lambda_h = \lambda_g = 0.5$.

B Annuities and the levelised capital cost

Investment costs and operating costs live on different clocks: a turbine is paid for once and produces for decades. To put both into one objective — as Section 7 does — the investment must be converted into an equivalent *annual* payment.

B.1 The annuity factor

Lend an amount I at interest rate r, to be repaid in n equal annual instalments aI. The present value of the instalments must equal the loan: $I = aI \sum_{k=1}^{n}(1+r)^{-k}$, and solving the geometric sum gives the *annuity factor*

$$a(r,n) \;=\; \frac{r}{1-(1+r)^{-n}}. \tag{49}$$

An investment of I euros per MW with lifetime n is thus economically identical to a rent of $a(r,n)\,I$ euros per MW per year: a stream of capacity services purchased annually. Adding fixed operating costs (FOM, usually quoted as a percentage f of the investment) gives the annual fixed cost the models use,

$$F \;=\; (a(r,n)+f)\,I. \tag{50}$$

For longer lifetimes $a \to r$, which is essentially the case e.g. for hydro damn costs with 60-year lifetimes. The discount factor r is incredibly important for such technologies, in particular when they are also capital-intensive (wind, solar, batteries).

B.2 Levelised cost, and what it leaves out

Dividing the annual fixed cost by annual production gives the familiar *levelised cost of electricity*,

$$LCOE \;=\; \frac{(a(r,n)+f)\,I}{8760\cdot CF} + c, \tag{51}$$

with c the marginal cost and CF the capacity factor. It measures the average costs of a technology. On the cost side, it requires assumptions on technical lifetime n, discount rate r, capacity factor CF, marginal costs - and full investment costs. The capacity factor is not simply read off some technical specification, it is an equilibrium object that requires knowing how often this specific generator will be active throughout a year.

B.3 Screening curves

Plotting total annual cost per MW against running hours, $F+c\cdot h$, gives each technology a straight line — intercept the fixed cost, slope the marginal cost (Figure 7.1). The lower envelope of the lines answers “which technology is cheapest for a duty of h hours per year”: capital-light, fuel-heavy plants win the short duties, capital-heavy, fuel-light plants the long ones. Classic screening-curve analysis intersects this picture with the load duration curve to read off an optimal mix; the LP of Section 7 is the same logic made exact, with intermittency, storage and trade — none of which fit in the two-dimensional picture — handled by the constraints.

C Provenance: data and the network

Every number in this note is meant to be traceable to a source and reproducible. This appendix records where the inputs come from, and how the bidding-zone network of Sections 6 to 9 was derived.

The models themselves are built in PyPSA (Brown, Hörsch, et al. 2018), an open framework for power system analysis. Every model in this note *can* be solved with a free and open-source linear programming solver, and at the instance sizes the course repository ships by default that is what we recommend: nothing here requires a licence. The largest instances behind the note's own figures — the capacity expansion problems of Section 7 to Section 8, which run to around half a million variables — were solved with a commercial barrier solver under a free academic licence, purely for speed. The formulation is identical either way, and so is the answer. The course repository holds every model, dataset and script, together with the instructions for reproducing each figure and table here.

C.1 Data sources

- **Danish market data** — Energi Data Service (Energinet), dataset `ProductionConsumptionSettlement`: hourly gross consumption and production by category for DK1 and DK2, 2019–2024. CC-BY 4.0, no API key. Availability profiles are production divided by its annual maximum — a classroom proxy that ignores within-year capacity growth. The same request also keeps production in *levels* by category (central thermal, local CHP, commercial) together with the cross-border exchange flows, which is what Figure 3.8 holds the modelled week against. All six of DK1's wires are read — Norway, Sweden, Germany, the Netherlands, Great Britain and DK2 — so that production plus net imports equals consumption hour by hour; omitting any of them silently breaks that identity.

- **Temperature** — Open-Meteo historical archive (ERA5-derived), hourly 2m temperature, Aarhus, 2024. CC-BY 4.0.

- **Zonal weather years** — the same archive, hourly wind speed at 100 m and surface radiation at each of the twelve zones' coordinates, 2015–2024. CC-BY 4.0, no account. The basis of Section 9's sweep; construction and caveats below.

- **Technology costs** — PyPSA's `technology-data` (PyPSA, Danish Energy Agency, et al. 2025) *fetched at a pinned tag* (`v0.13.2`) and reshaped, for each of the cost projection vintages 2025 to 2050 in five-year steps. It compiles the Danish Energy Agency's catalogue (Danish Energy Agency and Energinet 2024) together with several other published sources and converts everything to a common currency year. The small table of Table 1 is *generated* from the 2025 vintage; Section 7 solves on the 2030 vintage and Section 9.2 sweeps them all. The compiled files carry no single licence, so nothing from them is redistributed by the course repository — students re-fetch byte-identically.

- **Battery costs** — the one place where the note overrides the compilation's investment costs, and deliberately so: the compilation's battery trajectory has fallen well behind the world market, its 2040 vintage costing roughly what turnkey systems sold for in 2025. Every vintage's four-hour system cost is therefore capped at **165 €/kWh**, BloombergNEF's 2025 European average turnkey price (177 $/kWh in the *Energy Storage System Cost Survey 2025*, December 2025, press-release figures; converted at $\sim$1.07 USD/EUR; the global average was 117 $/kWh and the Chinese 73 \$/kWh). The split between inverter (€/kW) and cells (€/kWh) keeps the compilation's own ratio, and vintages already below the cap are left alone, so the 2025–2050 sweep of Section 9.2 stays monotone: the cap binds for the 2025 and 2030 vintages only. The currency conversion, and reading a nominal-2025 dollar price against the compilation's EUR-2020 basis, are part of the assumption; the processed data records the compilation's value beside the one used, in the same visible-override format as the fuel file.

All downloads land outside the published tree and every transformation is scripted, so the processed inputs are reproducible from the original sources.

C.2 One set of plant characteristics

The note's data arrives in three tiers: the small table of Sections 2 to 5, the shipped network of Sections 6 to 9, and the investment costs of Sections 7 and 8. Each comes from a different source, and sources of this kind routinely disagree on fuel prices, so that the same combined-cycle plant would cost one amount to run in one section and another in the next. Three rules keep the tiers on one basis.

- **One file holds the fuel prices and carbon contents**, and all three tiers read it. It records the compilation's own published value beside the one this note uses, so every override is visible rather than buried; the note's prices are stylised 2025 forward levels, because the compilation's are inherited from a study over a decade old and sit well below what European plants have paid recently.
- **The small table is generated, not hand-written.** Every efficiency and operating cost in Table 1 is read from `technology-data` at the pinned tag, so the machines named there are the same machines, with the same numbers, as in the course's companion note on generation technologies — which derives its catalogue from the same source at the same pin.
- **The shipped network is re-priced when it is loaded.** Each thermal generator keeps the efficiency PyPSA-Eur gave it — that heterogeneity is real, and worth having — and takes this note's fuel price and operating cost. Carrier emission factors are set from the same file at the same time.

Two carbon entries are accounting choices rather than measurements. Biogenic carbon is counted as zero. Municipal waste carries only its fossil, largely plastic, fraction at 0.15 t per $\mathrm{MWh_{th}}$ — not the zero it is often waved through with, and enough, divided by a low electrical efficiency, to make waste one of the more carbon-intensive entries in the table.

Sections 7 and 9 keep the 2030 vintage for *investment* costs, since they are asking what it costs to build for 2030 and Section 9.2 is about exactly that uncertainty. The residual efficiency difference between the two vintages is under 2%, and the pipeline writes it out rather than leaving it implicit.

C.3 The bidding-zone network

The network of Sections 6 to 9 was derived from **PyPSA-Eur** (Hörsch et al. 2018)[23] at release **v2026.08.0**, run once by us — the workflow's machinery (snakemake, atlite, a ∼7 GB weather cutout) is never a student dependency. The full configuration override is a few short blocks; everything else is the release default. The choices that matter:

- **Countries: DK, DE, SE, NO** — Denmark and its synchronous neighbours. The Dutch (COBRAcable) and British (Viking Link) interconnectors are thereby cut from the model.
- **Clustering: administrative, at bidding-zone level** — the transmission grid (built by PyPSA-Eur from OpenStreetMap data) is aggregated to the twelve actual market price areas rather than to synthetic clusters, so every zonal price in Section 6 has a direct Nord Pool/EPEX counterpart.
- **Weather and demand year: 2024** — the note's data year throughout, matched to the prebuilt `europe-2024-sarah3-era5` ERA5/SARAH-3 cutout, which is downloadable without a Copernicus account (2024 is a leap year; the workflow drops 29 February, leaving 8760 hours). Renewable profiles come from atlite on that cutout; loads are the realised 2024 hourly demand from an archived ENTSO-E Transparency snapshot; fleets from powerplantmatching and OPSD (today's capacities). Reservoir hydro carries the 2024 runoff, with one caveat: the annual statistics used to normalise inflow levels end at 2023, so the 2024 level — and with it the Nordic water values of Section 6 — rests on a regression of those statistics on annual runoff rather than on a published figure. The resulting Norwegian inflow, 116 TWh, is within the historical range.
- **No carbon price** enters the marginal costs, and the exported network is *unsolved*: all optimisation, and any policy, happens in the note's own pipeline.

The workflow output (the clustered, electricity-only network) is then trimmed by a short export script: the sector-coupling placeholders PyPSA-Eur attaches for its downstream stages (zero-capacity hydrogen and battery components) are dropped, every capacity is fixed at its existing value, the TYNDP zone codes are renamed to the market names (DK1, DK2, DE, SE1–SE4, NO1–NO5), branches are renamed by corridor, and time series are stored in single precision. The published artefact is ∼2 MB and loads with one line of PyPSA.

One adjustment and one honest limitation carry into Sections 6 to 9. The adjustment: PyPSA-Eur's AC capacities are thermal ratings, but a zonal market trades on commercial NTCs, which are much smaller on the borders — with the physical ∼10 GW on the Jutland–Germany border, DK1 would never decouple from the German price, which is not how the market behaves. The export therefore derates the two cross-border AC corridors (DK1–DE to 2.5 GW, DK2–SE4 to 1.7 GW) to the maximum NTCs published for 2024; intra-country AC and the DC cables keep

[23] `https://github.com/PyPSA/pypsa-eur`; code MIT, data CC-BY-4.0/CC0 under REUSE annotations.

their capacities. One wrinkle: the Øresund NTC is directionally asymmetric — 1.7 GW toward DK2 but 1.3 GW toward SE4 — and a symmetric AC line cannot carry that, so the export keeps the larger direction. The limitation is now a smaller one, and it is described in the next subsection.

C.4 Calibrating the network to its year

Two properties of the shipped network needed attention before any section solved it, and Section 6, which holds the network against the 2024 market, needed one more.

The hydro level. The regression above lands Norwegian reservoir inflow at 124 TWh together with run-of-river, about ten percent below the 137.6 TWh that NVE states as Norway's normal annual hydro production (reference period 1991–2020) and well below the 140 TWh Norway's plants produced in 2024 (Statistics Norway 2025, a record year, with reservoirs ending above their median, both figures are Statistics Norway's and NVE's). Every section's network is therefore put on the normal-year level when it is loaded: reservoir inflow is scaled so that inflow plus run-of-river over the year equals the normal production. That is the right level for the 2050 reference cases of Sections 7 to 9, which should not inherit either a dry regression or a record year. Solved at the shipped level, the model turns Norway into a net importer where the real Norway exported 18 TWh, and the whole Nordic plateau couples to the German thermal price.

The internal Nordic grid. The cross-border AC corridors were derated to commercial capacities on export, but the corridors inside Sweden and Norway and between them kept their thermal ratings, three to five times the transfer capacities the market trades on, so the model moves water south far more freely than the market can. Section 6 derates every corridor whose capacity exceeds TYNDP's reference-grid transfer capacity to that figure, never uprating (10 corridors change, all of them Nordic).

The year itself. Section 6 scales Norwegian inflow once more, by a single factor found by a few secant steps, such that the solved hydro output equals the 2024 production (the factor on top of the normal level is 1.13; inflow and output differ because a full reservoir spills). Table 9 shows what each step does, starting from the network at its normal-year level. Both steps are calibrations to an outcome, and the note says so where it uses them; they are confined to Section 6.

C.5 The weather of the expansion sections

The network arrives with one year of availability profiles, computed on the full reanalysis grid with real power curves and land-use weighting. Section 6 solves on them, because it is held against the 2024 market and wants the best profiles available. The expansion sections need something the network cannot supply: Section 9's sweep re-solves the capacity expansion problem on one weather year at a time, and needs every zone to see the *same* year. So Sections 7 to 9 all solve on a second source: hourly wind speed at 100 m and surface solar radiation from the **Open-Meteo historical archive** (ERA5-derived, CC-BY 4.0, no account required — the same source as the

Table 9: What the two calibrations do to the Nordic price level, at the 65 €/t of Section 6, against the 2024 market.

	NO2	SE3	SE4	NO hydro	NO net export
	mean price, €/MWh			TWh	TWh
network at the normal-year hydro level	61	59	59	125	-1
Norwegian inflow scaled	58	58	58	140	14
Nordic corridors at reference NTCs	67	60	60	125	-1
both (the section's instance)	59	58	58	140	14
the 2024 market	50	36	50	140	18

temperature series of Section 5), for 10 years, at each zone's own coordinate in the network. Wind speed is converted to availability by the textbook power curve (nothing below 3 m/s, cubic to a rated 12 m/s, flat to a 25 m/s cut-out); radiation is scaled by 1000 W/m^2. Section 7 and Section 8 solve on the archive's 2024, so the reference case of Section 7 is one of the columns of Section 9's sweep, not a separate construction of the same year.

Two things about this construction matter more than the details.

Levels come from the network, shapes from the archive. Each swapped profile is rescaled so that its annual mean equals the mean of the profile the network already carried. The capacity factors therefore remain the carefully derived ones — computed with real power curves and land-use weighting over a grid — while the archive supplies only the timing and, crucially, the correlation between zones. A single point and a stylised power curve is a good way to know when a calm week happened and where; it is not a good way to know a capacity factor, and it is not asked to be.

What it still simplifies. One coordinate stands for a whole bidding zone. For Denmark that is defensible; for Germany it is not really, since a country that wide smooths its own wind across its geography in a way a single point cannot. The derived German profile is therefore more volatile than the German fleet actually is, and the cross-border correlations in the sweep are, if anything, a little too low. Demand is held at the network's values throughout, so the sweep does not capture the way a cold still week raises load at the same time as it removes wind — a correlation that would make the tails worse, not better. Both effects point the same way: the spread reported in Section 9 is a lower bound on the risk.

C.6 How the year is aggregated

A year has 8760 hours, and a capacity expansion problem over twelve zones at full resolution runs to several million variables. The published instances therefore reduce the year to a smaller number of snapshots, and how this is done is an input like any other. Two schemes are implemented in the course repository, and the note uses one of them.

Every k-th hour. Keep one hour in k and weight it by k. Simple, and adequate for a dispatch problem with short-lived storage: at 1095 snapshots (every eighth hour) it reproduces Section 7's full-year carbon price, mix and battery build. It fails on a seasonal store: sampled eight hours apart, the intra-day flexibility that makes most of a hydrogen store unnecessary is invisible, and Section 8's store comes out 1.8 times its full-year size.

Chronological segments. Cut the year into 1095 contiguous segments of unequal length, between 2 and 26 hours (median 8), short where load and weather move and long where they do not, each entering the problem as one period at its hourly means and weighted by its duration. Chronology is intact, so a store of any duration sees the real sequence of the year; what is lost is the variation *within* a segment. This is the scheme every network section uses, Sections 6 to 9, at 1095 segments for the note's figures and 336 for the instances the repository ships by default.

The check. Both expansion sections were re-solved at all 8760 hours to grade the aggregation. The reference case of Section 7 builds 3.8 GW of battery in the full year against 2.7 on segments and 6.5 GW of offshore wind against 6.6, at a Danish bill net of trade of 2.34 against 2.35 bn€ and Danish emissions of 0.37 against 0.36 Mt. The hydrogen store of Section 8 is 18.93 TWh in the full year against 19.31 on segments, and its carbon price 663 against 663 €/t. Every number the note quotes from the network sections is the segmented one; the full-year figures are quoted where a section's headline depends on them. One consequence of segments is that every mean and every duration curve in Sections 6 to 9 weights each snapshot by the hours it stands for: a calm week that became one long segment counts for a week, not for one observation.

D What is actually inside the models

Every model in this note is a simplification, and by now you have met the individual simplifications one at a time, in the "which way does it bias the answer?" boxes. This appendix collects them. It answers, in one place, the question you should ask of *any* energy system model somebody puts in front of you: What is in it, what is not, and which way does each omission push the result? Read it as a checklist rather than as prose.

D.1 From the written models to the solved ones

From Section 6 onwards the simulations are assembled in a modelling framework from standard building blocks (a market, a generator, a storage unit, a line, a conversion between two markets) rather than written out equation by equation. Nothing is hidden by this: Every block expands into constraints of the kind written in the text, and every shadow variable the note reads is the dual of one of those constraints. The course repository documents the correspondence block by block; the note does not repeat it.

D.2 What the models contain

The network behind Sections 6 to 9 is derived from PyPSA-Eur (Hörsch et al. 2018); Appendix C documents the derivation. What survives into our models is listed in Table 10.

Table 10: What survives of the network into the models.

Ingredient	As represented here
Geography	12 real bidding zones, DK1, DK2, DE, SE1–SE4 and NO1–NO5, each with its own hourly price.
Generation fleet	67 generators covering CCGT, OCGT, coal, lignite, nuclear, oil, biomass, waste, geothermal, run-of-river, onshore and offshore wind, and solar, at today's capacities.
Weather	Hourly availability profiles $\gamma_{g,h}$ per zone and technology: the network's own reanalysis profiles in Section 6, and the zonal weather archive in Sections 7 to 9, all twelve zones on one year at a time and correlated across the map as the atmosphere correlates them (Appendix C.5).
Transmission, DC	9 named HVDC cables (Skagerrak, NordLink, Kontek, Konti-Skan, the Great Belt, Baltic Cable and others), each with a capacity limit and *controllable* flow in both directions.
Transmission, AC	17 AC circuits with reactances, so flows obey Kirchhoff's voltage law: Loop flow is represented, and power cannot be routed freely within the synchronous area. Cross-border AC capacities are derated to commercial NTCs, and in Section 6 the internal Nordic corridors as well (Appendix C.4).
Hydro	14 storage units, Nordic reservoirs and pumped hydro, with historical inflow, round-trip efficiencies, and durations from 19 hours up to several thousand hours. This is what makes the Nordic price duration curves flat, and the water value an equilibrium object the model computes rather than a number we assume.
Storage state	A cyclic condition: The state of charge must end the horizon where it began, so no energy is conjured by emptying the tank on the last day.
Curtailment	Represented, free, and unlimited (see below).
Sector coupling	Section 5: a heat market, a gas boiler, and a heat pump with an hourly, temperature-dependent COP. Section 8: a shiftable vehicle-charging block, and an industrial hydrogen demand served by reforming, imports, electrolysis and storage, with a turbine that may burn hydrogen back.
Demand curtailment	Sections 7 and 8: a step demand curve in every zone, with a last-resort block at the value of lost load and two cheaper blocks (Section 7.2).
Carbon policy	Section 7: a common carbon price on every zone, swept; and, as a scenario, a territorial CO_2 budget on Danish generation alone. Section 8: one CO_2 budget over all twelve zones, whose dual is the carbon price.
Investment	Section 7: the Danish fleet is wiped and rebuilt from continuous capacity variables priced at annualised capital cost, under stylised land caps; the neighbours keep their fleets and may add to them under published potentials. Section 8: every zone builds beside its existing fleet, interconnectors may be bought.

D.3 What the simulations add

The models written in the text are kept minimal, because every extra constraint is a page of algebra that buys no new economics. The *simulations* behind the figures of Sections 6 to 9 carry three further constraints, applied identically in every solve. Each is listed here with its parameters and its known limitations. None of the three changes the structure of any argument in the text, and their combined effect on the headline numbers is modest. One feature that is deliberately *not* here is an operational reserve requirement: Reserve and commitment detail belong to dedicated operational studies, and the capacity expansion literature this note builds toward omits them in comparable solves.

- **Ramping limits.** The slow thermal carriers get PyPSA-Eur's own hourly limits per unit of capacity: coal 0.9, lignite 0.6, nuclear 0.3 per hour. Biomass, which PyPSA-Eur does not rate, takes coal's 0.9 (the Danish units are converted coal boilers), and waste 0.6, from the Danish Energy Agency catalogue's note that the boiler regulates about 1% per minute; gas turbines traverse their whole range within an hour and get none. One wrinkle: Most figures solve on segments of the year several hours long (Appendix C.6), and there is no exact way to state an hourly ramp limit on a multi-hour step. Scaling the limit up by the step's length would say a coal plant can traverse its range between snapshots, but the aggregated series is already a smoothed version of the hourly path, so that reading compounds one optimism with another and every limit goes vacuous. The simulations instead impose the true hourly rates per step, deliberately conservative. Limitation: These are average-fleet numbers, and there are no start-up costs or minimum loads behind them (that would need integer variables, and the identity between dual and price would fail).
- **Forced outages.** Thermal availability is derated: 0.95 for gas and oil plants, 0.90 for coal, lignite, nuclear, biomass and waste. These are generic reliability assumptions, not statistics for these particular fleets. Limitation: Derating shaves every hour a little instead of removing whole plants for whole weeks, so the model never experiences the correlated loss of several units at once, which is the tail event reserve margins exist for.
- **Transmission losses.** Each HVDC crossing loses a flat 3% (converter pair plus cable, independent of length); AC flows lose energy by a piecewise-linear approximation of the quadratic resistive loss. Limitation: The linearised loss enters as an inequality, so in a zero-price hour the optimiser may "burn" surplus energy in a line. Harmless here, where curtailment is free, but a known artefact of the formulation.

D.4 What the models do not contain

Each row of Table 11 names an ingredient that is absent, says where it would enter, and states the *direction* in which its absence bends the answer.

Table 11: What the models do not contain, where it would enter, and the direction of the error its absence makes.

Absent	What it would be	Direction of the error
Start-up costs, minimum load, unit commitment	Integer on/off decisions per plant per hour	Makes the problem non-convex, so the identity between dual and price would fail. Their absence understates the cost of cycling plants and overstates how cleanly the market can be priced.
Value of lost load, in the dispatch models	In Sections 3 to 6 load is a fixed number that must be served, effectively $v = \infty$ (Section 2.4). The expansion models of Sections 7 and 8 price it, as a three-block step demand curve	Removes the top of the price distribution wherever it is absent. Understates average prices, peaker revenue, storage value and demand-response value in the dispatch sections. In the expansion sections the remaining coarseness (three blocks rather than a demand curve) overstates the carbon price needed to hit a target.
Uncertainty itself: forecast error and reserves	Stochastic availability and outages; a reserve requirement per hour, holding headroom against them	The simulations derate for outages (Appendix D.3), but that is a deterministic proxy: The optimiser still sees every realisation in advance, and no reserve is ever held. The model sees *variability*, never *uncertainty*; it understates system cost and overstates how well a renewables-heavy system can be operated.
Intra-zonal congestion	A nodal network inside each zone (Section 6.1)	All congestion rents reported are cross-border only. Understates the true cost of moving power, and hides redispatch entirely.
Reactive power, voltage, frequency	The full AC power flow	Ancillary services are outside the model, as is most of what a system operator actually worries about minute to minute.
Demand elasticity	A downward-sloping demand curve per hour	Only the heat pump of Section 5 and the three blocks of Section 7 respond to price, and as step functions. Overstates the carbon price needed to hit a target, because almost nobody is ever asked to consume less.
Must-run generation and negative bids	Subsidised or inflexible plants bidding below zero	The model's price floor is zero; the market's is not. This is one of the three tail mismatches in Section 3.6.
Market power	Strategic withholding by pivotal suppliers	Model prices are a competitive *benchmark*: a lower bound on what a strategic player could extract.
Lumpy investment, lead times, vintages	Integer capacity, build delays, existing-plant ages	The expansion sections build fractional gigawatts instantly. They cannot speak to the *pace* of transition, nor to stranded assets.
Technology learning	Costs falling with cumulative deployment	Costs are fixed at a catalogue year; the model cannot represent the feedback from deployment to cost that has driven the actual solar and wind cost decline.
CHP	A fixed electricity-to-heat ratio (Section 5.4)	Misses the technology that historically defined Danish district heating, and the one that *decorrelates* heat and power prices.
A hydrogen network	Pipelines letting one zone sell hydrogen to another (Section 8.5)	Each zone's hydrogen sector is self-contained, so a zone with cheap power and no domestic hydrogen demand (Norway, here) gets no hydrogen sector at all and cannot export. Understates the electrolytic

One that is not in the table, because it is not an omission. Section 7 and Section 8 use *different* land caps for the same Danish zones: the first uses this note's own stylised limits, the second the published per-zone potentials that its eleven other zones also use. Neither is wrong, and harmonising them would cost Section 7 the property that makes it teachable, namely caps that are round numbers a reader can hold in their head. But the two sections are therefore not two runs of one model, and no quantity should be carried from one to the other.

D.5 Three questions worth answering explicitly

Three entries above need a paragraph each.

Is curtailment in the model? Yes, and it is free. A renewable generator's constraint is an inequality, $q_{g,h} \leq \gamma_{g,h}\bar{q}_g$, so the planner may always use less than the weather offers, at no cost and with no limit. When it does, the spilled energy is curtailment (Section 3.3). Two consequences follow. First, this is why the model's price floor is zero rather than negative: The marginal generator in a surplus hour is a wind turbine that is indifferent, not one paying to stay on. Second, free curtailment is optimistic. Real curtailment is contested, sometimes compensated, occasionally physically constrained, and in some markets a subsidised generator would rather bid negative than be curtailed at all.

Are ramping or flexibility costs in the model? In the written models, no: Every generator can move from zero to full output between two consecutive hours, at no cost. Section 4.3 derives what changes when it cannot (the merit order becomes time-dependent and plants enter it at an effective marginal cost $c_g + \xi_{g,h} - \xi_{g,h+1}$), precisely so that you can reason about the size and sign of the omission. The simulations do impose ramp limits on coal, lignite, nuclear, biomass and waste, held at their hourly values even on the multi-hour segments they solve on, and Figure 4.3 sweeps the rate on DK1's own slow plant, but ramping carries no *cost* anywhere. The note's models remain systematically optimistic about how easily a renewables-heavy system can be operated, just a little less so than the written versions alone would be.

Do the storage plants have perfect foresight? Yes, completely. The optimisation is a *single* problem covering the whole horizon, so every battery and every Norwegian reservoir chooses today's charging knowing every price for the rest of the year. This matters more the longer the storage; Section 9.4 prices it for the battery of Section 4 and says what to expect for a reservoir.

D.6 The bias index

Finally, Table 12 collects the note's "which way does it bias the answer?" boxes. The habit they represent: Name the assumption, then work out the sign of its error before you argue about the number.

None of this is an argument against using the model. Every alternative to a transparent optimisation model is a less transparent one, and a model whose omissions are catalogued, with

Table 12: The bias index: every bias box of the note, collected.

Assumption	Where	Sign of the error
Load always served ($v = \infty$)	Section 2.4	Understates prices, peaker revenue, storage and demand-response value
One zone, one country, one direction of trade	Section 3.5	Understates the value of Danish wind (no exports); overstates the value of the wires; understates the network cost of VRE
Frictionless ramping, perfect foresight for storage	Section 4.3	Overstates fleet flexibility; understates the value of everything that supplies flexibility; overstates storage value
Constant COP (had we used one); a fixed fleet	Section 5.4	Would overstate heat pump value and understate the peak load electrification adds; overstates electrification's emissions
Zonal prices, fixed NTCs, perfect foresight over inflow	Section 6.4	Understates the cost of moving power and the Nordic cable rents; overstates Nordic prices in a wet year
Perfect foresight in investment	Section 7.5	Understates transition cost; overstates confidence in the mix
No hydrogen network; a small footprint; a biogenic backstop	Section 8.5	All three understate the case for hydrogen as electricity storage
One point per zone; demand held fixed across weather years	Appendix C.5	Understates cross-zone correlation and the cold-still-week coincidence of high load with low wind; the weather spread of Section 9 is a lower bound
A cost projection read as a probability	Section 9.2	Understates cost uncertainty: The database publishes one number per technology per year, not a distribution, and its past errors on solar and wind have run in one direction

signs, is far more useful than one whose author has not thought about them. The point of the catalogue is that it converts "the model is a simplification" from an excuse into an instrument: Given a result you are unsure about, you can now find the row that threatens it and say which way it moves.

E What the literature finds, and where this note sits

Section 8 tests two published claims and reports the result. This appendix does the wider job: It collects what a handful of recent studies of the European system actually find, says which of their conclusions this note's models are equipped to reproduce, and says which ones they are *not* equipped to reproduce, and in which direction that bends our answers.

Two warnings about how to read it. First, every study below is between five and fifteen times the size of the model of Section 8, and most of them cover all of Europe and every sector; where we disagree with them, the prior should be that they are right and we are small. Second, this is a short list chosen for what it teaches, not a systematic review. It is enough to place this note honestly and not enough to settle anything. Everything cited here is on the course reading list, with an open-access link wherever the authors have provided one. You are not expected to have read any of it before this appendix; you are expected, afterwards, to be able to open one of these papers and work out which ingredient produced its headline result.

E.1 Four strands

(i) The division of labour between storage technologies. Victoria et al. (2019) is the canonical statement, and it is the paper Section 8 is written against. Its findings, in its own terms: Electric batteries act as short-term storage counterbalancing solar, while hydrogen storage smooths wind's longer fluctuations; significant storage appears only once CO_2 is cut by more than about 80% of 1990 levels; and at a 95% cut the optimal system holds battery energy of about 1.4× and hydrogen storage of about 19.4× average hourly electricity demand. It also finds that flexibility from electric-vehicle batteries *avoids the need for additional stationary batteries* and reduces the use of pumped hydro.

(ii) Offshore wind is an infrastructure question. Once land binds, how much offshore wind a system can absorb depends on what it is allowed to build to bring the energy ashore. Glaum et al. (2024) put numbers on this for the North Sea: With point-to-point connections of the kind built today, 310 GW can be integrated; allow meshed offshore grids and offshore hydrogen and the figure rises to 420 GW, with savings up to 15 bn€ a year. Floating wind, about 75 GW, appears *only* when offshore hydrogen production is available, and the bulk of the offshore energy comes ashore as hydrogen rather than as electricity, by about two to one.

Lüth et al. (2023) ask the same question at Danish scale, for the planned energy islands, and find that distance decides: Close-to-shore farms serve the power system best through cables, while electrolysis earns its keep far from shore because it avoids expensive long-distance cable. Their sharpest result for our purposes is a different one: **Electrolyser investment is sensitive to the hydrogen price and much less so to the carbon price.**

(iii) Flexibility is a taxonomy, not a quantity. Chyong et al. (2024) classify what a 2050 European system needs and who supplies it: Hydrogen storage and production carry the inter-seasonal margin; a combination of electrical storage, hydrogen-based storage and hybrid heat pumps carries the intraday one; and *spatial* flexibility, national networks and cross-border interconnection, is a source in its own right rather than a way of delivering the others. Their headline comparative claim is that in deep-decarbonisation scenarios cross-border *electricity* transmission capacity comes to exceed that of natural gas, while the need for gas networks and storage falls substantially.

Odenweller et al. (2026) approach flexibility through prices, by coupling a long-horizon integrated assessment model to an hourly power system model. Their result is an asymmetry worth remembering: Flexible loads (electrolysers, smart-charging vehicles) pay *below*-average electricity prices, while heat pumps pay almost *twice* the average, because their demand peaks in winter when the system is tight. Remove demand-side flexibility and prices rise for every end-user.

(iv) Hydrogen wants a network. Neumann et al. (2023) price the thing this note most conspicuously lacks: A European hydrogen network is worth up to 26 bn€ a year, about 3.4% of system cost, and between 64% and 69% of it could be repurposed natural-gas pipeline rather than new build. Expanding the electricity and hydrogen grids together saves the most, around 10%.

E.2 Which conclusion becomes which ingredient

The table below is the design record for Section 7 and Section 8: For each conclusion above, the model ingredient it implies, and whether this note carries it. It is worth reading as an argument about *modelling* rather than about Europe. A conclusion in this literature is almost always the visible end of one ingredient, and the way to read any such study is to ask which ingredient produced it.

Table 13: Conclusions of the expansion literature, the model ingredient each implies, and whether this note carries it.

Conclusion	Ingredient it implies	In?	Where, or why not
Batteries do the diurnal work	Storage with a duration choice	✓	Section 4; a duration menu in Section 7
Hydrogen does the seasonal work	Electrolyser, store, turbine	✓	Section 8: built, and the turbine declined (Result 9)
EV batteries displace stationary ones	Shiftable charging block	✓	Section 8, rung (ii)
A biogenic fuel puts a plateau into the abatement curve	Biomethane peaker	✓	Result 7: the switching price in Section 7, and one of the margins the budget of Section 8 crosses
Spatial flexibility is its own category	Buyable interconnection	✓	Section 8: bought only once the budget binds
Demand elasticity shapes price formation	A step demand curve	✓	The three blocks of Section 7, in every zone
Firm low-carbon capacity competes for the same role	Nuclear candidate	✓	On the menu in both sections; never built
Hydrogen wants a network	Hydrogen transport between zones	—	**Absent.** The largest single omission; see Appendix D
Offshore integration is an infrastructure choice	Offshore hubs, meshed grid, offshore electrolysis	—	**Absent.** Our offshore wind lands directly in its zone
Costs fall with deployment	Endogenous learning	—	**Absent.** Costs are a catalogue year (Section 9.2 varies them exogenously)
Sector coupling deepens the storage threshold	Heat, transport, industry demand	partly	Heat in Section 5, transport and industrial hydrogen in Section 8; no aviation, shipping or industrial process heat

E.3 Where Section 8 agrees, and where it does not

Now the comparison, in Table 14. Our numbers are the ones Section 8 reports; theirs carry citations.

Table 14: Published claims of the expansion literature against what Section 8 finds.

Published claim	What we find	Reading
Storage matters only below ∼20% of 1990 emissions (Victoria et al. 2019)	Battery build goes 3.0 GW unconstrained → 63.9 at −75% → 102.6 at −99%	**Agrees.** Storage is a deep-end phenomenon in our model too
EV flexibility avoids the need for *additional* stationary batteries (Victoria et al. 2019)	54% of the battery displaced at −75%, 32% at −99%	**Agrees, and quantifies.** Substantial but partial, not a replacement
Hydrogen storage smooths wind's synoptic variability (Victoria et al. 2019)	0.0 GW of hydrogen turbine and 0.0 TWh burnt back, at every budget	**Disagrees**, and Result 9 says why. Our footprint is small and has a cheaper backstop
Cross-border electricity capacity grows and overtakes gas (Chyong et al. 2024)	19.9 of 30 GW of offered interconnection bought, and only once the budget binds	**Agrees in direction.** We cannot speak to the comparison with gas
Electrolyser investment tracks the hydrogen price more than the carbon price (Lüth et al. 2023)	Electrolysers enter as the *carbon* budget tightens, against a fixed import price	**Apparent tension**, and an untested one: We never varied the hydrogen price
A hydrogen network is worth up to 26 bn€/yr (Neumann et al. 2023)	No hydrogen trade between zones at all	**Untested.** Known sign: We understate the electrolytic route
Offshore integration reaches 420 GW with meshed grids and offshore hydrogen (Glaum et al. 2024)	Offshore lands in its zone; no offshore hubs exist	**Untested.** We cannot represent the question
Flexible loads pay below-average prices; heat pumps almost double (Odenweller et al. 2026)	Not computed: We report system prices, not per-technology ones	**Not computed.** Everything needed for it is in the solved model

Down the right-hand column, the pattern is this. Where this note carries the ingredient, it broadly reproduces the literature: the storage threshold, the EV substitution, the value of spatial flexibility. Where it disagrees outright, it disagrees for a reason that is itself a result (Result 9). And where it is silent, it is silent because an ingredient is missing, not because the question is hard: Three of the eight rows are omissions with a known direction of error, listed in Appendix D. A small model will not tell you the cost of decarbonising Europe. It will tell you which ingredient produces which conclusion, and that is what lets you read the studies that do.

E.4 Questions this leaves open

Each of the following is a claim above that this note's models could be made to test, in roughly increasing order of effort. They are here because they are the natural next things to do with the code, and because a reader who works through one of them will have done something closer to research than to an exercise.

i. *Find the break-even utilisation.* Result 9 argues that a round-trip hydrogen chain fails on running hours. Using the prices of Section 8, compute the electrolyser full-load hours at which the chain breaks even, and compare it with the 3,979 hours the industrial customer actually delivers.

ii. *Separate the two prices.* Lüth et al. (2023) find electrolyser investment more sensitive to the hydrogen price than to the carbon price. Vary each in turn, the import price at a fixed budget and the budget at a fixed import price, and see whether that holds here.

iii. *Who pays what?* Compute the load-weighted average price paid by each flexible block and by the electrolysers, against the simple average price, and test the asymmetry of Odenweller et al. (2026). Nothing new needs solving; the prices are already in the solution.

iv. *Price spatial flexibility.* Re-solve rung (i) with the interconnectors fixed instead of buyable, and read the difference as the value of the option to build wire.

v. *Give hydrogen a wire.* Allow hydrogen to move between two zones and see whether Norway, which has cheap power, no domestic hydrogen demand, and therefore no hydrogen sector at all, builds an electrolyser. This is the smallest edit in the list with the largest payoff against Neumann et al. (2023).